

1 Factors affecting inferences on natural mortality and
2 associated environmental effects in state-space
3 age-structured assessment models

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Abstract

Treatment of natural mortality is a major consideration in assessment models and state-space approaches allow estimation of temporal variation in this mortality rate as well as effects of specific covariates. However, there has been no investigation of the reliability of inferences made from state-space assessment models when there is process error or covariate effects on natural mortality. We conducted a large-scale simulation study that considers models fit to data simulated from operating models with alternative assumptions defined by several factors. We fit estimating models to simulated observations with alternative assumptions on the inclusion of covariate effects, estimation of the median natural mortality rate, and the source of temporal variability in the population demography. Our results suggest that estimation of covariate effects on natural mortality is possible and reliable even when the population process error source was mis-specified in some scenarios with lower observation uncertainty, contrast in fishing pressure, and higher covariate temporal variability. Estimation of assessment output such as spawning stock biomass can be reliable even when assumptions for variation in natural mortality are mis-specified.

keywords: state-space assessment models, time-varying natural mortality, bias, AIC

Introduction

State-space population models are now used widely for fisheries stock assessment in Europe, the United States, and Canada (Nielsen and Berg, 2014; Cadigan, 2016; Pedersen and Berg, 2017; Stock and Miller, 2021). Because application of these methods are considered best practice and recommended for the next generation of stock assessment models (Hoyle et al., 2022; Punt, 2023), it is expected their use will only grow globally. An appeal of state-space models lies in their separation of sources of biological and measurement variability by treating temporal variation in population characteristics as statistical time series with periodic observations measured with error. Through advances in computational capacity, we can use sophisticated numerical approaches to estimate parameters of these models as a combination of fixed and random effects (Thorson and Minto, 2015; Kristensen et al., 2016).

State-space stock assessment models, with non-linear functions of latent processes and numerous observation types with different probability distributions, represent one of most complex classes of state-space models. The literature on the effects of various factors on reliability of inferences from state-space assessment models is growing (Li et al., 2024; Miller et al., In review). The importance of contrast in population size and fishing mortality (F) and quality of data used to fit assessment models including the state-space variety is known (Magnusson and Hilborn, 2007; Miller et al., In review). Furthermore, estimation of natural mortality (M), and even temporal variability in M is possible in many scenarios (Lee et al., 2011; Cadigan, 2016; Miller and Hyun, 2018; Miller et al., In review).

The effects of temporal variation in recruitment via undefined or explicit environmental factors have been extensively investigated in both traditional assessment models and state-space models (Myers, 1998; Haltuch and Punt, 2011; Johnson et al., 2016; Miller et al., 2016). Reliability of estimating environmental and spawning biomass effects on recruitment in state-space assessment models requires a combination of strong effects, informative age composition data, contrast in the environmental covariate, and lower recruitment variability

(Britten et al., 2026; Miller et al., In review).

A critical aspect of fisheries assessment models and their use in management is short-term projections that are used to determine catch advice. While understanding drivers of recruitment is important, particularly for subsequent effects on reference points, recruitment in short-term projections typically has little impact on the exploitable biomass in the first few projection years. However, assumptions for M have immediate and larger effects on projected biomass because they affect the abundances at older age classes at the end of the data time series that constitute spawning biomass and catch (Brodziak et al., 2008; Stock et al., 2021).

Because of the effects of M on both biological reference points and short-term projections, better understanding sources of variation in M would provide more accurate estimation of abundance and productivity and, therefore, improved management. Temporal variation in M is less studied than recruitment, but its importance for explaining variability in observations has been demonstrated in state-space assessment models for Atlantic cod and yellowtail flounder (Cadigan, 2016; Stock et al., 2021). Deriso et al. (2008) also demonstrated the importance of several factors affecting M for Pacific herring.

Assessment models could include temporal variation in many aspects of population dynamics or how observations are related to the population. For example the Woods Hole Assessment Model (WHAM) can include process errors treated as random effects for transition in cohorts over time (hereafter referred to as apparent survival), catchability for indices of abundance, selectivity of fishing fleets or indices, movement between regions, or in M (Stock and Miller, 2021; Miller et al., 2025). However, mis-specifying the type of temporal variation could lead to biased estimation of population size and stock status and, therefore, poor fisheries management decisions (Legault and Palmer, 2016; Szuwalski et al., 2018). Studies of the reliability of inferences regarding the presence of temporal variability in M are limited. Miller et al. (In review) found AIC could accurately distinguish process errors in apparent survival

but not for those specifically due to M except when uncertainty in population observations (indices, catch, and age composition) was low and there was greater temporal variation in M . Li et al. (2024) found including more sources of process error than existed in the operating model (OM) was a better model-building approach than excluding them a priori.

Here, we present results from a simulation study with OMs varying by degree of observation-error uncertainty, sources of process error, fishing history, temporal variation in environmental covariates, and magnitude of the effect of the covariate on M . We fit the simulated observations from these OMs with estimating models (EMs) that make alternative assumptions for sources of process error and whether median M and covariate effects were estimated. We evaluated the effects of these factors on convergence of fitted models, whether Akaike’s information criterion (AIC) can determine the correct source of process error and correct assumption about covariate effects on M , estimation bias for median M and covariate effects, and the accuracy for estimators of important terminal year M and spawning stock biomass (SSB).

Methods

Our analyses used WHAM to construct both OMs and EMs (Miller and Stock, 2020; Stock and Miller, 2021; Miller et al., 2025). WHAM has been used extensively to configure OMs and EMs for several other simulation studies (Stock et al., 2021; Legault et al., 2023; Li et al., 2024; Britten et al., 2026; Li et al., 2026a) and is used to assess many commercially important stocks in the Northeast U.S. (e.g., NEFSC, 2022a,b, 2024). We used version 1.0.6.9000 (commit 77bbd94) to generate all results.

We completed a simulation study with 288 OMs. The factors defining the configuration of each OM, described in detail in subsequent sections, include source of process error (3 levels), index and catch observation uncertainty (2 levels), environmental covariate uncertainty (2 levels), temporal variation in the latent environmental covariate (4 levels), size of the

113 covariate effect on M (3 levels), and fishing history (2 levels). We simulated 100 data sets
 114 for each OM that included simulations of process errors.

115 For each simulated data set we fit a set of 12 EMs. The factors that distinguish the EMs, also
 116 described in detail below, include source of population process error type (3 levels) whether
 117 median M was estimated or assumed known (2 levels), and whether the environmental
 118 covariate effect on M was estimated or not (2 levels).

119 The sources of population process error that were used in the OMs or assumed in the EMs
 120 were on recruitment only (R), recruitment and apparent survival (R+S), or recruitment
 121 and M (R+M). We did not use the log-normal bias-correction feature for process errors
 122 or observations described by Stock and Miller (2021) for OMs and EMs (Li et al., 2026b).
 123 Simulations were all carried out on the University of Massachusetts Green High-Performance
 124 Computing Cluster. Code for completing the simulations and summarizing results can be
 125 found at https://github.com/timjmiller/SSRTWG/ecov_study/mortality.

126 **Operating models**

127 **Environmental covariate**

128 In WHAM, environmental covariates are modeled as state-space processes with annual ob-
 129 servations of the true latent covariate (Miller et al., 2016; Stock and Miller, 2021). In our
 130 simulations, we assumed the latent covariate to be a stationary first order autoregressive
 131 (AR1) process,

$$X_y|X_{y-1} \sim N\left(\mu_E(1 - \rho_E) + \rho_EX_{y-1}, (1 - \rho_E^2)\sigma_E^2\right),$$

132 with marginal mean $\mu_E = 0$ and variance σ_E^2 . The four configurations of the latent environ-
 133 mental covariate in the OMs assume one of two values for the marginal standard deviation
 134 $\sigma_E \in \{0.1, 0.5\}$ and for the autocorrelation parameter $\rho_E \in \{0, 0.5\}$.

WHAM treats the observations of the latent environmental covariate as unbiased and normal,

$$x_y|X_y \sim N(X_y, \sigma_e^2).$$

The standard deviation of the environmental observations in the OMs was one of two values $\sigma_e \in \{0.1, 0.5\}$. Figure S1 provides example simulations of the latent covariate and observations under the alternative configurations.

Population

Many of the characteristics of the population biology and structure including the age classes (10 age classes: 1 to 10+), time span (40 years), maturity (Figure S2, top left), growth (Figure S2, top right), time of spawning (1/4 of the year), and recruitment (Figure S2, bottom right) were identical to Miller et al. (In review). The maturity at age was a logistic function with age at 50% maturity (a_{50}) = 2.89 and slope = 0.88 and weight at age was derived from a von Bertalanffy growth function where $t_0 = 0$, $L_\infty = 85$, and $k = 0.3$, and a length-weight relationship

$$W_a = \theta_1 L_a^{\theta_2},$$

where $\theta_1 = e^{-12.1}$ and $\theta_2 = 3.2$.

In WHAM, the general model for M in year y is a log-linear function of both covariate effects X_y and process errors $\varepsilon_{M,y}$ and a parameter β_M that defines median M ,

$$\log M_y = \beta_M + \beta_E X_y + \varepsilon_{M,y},$$

where the process errors are modeled as random effects that may, in general, be autocorrelated Gaussian random variables,

$$\varepsilon_{M,y}|\varepsilon_{M,y-1} \sim N(\varepsilon_{M,y-1}, (1 - \rho_M^2)\sigma_M^2)$$

(Stock and Miller, 2021), but we assumed $\rho_M = 0$ in our R+M OMs. We assumed the median M , $e^{\beta_M} = 0.2$, is constant across ages. For R and R+S OMs and EMs, $\varepsilon_{M,y} = 0$. For all R+M OMs, we assumed the same standard deviation $\sigma_M = 0.3$, which was estimated in the R+M EMs. The covariate effect is one of 3 alternative values in the OMs, $\beta_E \in \{0, 0.25, 0.5\}$. The parameters defining the simulated covariate time series, size of the covariate effect, and any M random effects resulted in a range of different levels of variation in annual values (Figure S3).

We assumed expected recruitment each year is from a Beverton–Holt stock-recruit relationship (SRR)

$$R_y = \frac{aSSB_{y-1}}{1 + bSSB_{y-1}}.$$

All biological inputs to calculations of SSB per recruit (i.e., weight, maturity, and M at age) were constant in the R and R+S OMs without covariate effects on M . Therefore, steepness and equilibrium unfished recruitment were also constant over the time period for those OMs (Miller and Brooks, 2021). As in Miller et al. (In review), our assumed biological inputs and selectivity (defined below) with constant M result in equilibrium F that reduces SSB per recruit to 40% of the unfished level is $F_{40\%} = 0.348$. With an assumed unfished recruitment of $R_0 = e^{10}$, setting $F_{MSY} = F_{40\%}$ results in a steepness of 0.69 and $a = 0.60$ and $b = 2.4 \times 10^{-5}$. For R+M OMs and all OMs with covariate effects on M , steepness was not constant but we used the same a and b parameters as other OMs which equates to a steepness and R_0 at the median of the time series models for M and the covariate.

We also used the same two fishing scenarios as Miller et al. (In review) for OMs. In the first scenario, the stock experienced overfishing at $2.5F_{MSY}$ for the first 20 years followed by fishing at F_{MSY} for the last 20 years (denoted $2.5F_{MSY} \rightarrow F_{MSY}$). In the second scenario, the stock was fished at F_{MSY} for the entire time period (40 years). The magnitude of the overfishing assumptions was intended to reflect estimates of overfishing for Northeast US groundfish stocks from Wiedenmann et al. (2019).

We configured all R, R+S, and R+M OMs with uncorrelated random effects on recruitment with standard deviation on $\log(\text{recruitment})$ $\sigma_R = 0.5$. Miller et al. (In review) used this same assumption for R+M OMs and other OMs with fishery selectivity and index catchability process errors. For R+S OMs, apparent survival process errors were uncorrelated with $\sigma_{2+} = 0.3$.

Catch and index observations

We defined the generation of observations of aggregate (total combined across ages) catch and indices and corresponding age composition identical to Miller et al. (In review). There was a single fleet operating year round for catch observations with logistic selectivity defined with $a_{50} = 5$ and slope = 1 (Figure S2, bottom left). Observations were generated for all 40 years of the model. There were two index time series intended to represent fishery-independent surveys occurring in the spring (0.25 way through the year) and the fall (0.75 way through the year). We assumed catchability of both surveys to be 0.1.

WHAM treats the aggregate catch and index observations as log-normal and we assumed the standard deviation parameter was 0.1 for all annual catch observations. We assumed catch and index age-composition observations were generated from a logistic-normal distribution where errors on the multivariate normal scale were independent (Aitchison and Shen, 1980; Stock and Miller, 2021; Trijoulet et al., 2023). The standard deviation parameter was also constant across ages. There were two levels of observation-error variance for aggregate indices and age composition for both indices and fleet catch. A low-uncertainty specification assumed standard deviation parameter equal to 0.1 for all annual aggregate index observations and the standard deviation of the logistic-normal for all age-composition observations was 0.3. In the high-uncertainty specification the standard deviation parameter for the annual aggregate index observations was 0.4 and that for the age-composition observations was 1.5.

Estimating models

There were three factors defining the configuration of each EM: 1) whether β_M was estimated or assumed known; 2) whether an environmental effect β_E was estimated or not; and 3) whether the process errors were assumed on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and M (R+M). We fit each EM to each of 100 simulated data sets from each OM. The configuration of the process errors in the EMs generally matched the corresponding options in the OMs. For example, random effects were assumed uncorrelated for R EMs and R OMs and for R+S EMs and R+S OMs. However, R+M EMs did not assume M random effects were uncorrelated (ρ_M was estimated) although they were simulated without correlation in the R+M OMs. The environmental covariate observations were included in all EMs, whether effects on M were estimated or not, to ensure comparability of AIC. All fixed-effects parameters for selectivity, catchability, fully-selected F , mean recruitment, initial abundance at age, and variances for logistic-normal age composition distributions were estimated. Any process-error variance parameters for recruitment, apparent survival, and M were also estimated. For all EMs, the standard deviation parameter for aggregate catch and index observations was assumed known at the true value whereas that for the logistic-normal age-composition observations was estimated. Finally, the observation-error variance of the environmental observations and aggregate catch and indices were all assumed known at the true values.

Measures of reliability

EM convergence

We measured the frequency of convergence when fitting each EM to the simulated data sets for each OM. There are various ways to assess convergence of the fit (e.g., Carvalho et al., 2021) but we defined successful convergence as the Hessian of the marginal log-likelihood

being invertible and providing variance estimates for the fixed effects parameters as recommended by Miller et al. (In review).

AIC for model selection

We measured the frequency of correct model selection using marginal AIC. Using the marginal log-likelihood is known to be appropriate for mixed-effects models specifically when the analyst is interested in fixed effects rather than random effects (Vaida and Blanchard, 2005). Other studies have shown the utility of marginal AIC in distinguishing process error and other model structure for state-space assessment models applied to a particular stock (Miller and Hyun, 2018; Stock and Miller, 2021).

For a given OM, the set of EMs that were compared all made the same assumption about β_M (estimated or treated as known at the true value). For EM m , the marginal AIC is a function of the marginal log-likelihood maximized with respect to the estimated fixed effects in the EM θ and the number of estimated fixed effects $n(\theta)$,

$$\text{AIC}_m = -2 [\text{argmax}_{\theta} \log L_m(\theta) - n(\theta)].$$

All EM fits that successfully completed the optimization were used for this set of analyses. We did not condition on convergence as defined above because some lack of convergence would be expected for the correct behavior of more complicated models that include process errors that did not exist in the OM. For example R+M EMs fit to R OMs would be expected to estimate no variance in the M random effects and the estimated variance parameter going to zero would cause poor convergence but have the same marginal log-likelihood (with more fixed effects) and therefore higher AIC as expected. The correct EM made the correct assumption for the source of process errors (R, R+S, R+M) and either included a covariate effect on M when an effect was simulated ($\beta_E \in \{0.25, 0.5\}$) or did not include an effect when one was not simulated ($\beta_E = 0$).

Parameter estimation bias and accuracy

All results here used OM simulations with fits that satisfied the Hessian-based convergence criterion described above. We used this conditioning to reflect how practitioners would proceed in analyses of model fits with real assessment data. That is, practitioners would ensure models converged such that Hessian-based standard errors were available for all estimated model parameters.

We focused on statistical behavior of estimators of the covariate effect on M ($\hat{\beta}_E$); the median M parameter ($\hat{\beta}_M$); and terminal year M , SSB, and fully-selected fishing mortality (F). In preliminary analyses we examined results for the estimators of all the annual values for M , SSB and F over the whole time series, but we found no appreciable differences in patterns across the various factors defining the OMs and EMs. Furthermore, because results for terminal year F were generally inversely related to those for SSB, they are not discussed further but provided in the Supplementary Materials.

We calculated errors of $\hat{\beta}_E$ and $\hat{\beta}_M$, and the Hessian-based standard error estimators of these parameters ($\widehat{SE}(\hat{\beta}_E)$ and $\widehat{SE}(\hat{\beta}_M)$) as

$$\delta_i = \hat{\theta}_i - \theta_i$$

where θ_i is the true value of a given parameter or attribute for simulation i and $\hat{\theta}_i$ is the estimate from an EM fitted to data from simulation i . For standard errors the true standard error was determined as the standard deviation of estimates across converged fits of a given EM and OM combination.

We calculated the relative errors for terminal year M , SSB, and F as

$$\text{RE}_i(\theta_i) = \frac{\hat{\theta}_i - \theta_i}{\theta_i}. \quad (1)$$

268 To measure accuracy we also calculated the root mean square error (RMSE),

$$\text{RMSE}(\hat{\theta}) = \sqrt{\frac{1}{n} \sum^n (\hat{\theta}_i - \theta_i)^2},$$

269 where n is the number of simulations with converged fits of a given EM and OM. The true
 270 values for terminal year SSB, and in some OM M (R+M OM and any OM with covariate
 271 effects on M), varied among simulations. Finally, we calculated 95% confidence intervals for
 272 $\hat{\beta}_E$ and $\hat{\beta}_M$ using Hessian-based standard error estimates for converged EMs that estimated
 273 these parameters in each OM simulation and recorded whether they included the true value.
 274 Estimation of terminal year M could appear to be unbiased even for EMs that assumed
 275 no temporal variation in M because the assumed or estimated β_M parameter can be equal
 276 or near the median of the true terminal year values. This centering of the true values at
 277 the constant value of M in the EM happens because the mean of the M process error and
 278 covariate distributions are assumed equal to 0. Therefore, we calculated the correlation of
 279 estimates and simulated values for terminal year M for a given EM and OM combination,

$$\text{Cor}(\hat{M}, M) = \frac{\text{Cov}(\hat{M}, M)}{\text{SD}(\hat{M}) \text{SD}(M)}.$$

280 For R+M OM and any where there were covariate effects on M , high correlation (approach-
 281 ing 1) along with median relative errors close to 0 indicates ability of the EM to estimate
 282 annual M . If correlation and median relative errors are both near 0, this indicates no ability
 283 to estimate annual M . Note, also that the correlation is undefined when either the estimated
 284 or true value are constant across simulations because the corresponding standard deviations
 285 are 0. This applies to R and R+S EMs that assumed β_M to be known and $\beta_E = 0$, and R
 286 and to R+S OM where $\beta_E = 0$. We also did not use calculated correlations for EMs when
 287 there were only 2 converged fits because they are uninformative. Because true terminal year
 288 SSB values also varied across simulations we made the same correlation calculations for them

as well.

Summarizing results across OM and EM attributes

There were numerous OM and EM attributes across all simulation scenarios and, like Miller et al. (In review), we used two methods to summarize the major factors explaining differences in results for each measure of reliability. First, we fit regression models with the responses as the measures of reliability described above and the predictor variables were defined based on OM and EM attributes (e.g., MacKinnon et al., 1995; Wang et al., 2017; Harwell et al., 2018). We used logistic regression for binary indicators of convergence and AIC-based selection of the appropriate assumption of the covariate effect on M . For indicators of AIC-based selection of EM process error source (multiple categories) we performed multinomial regressions. For other measures of reliability we fit linear regression models. Relative errors of terminal year M , SSB, and F (Eq. 1) are bounded below at -1, so we used a transformation of these values

$$f(\hat{\theta}_i, \theta_i) = \log [\text{RE}(\hat{\theta}_i, \theta_i) + 1]. \quad (2)$$

We omitted fits where estimated terminal year SSB, M , or F was equal to zero because of the undefined transformation. RMSE is bounded below at 0 so we used a log transformation for responses in corresponding regressions. We did not use any transformation for the correlation of estimated and true values for terminal year M and SSB. Note, that there was only one RMSE and correlation calculated for an EM from all simulation fits from a given OM scenario. For all regressions we fit separate models with just individual OM and EM factors included, all factors combined, with all second order interactions, and with all third order interactions. For the multinomial regression, we used the `vglm` function from the VGAM package (Yee, 2008, 2015). We calculated percent reduction of residual deviance (from the null model with no factors) for each of the regression fits. As Miller et al. (In review) noted, there is a lack of independence of the results for each EM fitted to the same simulated data set of a given

313 OM and therefore we did not perform formal statistical analyses of effects of OM and EM
314 attributes.

315 For the second summarization method we used classification and regression trees (Breiman
316 et al., 1984) to illustrate the primary OM and EM attributes and interactions that partition
317 the values for each measure of reliability (e.g., Gonzalez et al., 2018; Collier et al., 2022).
318 We used classification trees for categorical measures (convergence and AIC model selection)
319 and regression trees for the other measures with continuous scales (errors, relative error, and
320 RMSE). The response variables were the same as the regressions for the deviance reduction
321 analyses. We used the `rpart` function in the `rpart` package (Therneau and Atkinson, 2025)
322 to fit trees. Full trees were determined using default settings except that we increased the
323 number of cross-validations to 100. For one classification tree fitted to AIC selection of
324 process error configuration results for R+S OMs, we had to make a small changes to the
325 default prior probabilities (the proportions of each class in the in the data) used in the
326 criterion on whether to allow branches from a given node because the default priors would
327 not produce any branches. For clarity, we manually pruned the full trees to show just the
328 primary branches.

329 All OM and EM factors we used to examine reductions in deviance and regression trees are
330 given in Table S1. We provide more detailed results for each EM and OM configuration
331 in the Supplementary Materials using similar methods to Miller et al. (In review). The
332 detailed results present probability of convergence, probability of each EM having lowest
333 AIC, median error and RMSE of β_M and β_E and corresponding standard error estimates,
334 and median relative error and RMSE for terminal year SSB, F , and M . There are also
335 detailed results for correlation of estimates and true values for terminal year M and SSB.

Results

EM Convergence

For fitted logistic regression models to indicators of convergence, the EM process error assumption provided the largest percent reduction in deviance (9-25%) of all OM and EM attributes for all three OM process error types (Table 1). Including second order interactions of OM and EM factors also provided further reduction of residual deviance (between 7 and 11%), indicating successful convergence depended on a combination OM and EM attributes.

Classification trees for each OM process error source all had the primary branch defined using the same attribute (EM process error assumption) that provided the largest reduction in deviance (Figure 1). Across all fits for each OM process error configuration, convergence was best for R+S OMs, but better convergence occurred when the process error was correct for R and R+S OMs. For R+M OMs, R EMs converged well (88%) and good convergence of correctly specified R+M EMs (80.2%) required constant fishing pressure and median M parameter (β_M) to be known. All branches based on observation error of covariates or other assessment data (indices and age composition) showed better convergence with more precise observations. Better convergence also occurred for EMs that did not estimate covariate effects regardless of whether the OM simulated them, EMs that assumed median M was known, and when OMs assumed greater temporal variability in the true environmental covariate. EMs assuming R+M process error (either correctly or not) converged better for OMs with constant fishing pressure.

AIC performance

Process error source selection

For all OM process error types, the level of error in both the population and covariate observations were the attributes that resulted in the largest percent reductions in deviance for multinomial regression fits to indicators of AIC selection of the process error configuration (Table 2). For R+S and R+M OMs, level of error in population observations provided deviance reduction (19% and 13%, respectively) much larger than any other OM and EM factors whereas error in covariate observations provided the largest reduction (8%) for R OMs. Lesser deviance reduction (2-3%) occurred for the true covariate effect size and whether the EM made the correct assumption about including a covariate effect for R OMs and error in covariate observations for R+M OMs. Including second and third order interactions, provided the largest reductions in residual deviance for R OMs (24%) and little reduction for R+S and R+M OMs (4% and 6%, respectively).

The attributes defining the primary branches of classification trees matched those that provided the largest reductions in deviance for all OM process error types (Figure 2). AIC performed poorly in selecting the correct process error for R+M OMs where lowest AIC occurred for the simpler R EMs. However, accuracy of AIC-based selection was high for R and R+S OMs. In R+S OM simulations where mis-classification occurred, the simpler R EMs had lowest AIC. For R OMs, there was a small decrease in accuracy when the OMs included the largest covariate effect on M , but in those scenarios, AIC was more accurate for the process error assumption when the EMs also incorrectly assumed no covariate effect.

Covariate effect selection

The factors explaining the largest reduction in deviance for logistic regression models fitted to indicators of AIC selection of the correct covariate effect assumption depended on the

381 magnitude of the true effect assumed in the OM (Table 3). When OMs had no covariate
 382 effect ($\beta_E = 0$) the OM process error type provided the largest reduction in deviance (4%).
 383 When OMs assumed a covariate effect ($\beta_E > 0$), level of error in population observations and
 384 magnitude of temporal variability in the true covariate provided the largest reductions in
 385 deviance (4% and 8-21%, respectively). The reduction in deviance provided by the magnitude
 386 of covariate temporal variability also increased with larger true covariate effect. Including
 387 second and third order interactions provided a modest reduction in deviance (5-7%).

388 The factors defining the primary branches of classification trees for indicators of AIC se-
 389 lecting the correct covariate effect assumption matched the factors providing the largest
 390 deviance reductions (Figure 3). Accuracy of AIC-based model selection for the covariate
 391 effect assumption was high overall for OMs with no covariate effect (88%) (low Type I error)
 392 and low when there was a covariate effect (23% and 35% for OMs with $\beta_E = 0.25$ and 0.5,
 393 respectively) (high Type II error), but higher accuracy occurred for certain combinations
 394 of OM attributes. When the true covariate effect was greatest ($\beta_E = 0.5$), AIC accuracy
 395 was 91% for EMs fit to OMs with high temporal variability in the covariate and low error
 396 in population and covariate observations regardless of whether the EM process error was
 397 correct. However, when the covariate effect was weaker ($\beta_E = 0.25$), high accuracy only
 398 occurred for a subset of those OMs with R or R+M process errors (76%), and the further
 399 conditioning on OMs with high autocorrelation of the covariate provided higher accuracy
 400 (86%). When there was no covariate effect, AIC accuracy was best for R OMs (95%), and for
 401 R+S and R+M OMs, accuracy was improved if the OMs also had lower error in population
 402 observations and the EM process error assumption was correct (85%).

Covariate effect inferences

Estimation bias

For regression models fit to errors in estimation of the covariate effect β_E , we found very little reduction in deviance for any of the OM and EM factors (Table 4). For the normal model in these regressions, deviance is the sum of squares and there are extremely large estimates of β_E (and errors) for many simulations resulting in extremely large sums of squares and relative differences from the null model are therefore small. Nevertheless, observation error of the covariate provided either the largest or second largest reductions in deviance of any of the single OM and EM attributes for OM process error configuration types (0.04% to 0.08%). OM fishing history provided reductions of similar scale for R and R+S OMs (0.03 and 0.06%, respectively). Other factors providing similar reductions were true covariate effect size for R OMs (0.05%) and level of error in population observations and EM process error assumption for R+S OMs (0.06% for each). Including second and third order interactions also provided relatively large reductions in residual deviance (0.2% to 1.5%).

Median errors for $\hat{\beta}_E$ across all simulations for each OM process error configuration were negative but close to 0 (-0.1 to -0.07) and primary branches were based on factors that matched those providing the largest reductions in deviance (Figure 4). Branches based on level of error in population or covariate observations provided better median estimation error (closer to 0) with lower uncertainty. Branches based on level of covariate temporal variation provided better median estimation error with higher temporal variation. For R OMs with the largest true covariate effect, estimation error was poor unless there was also high temporal variation and low observation error for the covariate. For R+S OMs, we also found lower estimation error when the EM process error assumption matched. We observed low median error across all simulations for R+M OMs with low error in covariate observations. The detailed results for all OM and EM configurations demonstrate that the median errors get worse with increased values for the true β_E for many scenarios without low

error in population and covariate observations and contrast in fishing pressure, indicating that the estimates remain close to zero when the true $\beta_E > 0$ (Figures S16 to S18). The detailed results also indicated little bias in covariate effect estimation regardless of the EM process error configuration when OMs had low population and covariate observation error and contrast in fishing pressure.

Standard error estimation bias

Factors providing the largest reductions in deviance for errors in estimation of the standard error of the covariate effect size ($\widehat{\text{SE}}(\hat{\beta}_E)$) were generally similar to those that were most important for estimation of β_E itself (Table S2). Level of error in covariate observations and EM process error assumption was important for all OM process error configurations (6-13% and 3-4%, respectively). Level of error in population observations provided relatively large reductions in deviance for R+S and R+M OMs (3% and 8%, respectively). The OM fishing pressure history provided a lesser reduction in deviance for all OM process error types (2-4%). Like estimation of β_E , relatively large further reductions in deviance also occurred when second and third order interactions of OM and EM attributes were included in the regression models (11-31%).

Median errors in estimation of the standard error for $\hat{\beta}_E$ across all simulations for a given OM process error configuration were strongly negative (-0.8 to -0.6), but combinations of OM and EM attributes resulted in improved values closer to zero (Figure S4). For all OM process error types, branches with lower error in covariate observations provided better median errors in the standard error estimator (closer to 0). Branches where EMs assumed median M was known also generally improved standard error estimation except for R OMs where error in covariate observations was high. However, branches with higher temporal variability in the covariate in R and R+S OMs provided improved median errors. For some configurations of R+S OMs, EMs with matching process error assumptions provided improved standard error

estimation, but some EMs with the incorrect process error assumptions provided median errors close to 0 for some R and R+M OM.

Examining the detailed results for each OM and EM configuration shows little bias of standard error estimation in the same OM scenarios where there was little bias in estimation of β_E itself was close to zero (Figures S19 to S21). However, in these high information OM, bias of standard error estimation was observed for several EMs that did not have the matching R+S and R+M process error configuration.

Confidence interval bias

For logistic regressions of indicators of confidence intervals including the true value of β_E , factors that provided the largest reductions in deviance were true covariate effect size for R and R+M OM (11% and 3%, respectively) and EM process error configuration for R+S OM (7%) (Table 5). Covariate observation error also provided relatively large reduction in deviance for R OM (8%). Lesser reductions of 1-2% occurred for level of error in population observations and covariate temporal variability for all OM process error types. Including second and third order interactions also provided relatively large further reductions in deviance (5-9%).

Over all simulations, rates of 95% confidence interval coverage indicated negative bias with rates ranging 85-88% for the different OM process error configurations (Figure 5). Factors defining the primary branches matched the factors that made the largest reductions in deviance for fitted logistic regression models. For R OM, the primary branch based on error in covariate observations indicated higher rates of confidence interval coverage with lower error, but when that error was higher, higher rates were associated with OM where no covariate effect was assumed ($\beta_E = 0$). For R OM with some covariate effect and higher error in covariate observations, higher rates of coverage occurred with higher error in the population observations. For R+S OM, higher rates of confidence interval coverage were

observed for EMs that assumed R+S or R+M process errors.

The more detailed results for each OM and EM configuration show that good confidence interval coverage can be obtained for all of the investigated covariate effect sizes when there is low error in covariate and population observations, high covariate temporal variability and contrast in fishing pressure for R and R+S OMs. The EM process error configuration was not a factor for the R OMs, but R+S or R+M EM configurations were necessary for R+S OMs (Figures S22 to S23). However, confidence interval coverage was negatively biased for all EM process error configurations fit to R+M OMs with these conditions even though little bias was observed in estimation of β_E or its standard error (Figure S24). Examining the estimates for each simulated data set when R+M EMs were fit to R+M OMs where we observed little estimation bias, there appears to be a positive correlation of the covariate and standard error estimates such that estimates less than the true value had negatively biased standard error estimates which would make confidence intervals too small (Figure S29).

Median natural mortality rate inferences

Estimation bias

Across all OM process error types, regression models fit to errors for median M parameter estimation ($\hat{\beta}_M$) showed largest reductions in deviance from the type of OM fishing pressure (4-17%), EM process error assumption (2-12%), and the EM covariate effect assumption (3-7%, Table 6). The level of observation error also provided a similar reduction in deviance for R OMs (3%). Relatively large reductions in deviance also occurred with second and third order interactions (7-16%).

Like $\hat{\beta}_E$, median errors for $\hat{\beta}_M$ were negative but close to 0 (-0.1 to -0.07) across all simulations for each OM process error configuration (Figure 6). The primary branches for all OM process error types were defined by the type of OM fishing pressure, with median errors closer to zero when there was a change in fishing pressure for R+S and R+M OMs. Branches based

on the level of error in population observations showed median errors closer to zero with more precise observations. For R+S and R+M OM, branches that included the matching EM process error assumption provided better median errors.

Detailed results reveal that bias of $\hat{\beta}_M$ is negligible for all EM process error assumptions fitted to R and R+M OM when there is contrast in fishing pressure and low population, but the matching EM process error assumption was also important for EMs fit to R+S OM (Figures S30 to S32). For those OM, any attributes related to covariate process error and effects on M had little effect on bias.

Standard error estimation bias

The EM process error assumption provided the largest reduction in deviance for regression models fitted to errors in estimation of the standard errors of $\hat{\beta}_M$ from R (3%) and R+M (7%) OM simulations (Table S3). The type of OM fishing pressure provided the largest reduction (13%) for R+S OM and a lesser reduction in deviance for R+M OM (4%). The EM covariate effect assumption also provided similar reductions in deviance for R+S (9%) and R+M (4%) OM.

The factors defining the primary branches of regression trees matched those providing the largest reductions in deviance for errors in $\widehat{SE}(\hat{\beta}_M)$, but median errors indicated strong negative bias across all OM process error types (Figure S5). Branches with lower error in population observations provided better median errors (closer to 0). For R+S OM, branches where the EM process error assumption was correct provided better median errors.

Detailed results for each OM and EM configuration indicate little bias in certain scenarios where OM had temporal contrast in fishing pressure and low error in population observations (Figures S33 to S35). For R and R+M OM with those conditions, all EM process error assumptions provided median errors close to zero except for OM with the true covariate effect strongest and the true covariate time series was autocorrelated and had high temporal

variation. For R+S OMs with those conditions, only EMs with the correct process error assumption exhibited low (absolute) median error.

Confidence interval bias

For logistic regressions of indicators of confidence intervals including β_M , factors that provided the largest reductions in deviance were the OM fishing history and level of error in population observations for all OM process error types (0.3-6%) (Table 7). Including second and third order interactions provided the largest relative increase in deviance reduction for R+S (8-11%) and R+M OMs (5-8%) and lesser increases for R OMs (3-5%).

The factors defining the primary branches of classification trees for indicators of confidence interval inclusion of β_M matched those providing the largest reductions in deviance for each of the OM process error types (Figure 7). Across all simulations, rates of 95% confidence interval coverage indicated negative bias with rates ranging 83-87% for the different OM process error types. Branches based on contrast in OM fishing history and lower error in population observations indicated less bias in confidence interval coverage except R+S OMs in where the the R EM configurations was incorrectly assumed.

The more detailed results for each OM and EM configuration show that good confidence interval coverage can be obtained for many R and R+S OM scenarios with low error in population observations and contrast in fishing pressure for R and R+S OMs, but coverage was generally unreliable for all R+M OM scenarios (Figures S22 to S24).

Terminal year natural morality rate

Estimation bias

Regressions of errors in estimation of terminal year M show largest reductions in deviance from whether the EM treats β_M as known (Table 8). For R and R+S OMs, annual M is

constant, and therefore, terminal M is known when β_M is assumed known and there will be no error in any of those EMs. Beyond the β_M assumption, the OM fishing history, level of error in population observations, and EM assumptions for covariate effect and process error all provided notable deviance reductions.

Regression trees for errors in estimation of terminal year M have primary branches based on the whether β_M was assumed known or estimated for all OM process error types (Figure 8). For R and R+S OMs, the branches where β_M is assumed known have no error as expected because yearly M is equal to e^{β_M} , but there was also very little error in terminal year M for the same branch in R+M OMs. When β_M was estimated, median errors were near zero for many EMs in R OMs, but median error was problematic for this subset of EMs in R+S OMs. On the other hand, detailed results for each OM and EM attribute show that there was little evidence of bias for R+S OMs when error in population observations was low and there was contrast in fishing pressure and the EM process error configuration was correct (Figure S45). In R+M OMs, median errors indicated little bias overall with temporal contrast in fishing pressure, although detailed results demonstrated best reliability when there was also lower error in population observations (Figure S46).

Estimation accuracy

Like bias of terminal year M , whether the EMs treated β_M as known or estimated provided the largest reduction in deviance (22-45%) for accuracy of estimates as measured by RMSE for all OM process error types (Table S4). Including second and third order interactions also provided large further reductions in deviance (21-38%). Also like bias, regression trees fit to log-RMSE of terminal M showed better accuracy when EMS assumed β_M was known rather than estimated (Figure S6). Better accuracy was also generally associated with lower error in population observations and temporal contrast in fishing pressure. For R+S OMs, R+S EMs provided better accuracy than incorrect process error assumptions.

Using correlation to measure accuracy of terminal year M estimation, provides a different perspective. Treatment of β_M provided relatively large reductions in deviance for all OM process error types (3-12%), but level of variability in the covariate was equally or more important (3-15%) as was whether the covariate effect was included in the EM (12%) for R and R+S OMs (Table S5). Including second and third order interactions also provided relatively large further reductions in deviance (15 to 24%). Regression trees demonstrated better accuracy with higher covariate variability, lower error in population and covariate observations, and when EMs assumed β_M was known (Figure S7).

Terminal year spawning stock biomass

Estimation bias

Fits of regression models to log-scale errors of terminal SSB resulted in largest reductions in deviance for OM fishing history (2-3%), error in population observations (1% for R and R+M OMs), and EM β_M assumption (2%; Table 9). The type of EM process error assumed also provided a relevant reduction in deviance for R+S OMs (1%). Including second and third order interactions of OM and EM attributes provided further relevant deviance reductions of 5-10%.

The primary branches of regression trees fitted to log-scale errors in terminal SSB were based on type of OM fishing history for all OM process error types, but differences in median errors between branches were small (Figure 9). For R+S and R+M OMs, branches that assumed β_M was known indicated less bias than those where it was estimated. For R+S OMs with constant fishing pressure and β_M estimated, less bias was indicated when the EM process error assumption was correct.

Estimation accuracy

Largest reductions in deviance for accuracy of terminal year SSB as measured by log-RMSE were provided by whether β_M was known or estimated (20-21%) and whether there was temporal contrast in fishing pressure (20-33%) across all OM process error types (Table S6). Lesser reductions were also provided by the EM process error assumption for R and R+M OMs. Inclusion of second and third order interactions of OM and EM attributes also provided relatively important reductions in deviance (26-40%).

Primary branches of regression trees fit to log-RMSE of terminal year SSB were based on the same factors that provided the largest reduction in deviance (Figure S8). All branches with temporal change in fishing pressure provided better accuracy than branches with constant fishing pressure. Similarly, better accuracy occurred for branches with β_M assumed known rather than estimated, and lower error in population observations.

Using correlation to measure accuracy of terminal year SSB estimation, temporal contrast in fishing pressure and EM treatment of β_M provided relatively important reductions in deviance like RMSE, but level of error in population observations was important rather than EM process error type (Table S7). Including second and third order interactions also provided relatively large reductions in deviance (6 to 20%). Regression trees demonstrated better accuracy with contrast in fishing pressure, lower error in population observations, and when EMs assumed β_M was known (Figure S9).

Discussion

Our simulation study demonstrated the importance of low observation error, contrast in fishing pressure, and large temporal variability in the environmental covariate for making reliable inferences about effects on natural mortality. Under these conditions, reliable AIC-based selection, estimation of environmental effects on M can be possible even when the

process error was mis-specified (e.g., R and R+S EMs fit to R+M OM). Under those same conditions, reliable estimation of standard errors and confidence interval coverage was also possible for R and R+S OMs especially when the EM process error type was correct. However, there was poor confidence interval coverage for R+M OMs even though there was little bias in estimation of the effect or standard errors, possibly related to correlation of the estimated effects and standard errors. Cadigan et al. (2024) found confidence interval coverage to be biased for SSB and F estimation in a state-space model in some scenarios but they attributed the poor coverage to bias in Hessian-based standard error estimation and their simulations held any process error random effects constant. Consideration of other methods of calculating confidence intervals may be warranted (e.g., Boyko and O'Meara, 2024). Previous research has shown that estimation of a constant M parameter requires contrast in time series and informative data (Lee et al., 2011), so it is no surprise that estimation of covariate effects on M also requires relatively good information via more precise observations and higher contrast in the covariate time series.

(e.g., those based on profile likelihood and/or Monte Carlo sampling of the log-likelihood surface)

We found AIC unreliable for determining the source of process error for EMs fit to R+M OMs. Miller et al. (In review) investigated R+M OMs with two levels of M process-error variability ($\sigma_M \in \{0.1, 0.5\}$) and only found AIC able to accurately distinguish R+M process errors with the higher level of process-error variability ($\sigma_M = 0.5$). Our R+M OM simulations assumed an intermediate level of variability ($\sigma_M = 0.3$), indicating that reliable detection requires $0.3 < \sigma_M \leq 0.5$. In our results and those by Miller et al. (In review), AIC typically chose R EMs which would indicate the fitted R+M EMs would estimate no variability in the M process errors. Future studies like ours where R+M OMs are simulated with greater variability in M process errors would better inform reliability of covariate effect inferences under such scenarios.

Deriso et al. (2008) attempted to estimate process errors as well as covariate effects with M for Pacific herring but similarly found no variability in M , suggesting there was too much uncertainty in the available observations relative to the true temporal variability in M . Given that we found covariate effect inferences using R EMs was reliable in R+M OMs with apparently little variability in M process errors, the findings of Deriso et al. (2008) on covariate effects for Pacific herring would presumably also be robust to true low variability in M . However, they did not account for uncertainty in covariate observations, some of which would probably have substantial uncertainty (e.g., competition and predation covariates). We found higher covariate observation uncertainty to cause true covariate effects to be less detectable using AIC but we did not investigate the implications for incorrectly assuming no covariate uncertainty for covariate inferences.

It was important to consider RMSE and correlation of estimates with the true values for reliable estimation of terminal year M because summaries of relative errors could suggest unbiasedness that is not relevant because the distribution of true values is centered at an assumed constant value or near an estimated constant value in the EM (e.g., R and R+S EMs that have a temporally constant M fit to OMs with temporally varying M). Despite the inability of AIC to distinguish M process errors, reliable estimation of annual M was possible in some scenarios regardless of the EM process error assumption. To achieve little bias and high correlation of terminal M estimates required larger variability in the covariate (when OMs and EMs included the effect on M) along with lower population and covariate observation error. It was also more difficult when EMs estimated β_M rather than treating it as known. However, there was little correlation of terminal year M estimates and true values for R+M OMs without covariate effects for any combination of OM and EM attributes (Figure S52). Therefore, our results suggest estimating annual M is easier when the temporal variation is due to effects of highly variable covariates than unexplained variation in M . Like the results for AIC-based selection, this may be a function of the level of variability in M we assumed for R+M OMs.

Accurate estimation of terminal year SSB, was more robust than terminal year M . When OMs had contrast in fishing pressure and low error in population observations, there was little bias and high correlation of terminal year SSB estimates regardless of whether OMs and EMs included covariate effects. However, for R+S OMs, it was important for the EM process error configuration to be correct. Fortunately, our results and those by Miller et al. (In review), suggest AIC-based model selection is accurate for this type of process error. However, the reliability of the SSB estimation does break down in the less ideal scenarios when there is higher error in population observations, and lack of contrast in fishing pressure (e.g., Figures S53 to S55).

The R+S and R+M EMs both include process error for the (apparent) survival of cohorts and would be expected to perform similarly, and they did in our simulations when the OM and EM matched. However, we found that the accuracy for model output like SSB and F that are important for management was generally worse using R+M EMs for R+S OMs than using R+S EMs for R+M OMs in the high information OMs with contrast in fishing pressure and low error in population observations. Additionally, there are implications for biological reference points for R+M EMs (e.g., Legault and Palmer, 2016) that are not present with R+S EMs. So, we recommend following the suggestion from Li et al. (2024) to prefer R+S EMs over R+M OMs unless strong biological evidence is present to support a particular R+M OM. Such support could be found through both biological understanding (e.g., Trijoulet et al., 2020) as well as statistical properties such as a large delta AIC for R+M compared to R+S associated with greater temporal variability in M (Miller et al., In review).

The pattern of improved inferences using state-space models in the presence of higher contrast in fishing pressure, lower observation error and greater process error variability that emerges from our work here and by ? is consistent with expectations from statistical theory. State-space models are more reliable when observation error is lower than process error (Auger-Méthé et al., 2016), and, more generally, maximum likelihood estimation is less bi-

ased and more accurate as the information in the data increases (Neyman and Scott, 1948; Kiefer and Wolfowitz, 1956). However, the greater complexity of state-space assessment models over, the more well-studied, linear and Gaussian state-space models makes determining appropriate levels of uncertainty in observations and population processes difficult. Statistical information available to the state-space assessment model could be increased with longer time series of observations, more time series (e.g, multiple indices of abundance and corresponding age composition data sets), or more precise observations of a given time series of observations (e.g., lower index observation precision or more sampling for age composition). Therefore, it will be important to conduct self-test simulations (where the OM and EM are the same) on a case-by-case basis to ensure the reliability of models used to manage commercially important fish stocks.

The ability to accurately infer covariate effects on M in some realistic situations indicates that such investigations may be fruitful. Ability to make inferences could improve further when WHAM is extended to incorporate tagging data (Miller et al., 2025). Tagging data can greatly inform M estimation (Pollock et al., 1991; Hampton, 2000) and this impact on M estimation should also apply to estimation of covariate effects or unexplained temporal variation in M . Given our findings and planned future WHAM development, we expect investigations of and accounting for covariate effects on M to become more common within the fisheries stock assessment process. At the same time, it will be equally important to conduct research that will improve our understanding of how best to measure the depletion of stocks and determine catch advice for these stocks with covariate effects on M .

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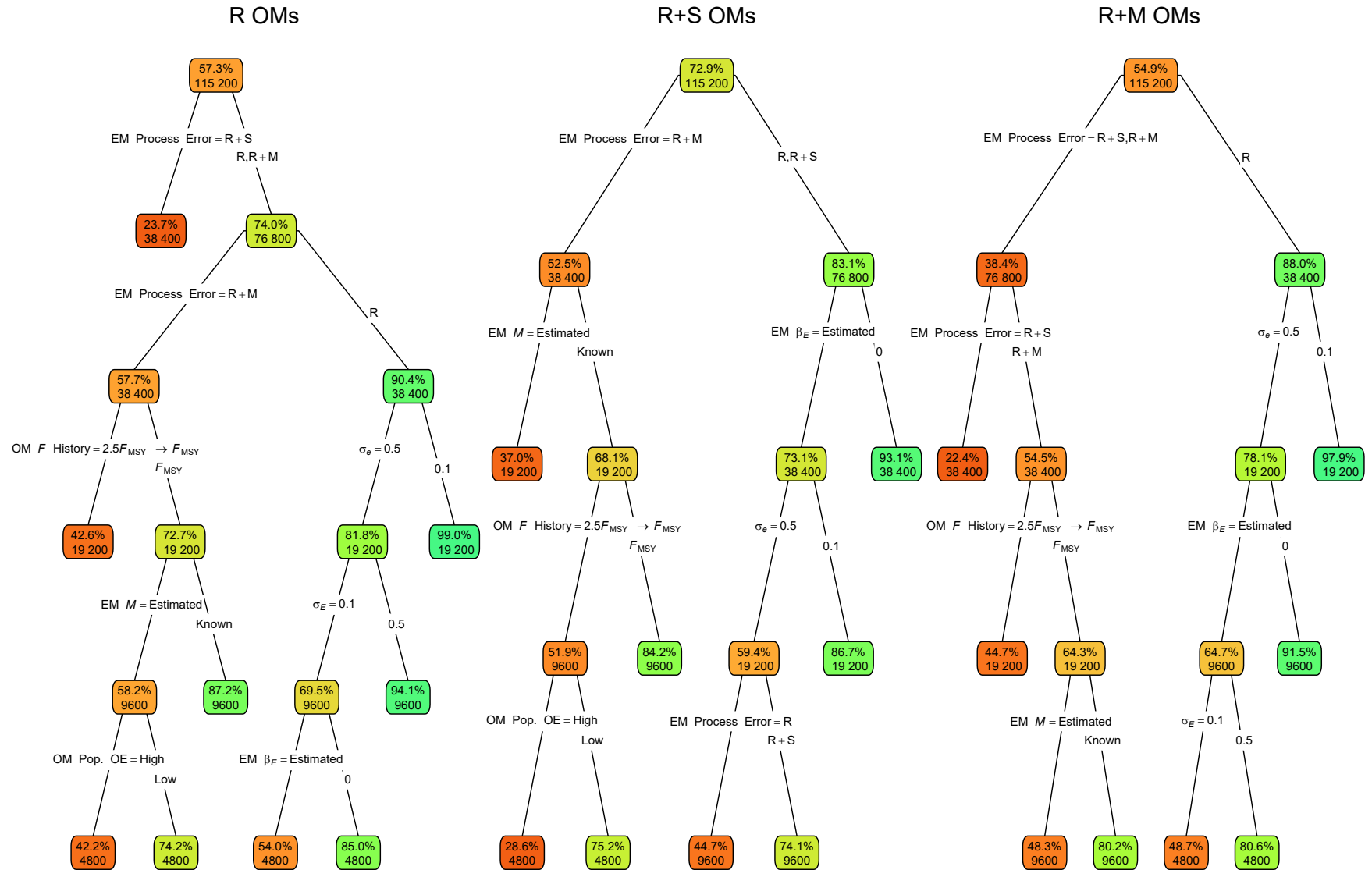


Fig. 1. Classification trees indicating primary factors determining convergence as defined by providing Hessian-based standard errors for operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Nodes denote percent convergence (top) and number of fits (bottom) for the corresponding subset. Lower or higher convergence rates are indicated by more red or green polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.

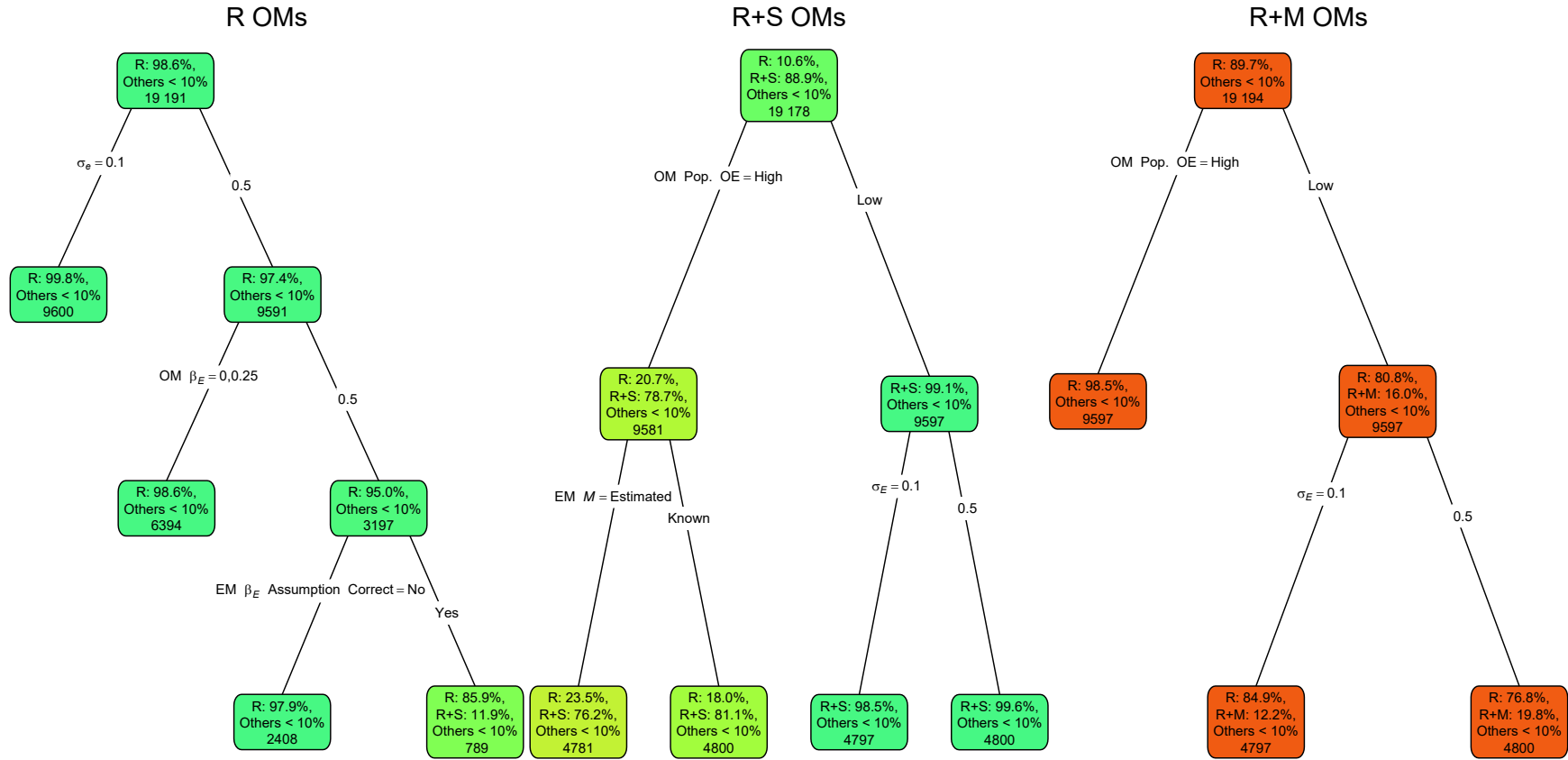


Fig. 2. Classification trees indicating primary factors determining which process error assumption in estimating models (EMs) provides the lowest AIC for operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the percentage of EM fits with a given process error assumption having the lowest AIC and the number of observations for the corresponding subset of fits. Lower or higher accuracy of the process error assumption are indicated by more red or green polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.

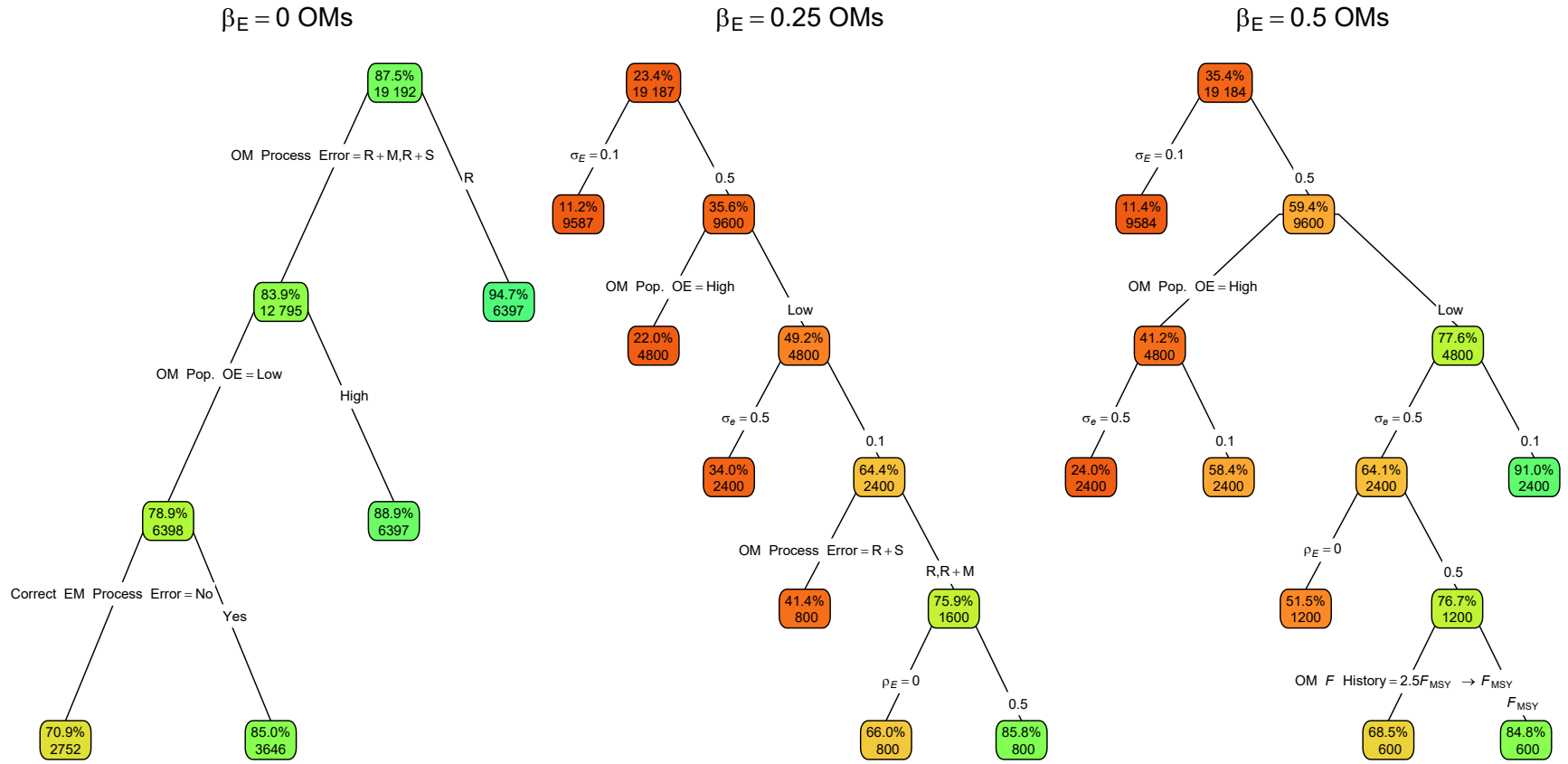


Fig. 3. Classification trees indicating primary factors determining whether the correct EM assumption for covariate effect on M (no effect with OM $\beta_E = 0$ or estimated effect when OM $\beta_E > 0$) provides the lowest AIC for OMs assuming values for β_E of 0, 0.25, and 0.5. Nodes denote the percentage of estimating models (EMs) with correct assumption and lowest AIC (top), and number of observations (bottom) for the corresponding subset. Lower or higher accuracy of the process error assumption are indicated by more red or green polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.

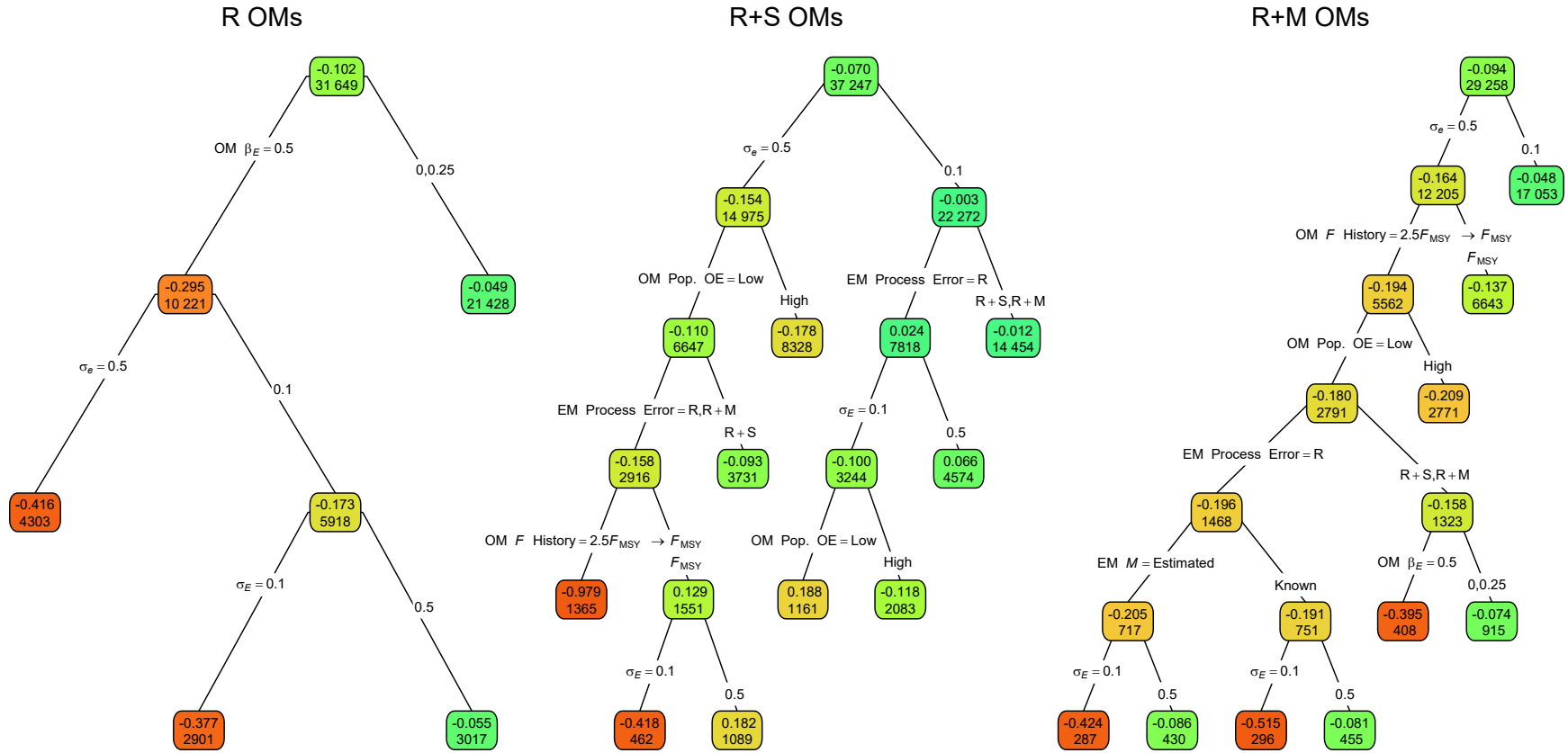


Fig. 4. Regression trees indicating primary factors determining reductions in sums of squares of errors in estimation of the covariate effect on M for operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.

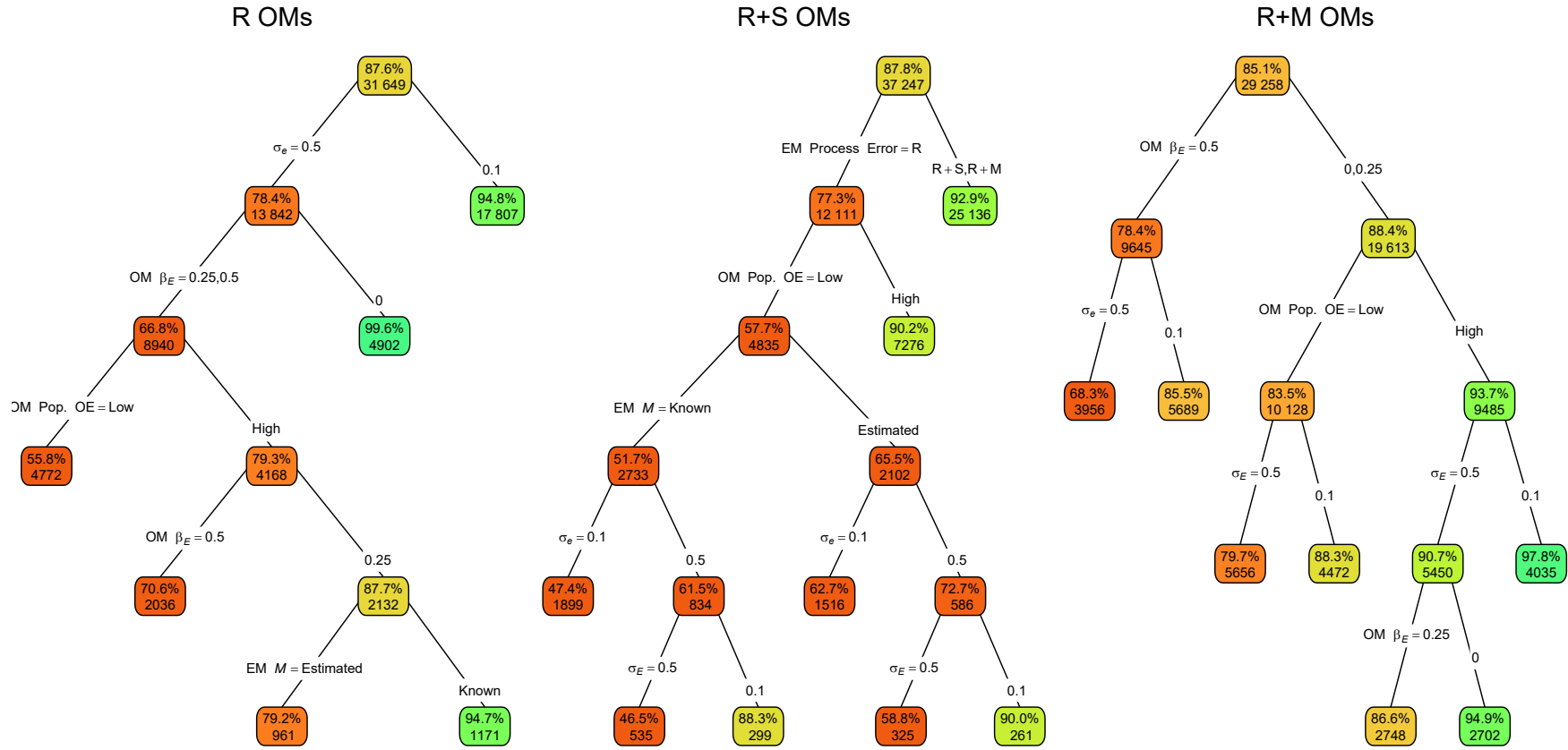


Fig. 5. Classification trees indicating primary factors determining whether confidence intervals for estimated covariate effects on M in estimating models (EMs) fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) included the true value. Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.

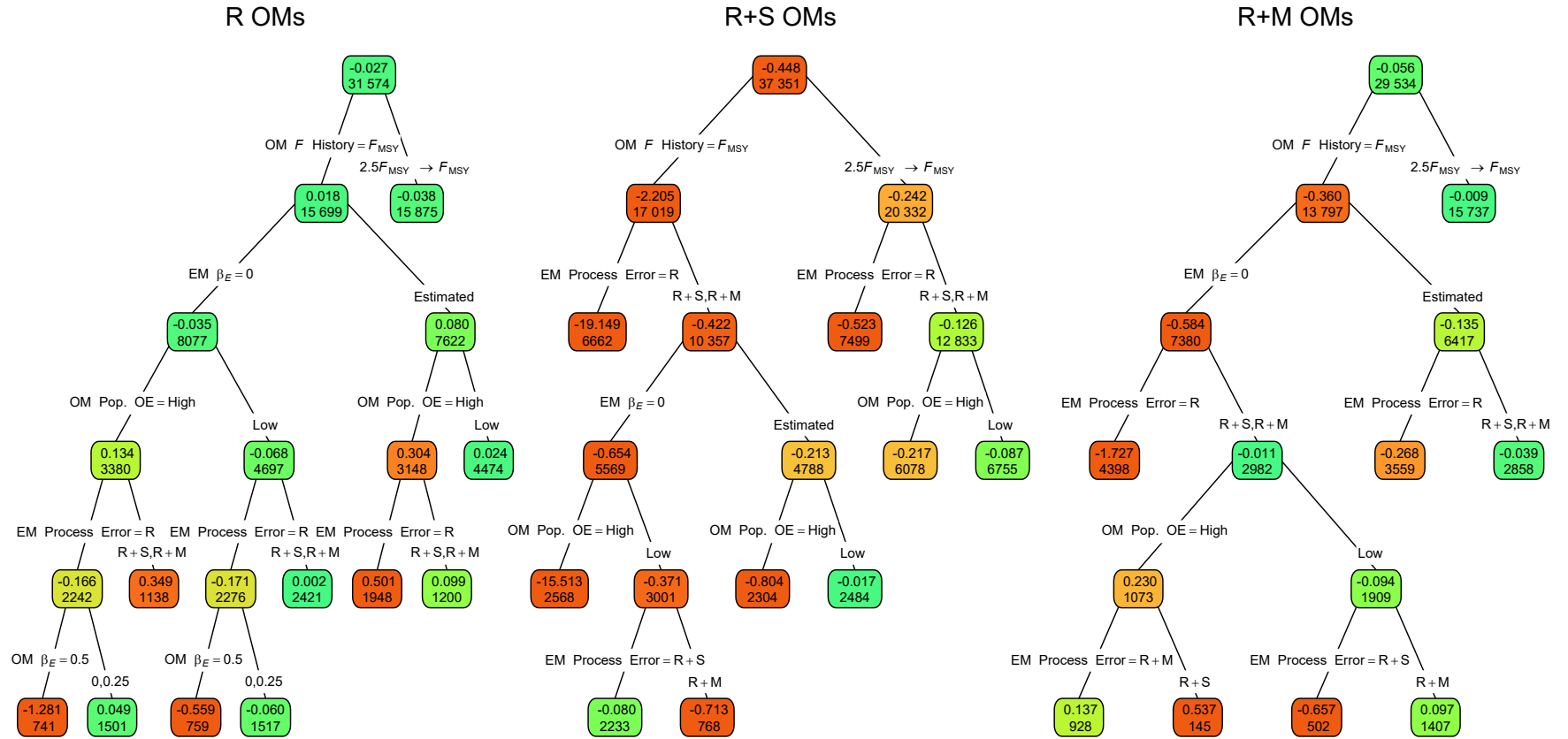


Fig. 6. Regression trees indicating primary factors determining reductions in sums of squares of errors in estimation of the median M parameter (β_M) in estimating models (EMs) fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.



Fig. 7. Classification trees indicating primary factors determining whether confidence intervals for estimated median M parameter (β_M) in estimating models (EMs) fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) included the true value. Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.

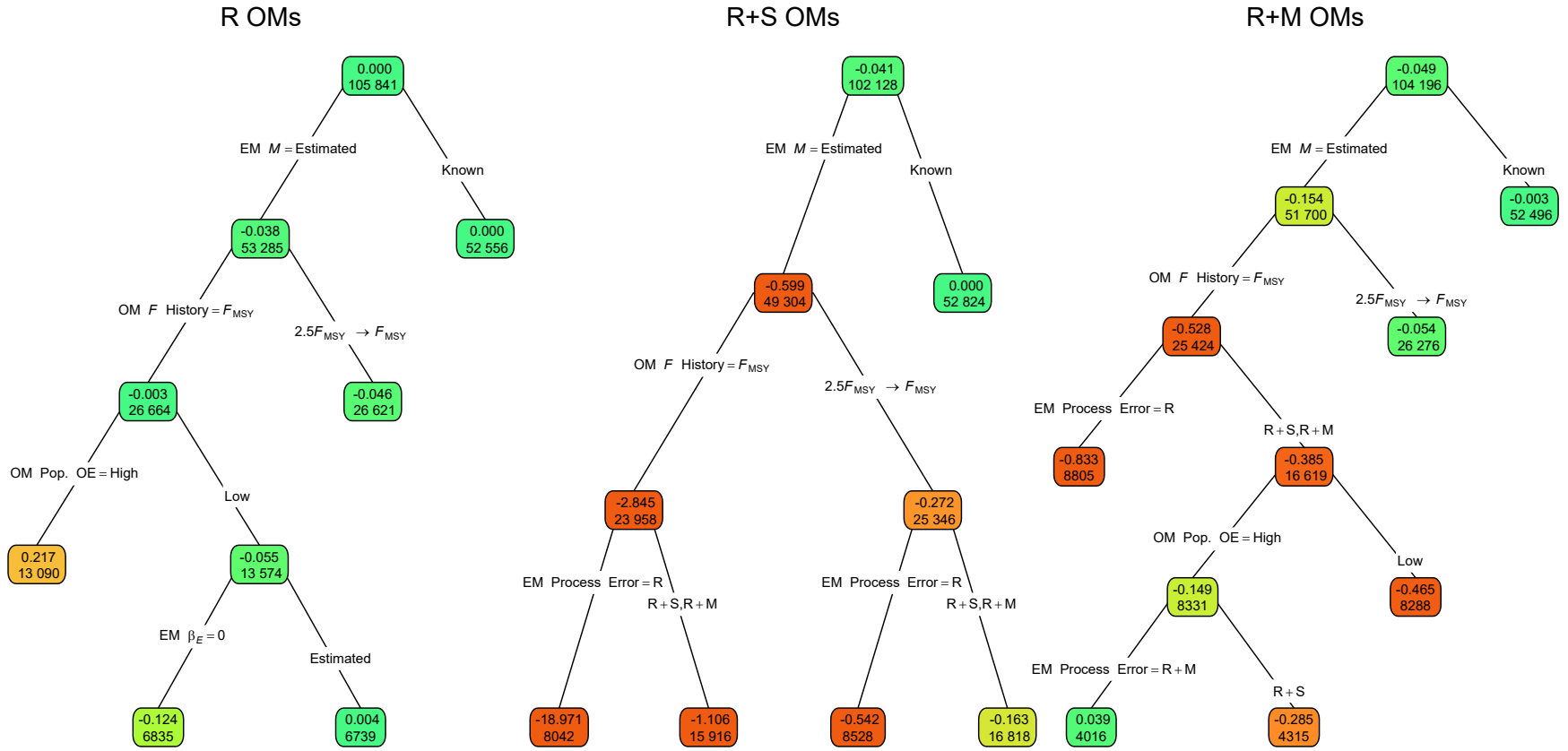


Fig. 8. Regression trees indicating primary factors determining reductions in sums of squares of errors in estimation measured by Eq. 2 for the terminal year M in estimating models (EMs) fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.

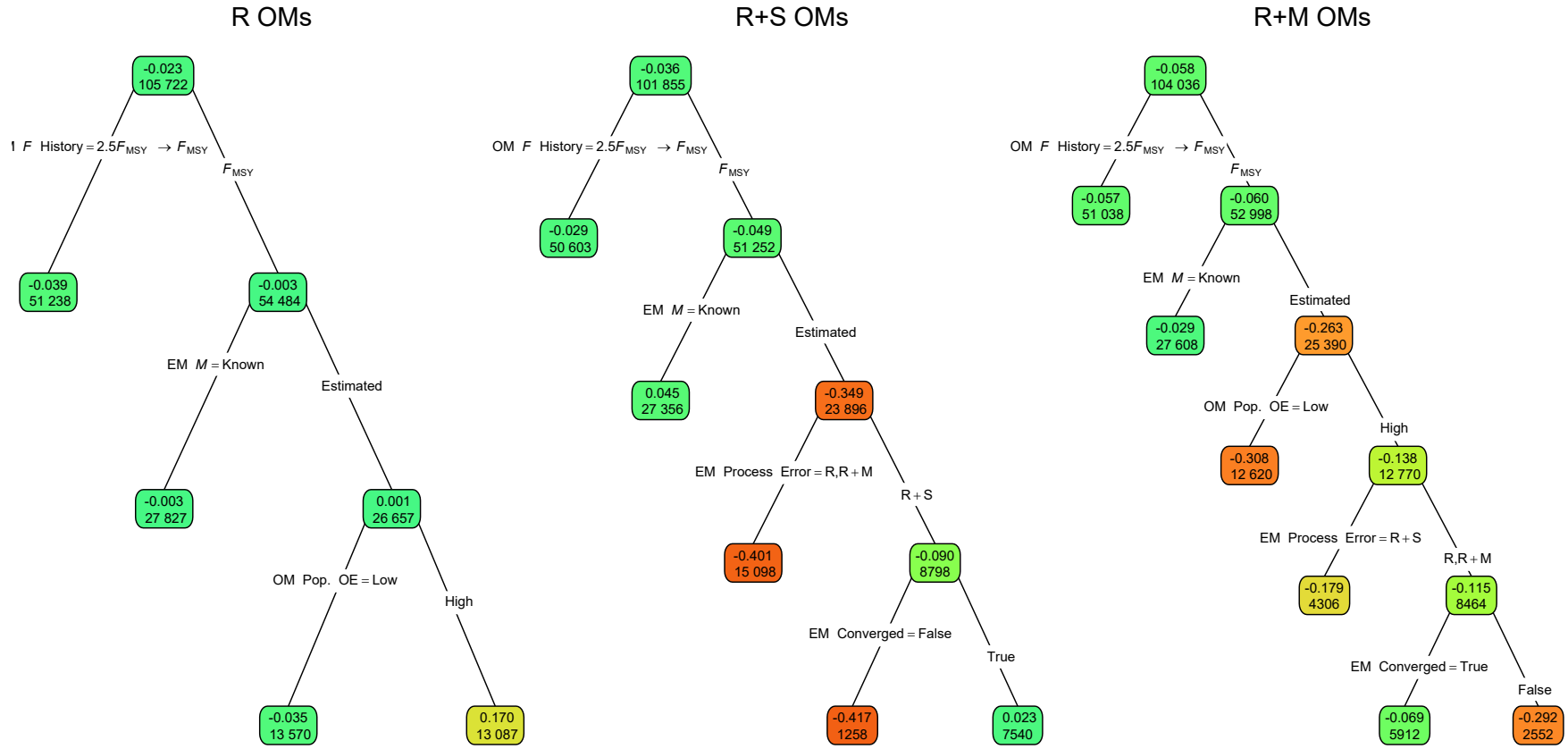


Fig. 9. Regression trees indicating primary factors determining reductions in sums of squares of errors in estimation measured by Eq. 2 for the terminal year SSB in estimating models (EMs) fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively. See Table S1 for description of OM and EM factors defining nodes and branches.

Table 1. For operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) (columns), the percent reduction in deviance for logistic regression models fit to indicators of convergence (providing Hessian-based standard errors) with each OM and estimating model (EM) factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	0.86	<0.01	0.24
OM Obs. Error	0.56	0.03	0.26
OM σ_e	0.66	2.99	0.93
OM σ_E	0.53	2.22	0.62
OM ρ_E	<0.01	0.02	<0.01
OM β_E	0.02	<0.01	<0.01
EM Process Error	24.53	9.33	23.06
EM β_E assumption	0.16	2.96	0.51
EM M Assumption	0.18	2.83	0.40
All factors	28.92	22.67	27.34
+ All Two Way	36.15	33.54	34.03
+ All Three Way	37.95	36.58	35.57

Table 2. For eoperating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) (columns), the percent reduction in deviance for multinomial logistic regression models fit to indicators of process error assumption in the estimating models (EMs) with lowest AIC with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	0.23	0.06	0.30
OM Obs. Error	0.54	18.58	13.43
OM σ_e	8.41	0.05	2.23
OM σ_E	0.17	0.08	0.59
OM ρ_E	1.29	0.03	0.06
OM β_E	3.42	0.04	0.38
EM M Assumption	0.35	0.34	0.48
EM β_E Assumption Correct	2.35	0.22	0.78
All factors	21.97	19.29	19.07
+ All Two Way	38.25	21.13	24.73
+ All Three Way	46.26	22.85	26.97

Table 3. For each operating model (OM) covariate effect assumption (columns), the percent reduction in deviance for logistic regression models fit to indicators of correct estimating model (EM) covariate effect assumption (no effect with OM $\beta_E = 0$ or estimated effect when OM $\beta_E > 0$) with lowest AIC with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	$\beta_E = 0$	$\beta_E = 0.25$	$\beta_E = 0.5$
OM F history	<0.01	0.09	0.18
OM Obs. Error	1.19	4.06	4.64
OM Process Error	4.11	0.71	0.25
OM σ_e	0.66	2.00	1.97
OM σ_E	0.35	7.97	20.66
OM ρ_E	0.21	1.23	1.48
EM M Assumption	0.23	0.15	0.15
EM Process Error Assumption Correct	1.95	0.39	0.06
All factors	6.95	17.65	33.44
+ All Two Way	10.58	23.21	38.36
+ All Three Way	11.83	24.95	39.79

Table 4. For operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) (columns), the percent reduction in deviance for linear regression models fit to errors in estimation of the covariate effect on natural mortality rate ($\hat{\beta}_E$) with each OM and estimating model (EM) factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	0.03	0.06	0.02
OM Obs. Error	<0.01	0.06	0.02
OM σ_e	0.04	0.08	0.06
OM σ_E	<0.01	0.02	0.01
OM ρ_E	<0.01	0.01	<0.01
OM β_E	0.05	<0.01	<0.01
EM Process Error	0.01	0.06	0.01
EM M assumption	0.02	0.02	<0.01
All factors	0.18	0.37	0.14
+ All Two Way	0.50	1.11	0.35
+ All Three Way	1.38	1.88	0.66

Table 5. For operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) (columns), the percent reduction in deviance for logistic regression models fit to indicators of whether confidence intervals for covariate effect estimates includes the true value with each OM and estimating model (EM) factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	0.24	0.05	0.01
OM Obs. Error	1.36	1.91	1.78
OM σ_e	8.26	0.24	0.92
OM σ_E	0.92	1.70	1.56
OM ρ_E	0.03	0.05	0.06
OM β_E	11.10	0.68	2.72
EM Process Error	0.01	6.51	0.27
EM M assumption	0.25	0.29	0.06
All factors	23.61	12.16	7.78
+ All Two Way	30.17	18.08	12.63
+ All Three Way	32.58	20.62	14.87

Table 6. For operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) (columns), the percent reduction in deviance for linear regression models fit to errors in estimation of the median natural mortality rate parameter ($\hat{\beta}_M$) with each OM and estimating model (EM) factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	4.03	16.52	6.51
OM Obs. Error	2.74	0.54	0.64
OM σ_e	0.01	<0.01	0.01
OM σ_E	0.18	0.02	0.15
OM ρ_E	0.07	0.05	<0.01
OM β_E	0.37	0.04	0.10
EM Process Error	2.31	11.99	2.65
EM β_E assumption	2.83	7.20	3.25
All factors	12.14	33.71	12.81
+ All Two Way	21.74	46.46	19.84
+ All Three Way	26.38	49.41	22.24

Table 7. For operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) (columns), the percent reduction in deviance for logistic regression models fit to indicators of whether confidence intervals for estimates of the median natural mortality rate parameter ($\hat{\beta}_M$) includes the true value with each OM and estimating model (EM) factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	6.08	1.46	0.14
OM Obs. Error	1.48	0.41	0.29
OM σ_e	0.01	0.06	0.01
OM σ_E	0.13	0.02	0.04
OM ρ_E	<0.01	<0.01	0.02
OM β_E	0.08	0.11	0.09
EM Process Error	0.53	0.08	0.04
EM β_E assumption	<0.01	0.19	0.01
All factors	8.46	2.32	0.62
+ All Two Way	11.97	10.57	5.99
+ All Three Way	13.35	13.51	8.44

Table 8. For operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) (columns), the percent reduction in deviance for linear regression models fit to errors in estimation measured by Eq. 2 for the terminal year natural mortality rate (M) with each OM and estimating model (EM) factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
Convergence	<0.01	0.18	0.01
OM F history	0.66	2.69	1.00
OM Obs. Error	0.29	0.43	0.16
$OM\sigma_e$	<0.01	<0.01	<0.01
$OM\sigma_E$	0.02	<0.01	<0.01
$OM\rho_E$	<0.01	0.01	<0.01
OM β_E	0.03	0.01	0.01
EM Process Error	0.22	1.42	0.43
$EM\beta_E$ assumption	0.35	1.07	0.39
EM M assumption	0.84	5.10	1.22
All factors	2.56	10.83	3.37
+ All Two Way	5.41	18.75	6.38
+ All Three Way	7.93	22.49	8.76

Table 9. For operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M) (columns), the percent reduction in deviance for linear regression models fit to errors in estimation measured by Eq. 2 for the terminal year spawning stock biomass (SSB) with each OM and estimating model (EM) factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
Convergence	0.02	0.13	<0.01
OM F history	2.08	2.74	1.75
OM Obs. Error	1.09	0.18	1.19
$OM\sigma_e$	0.01	<0.01	<0.01
$OM\sigma_E$	<0.01	<0.01	<0.01
$OM\rho_E$	<0.01	<0.01	<0.01
OM β_E	<0.01	0.01	<0.01
EM Process Error	0.52	1.05	0.57
$EM\beta_E$ assumption	<0.01	0.05	0.01
EM M assumption	1.76	2.12	1.51
All factors	5.97	6.58	5.21
+ All Two Way	12.81	13.00	11.97
+ All Three Way	16.41	15.57	15.73

888 **Supplemental Materials**

889 **Referenced Figures and Tables**



Fig. S1. Example simulations of environmental covariate latent processes and observations with different levels of observation error, and different assumptions about variability of the latent process.

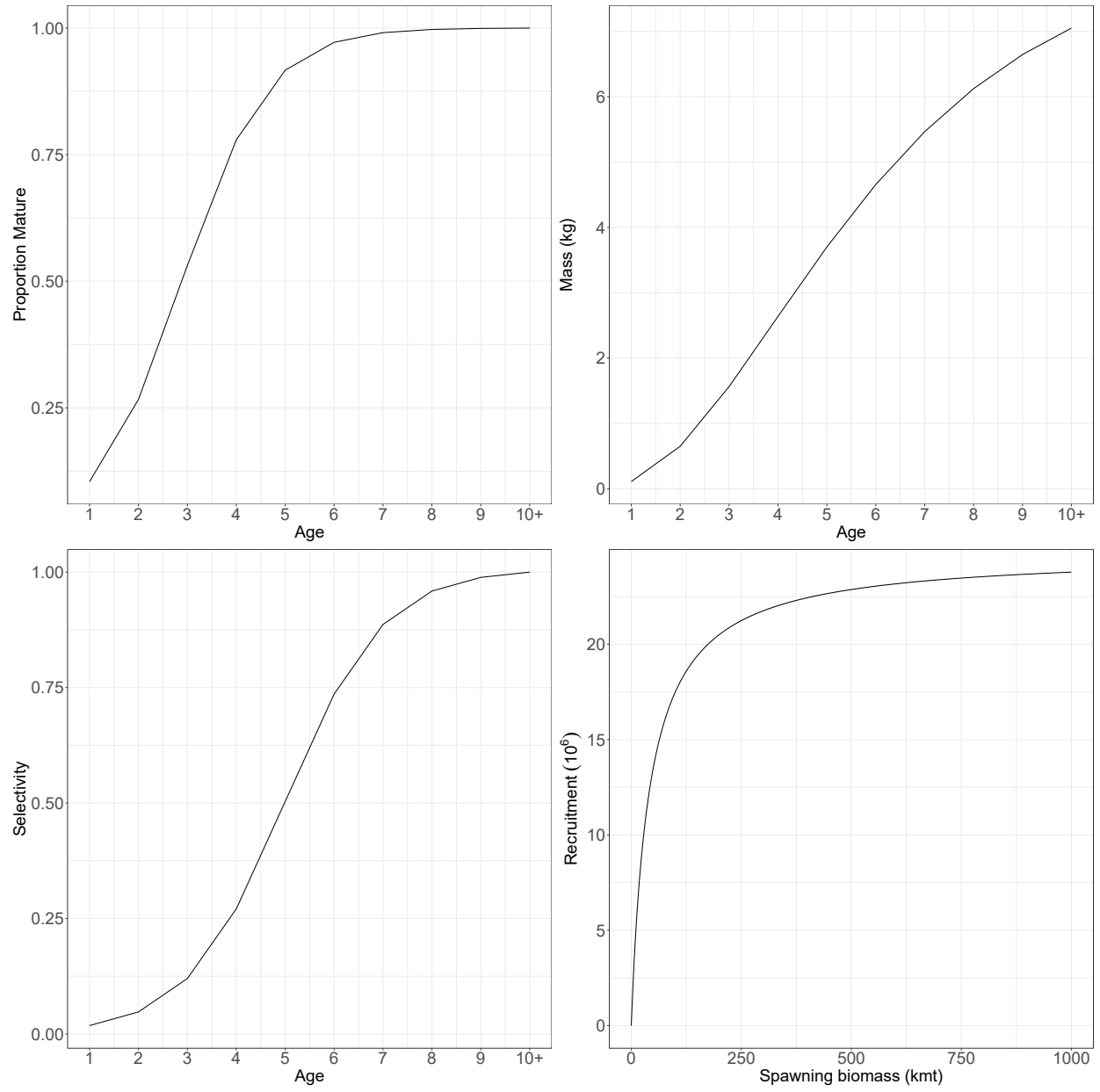


Fig. S2. The proportion mature at age, weight at age, fleet and index selectivity at age, and Beverton-Holt stock-recruit relationship assumed for the population in all OMs. For OMs with random effects on fleet selectivity, this represents the selectivity at the mean of the random effects.

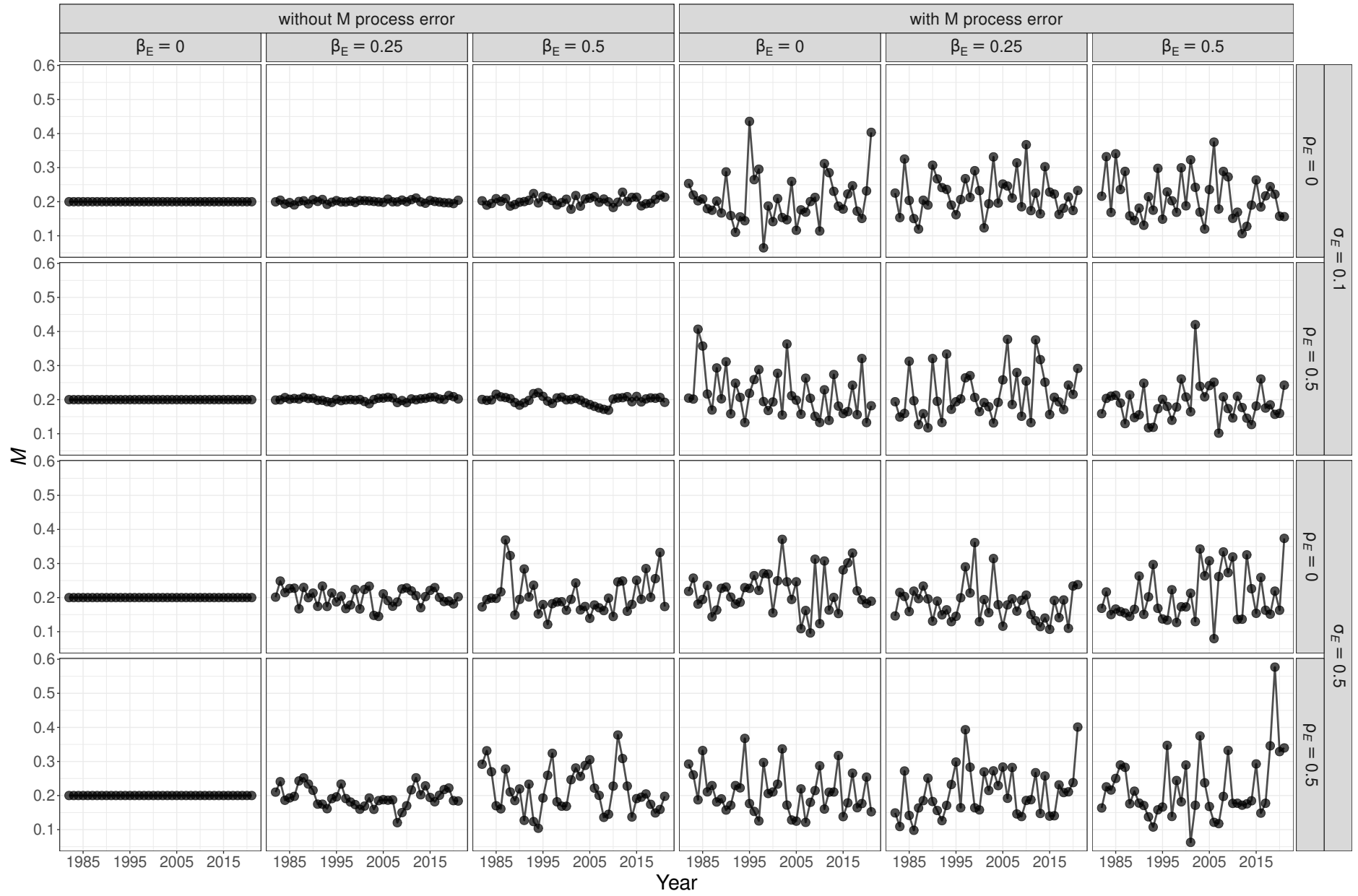


Fig. S3. Example simulations of annual M that may be a function of a temporally varying environmental covariate and autoregressive random effects.

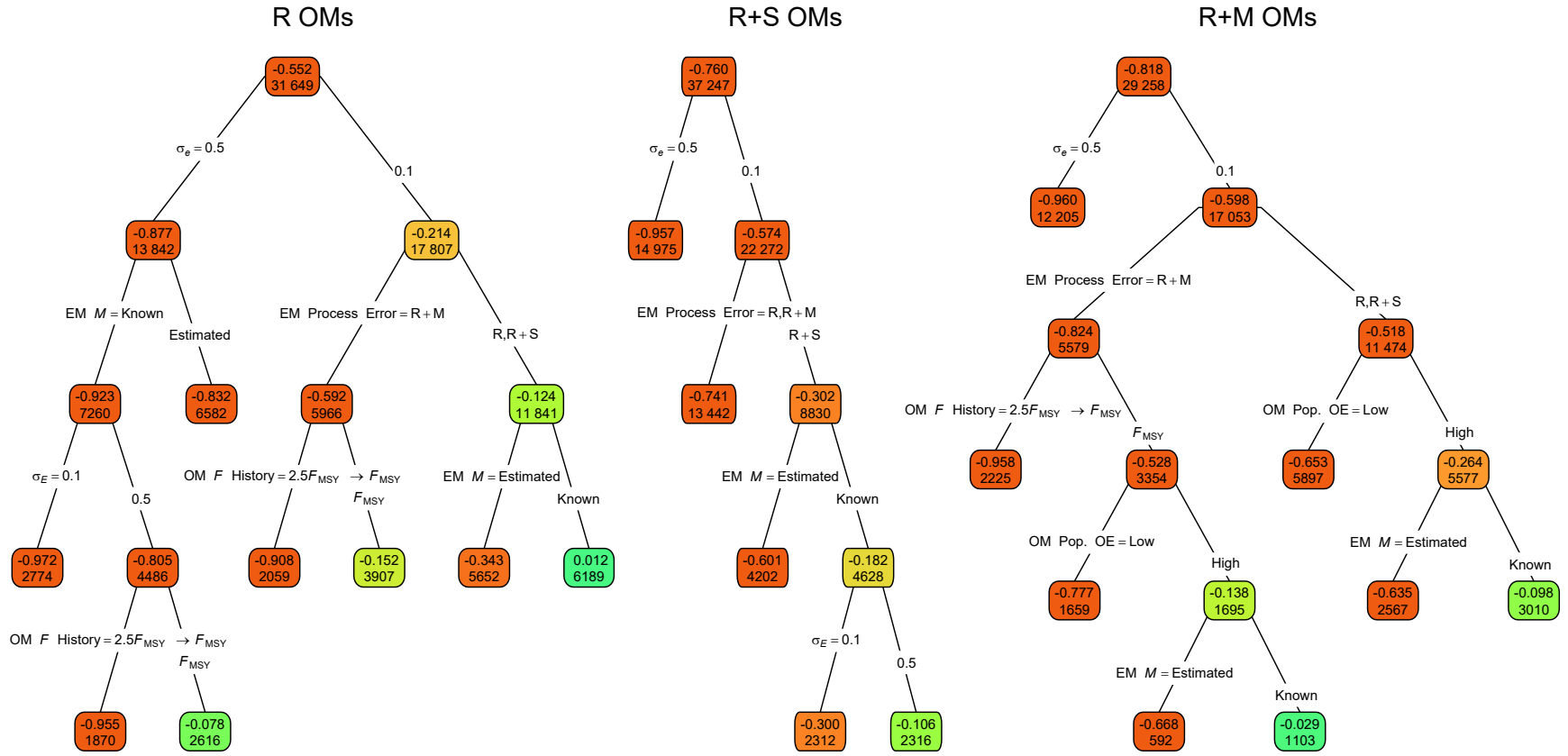


Fig. S4. Regression trees indicating primary factors determining reductions in sums of squares of errors in estimation of the Hessian-based standard error estimates for covariate effect on M ($\widehat{SE}(\widehat{\beta}_E)$) for operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively.

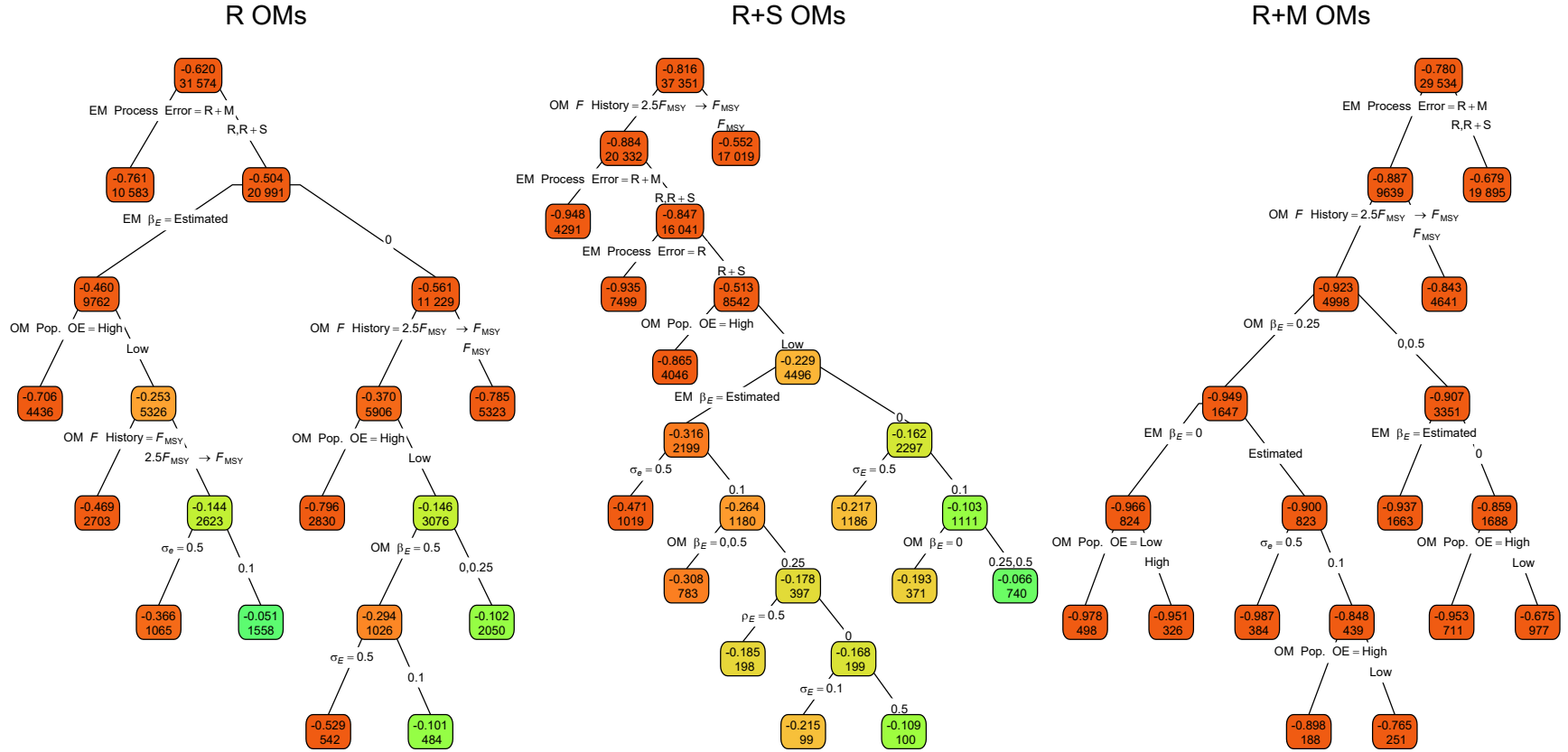


Fig. S5. Regression trees indicating primary factors determining reductions in sums of squares of errors in estimation of the Hessian-based standard error estimates for median M parameter ($\widehat{SE}(\hat{\beta}_M)$) in EMs fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively.

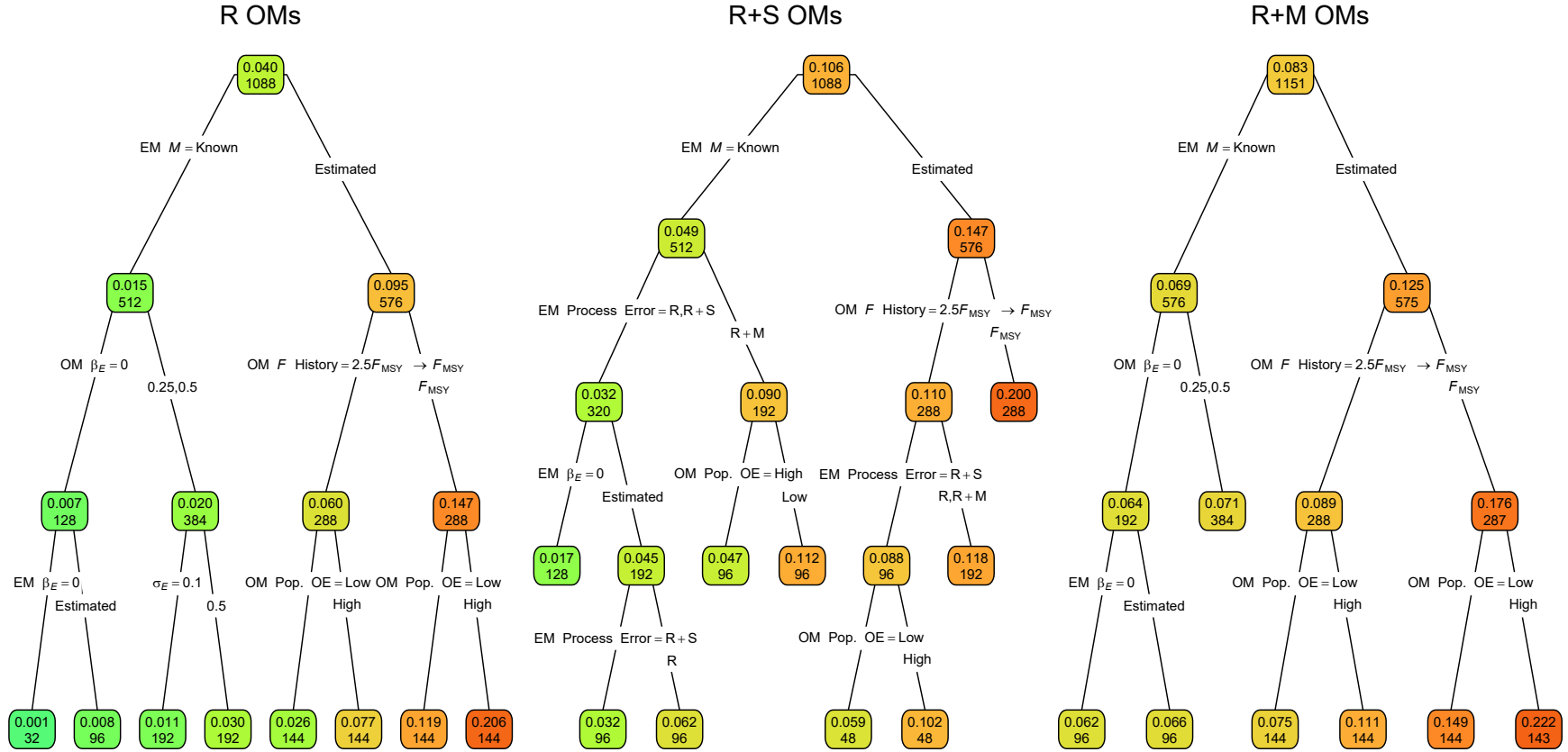


Fig. S6. Regression trees indicating primary factors determining reductions in sums of squares for the log-transformed RMSE of terminal year M in EMs fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more red or green polygons, respectively.

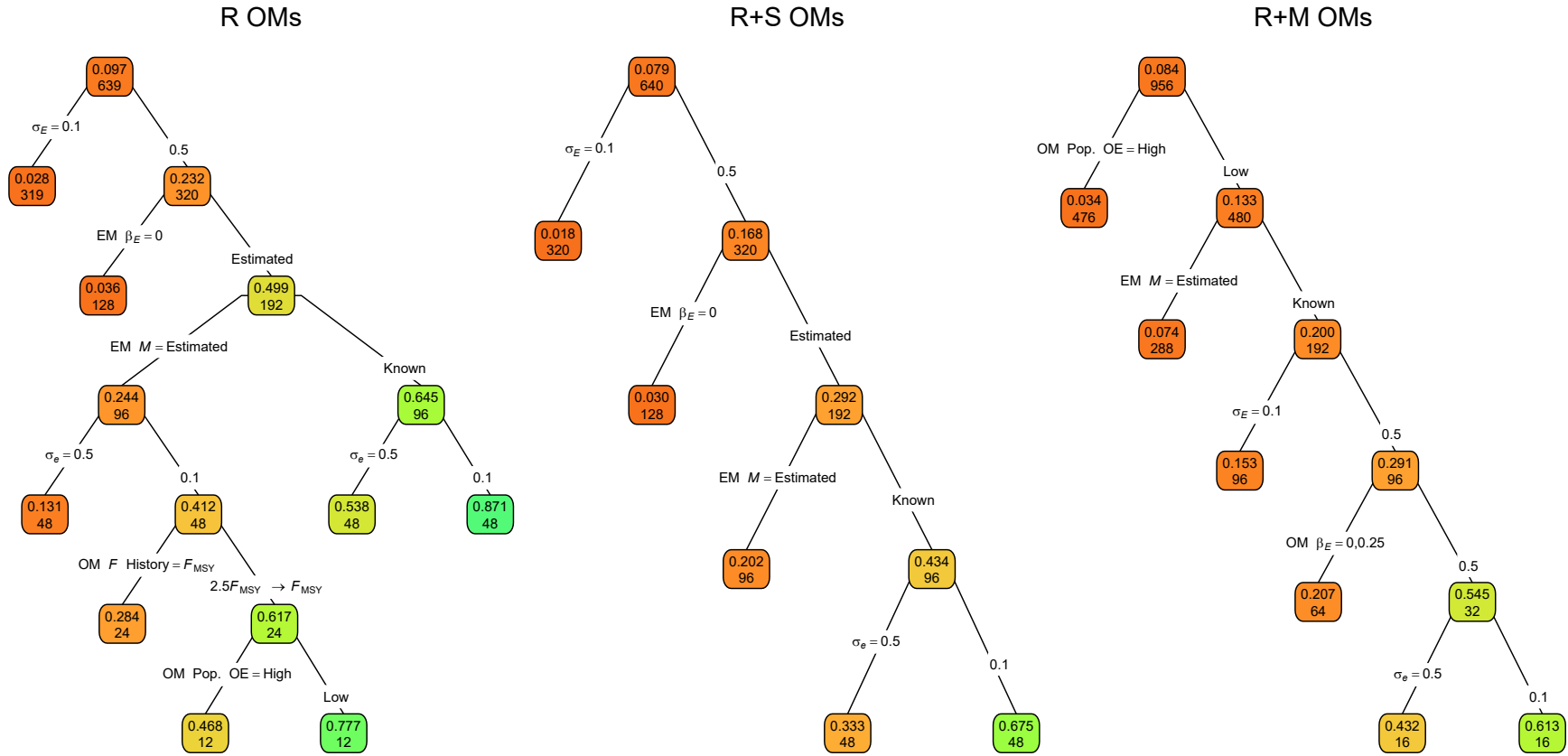


Fig. S7. Regression trees indicating primary factors determining reductions in sums of squares for the correlation of terminal year M estimates and true values in EMs fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median correlations are indicated by more red or green polygons, respectively.

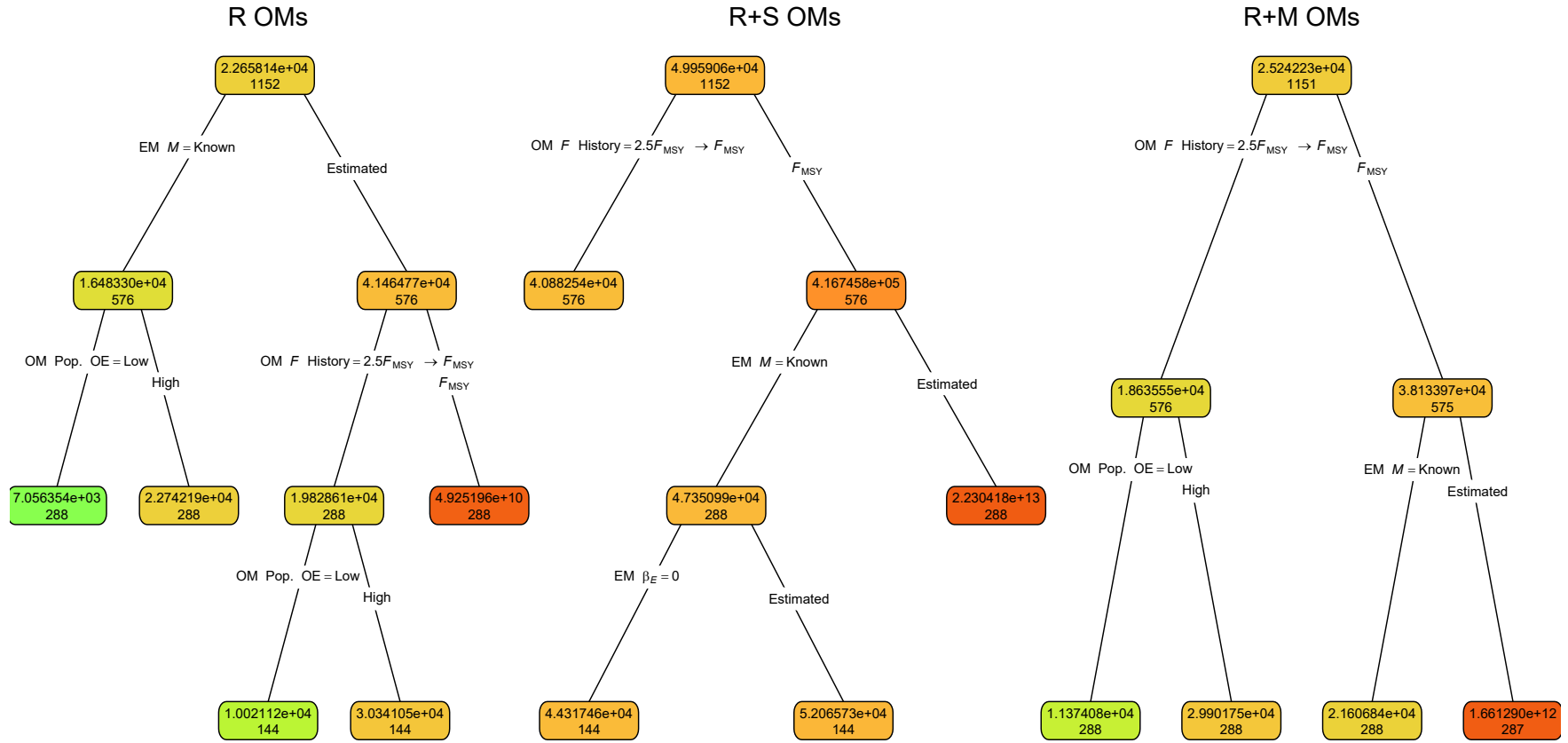


Fig. S8. Regression trees indicating primary factors determining reductions in sums of squares for the log-transformed RMSE of terminal year SSB in EMs fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more red or green polygons, respectively.

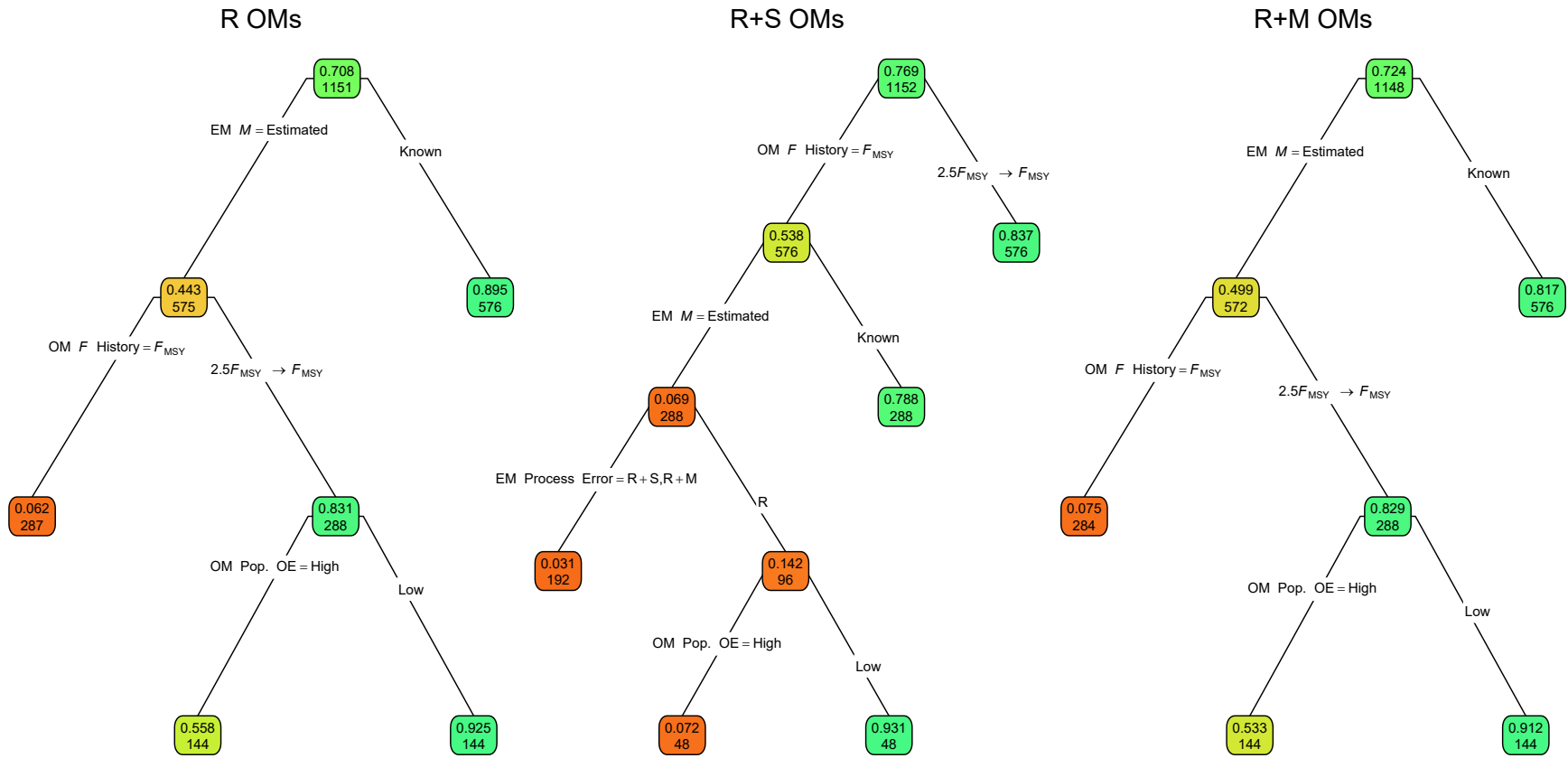


Fig. S9. Regression trees indicating primary factors determining reductions in sums of squares for the correlation of terminal year SSB estimates and true values in EMs fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median correlations are indicated by more red or green polygons, respectively.

Table S1. Operating (OM) and estimating model (EM) attributes used in analyses of deviance reduction and classification and regression trees. R, R+S, and R+M, denote models (OM or EM) with process error on recruitment only, recruitment and apparent survival, and recruitment and natural mortality, respectively. SD denotes standard deviation and F_{MSY} is the fishing mortality rate that achieves maximum sustainable yield.

Factor	Levels
OM Process error	R, R+S, R+M
OM Fishing History	$F_{\text{MSY}}, 2.5F_{\text{MSY}} \rightarrow F_{\text{MSY}}$
OM Covariate effect size (β_E)	0, 0.25, 0.5
OM Population Observation Error	Low, High
OM Covariate Observation Error	Low ($\sigma_e = 0.1$), High ($\sigma_e = 0.5$)
OM Covariate Process Error SD (σ_E)	0.1, 0.5
OM Covariate Process Error Correlation (ρ_E)	0, 0.5
EM Process Error	R, R+S, R+M
EM Covariate Effect	None ($\beta_E = 0$), β_E estimated
EM Median Natural Mortality Rate Parameter (β_M)	Known, Estimated

Table S2. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to errors in estimation of the Hessian-based standard error of the estimates of the covariate effect on M ($\widehat{\text{SE}}(\hat{\beta}_E)$) with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	1.84	2.68	4.00
OM Obs. Error	0.06	2.74	8.04
OM σ_e	6.27	7.28	12.83
OM σ_E	0.48	1.05	0.46
OM ρ_E	0.17	0.01	0.02
OM β_E	0.37	0.01	0.05
EM Process Error	2.90	3.89	2.98
EM M assumption	0.15	0.34	0.47
All factors	13.01	20.95	29.24
+ All Two Way	28.58	32.25	42.77
+ All Three Way	44.44	42.83	51.84

Table S3. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to errors in estimation of the Hessian-based standard error of the estimates of the median M parameter ($\widehat{\text{SE}}(\hat{\beta}_M)$) with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	0.55	12.91	3.65
OM Obs. Error	0.98	0.01	0.07
OM σ_e	<0.01	0.03	0.11
OM σ_E	0.13	0.03	0.19
OM ρ_E	<0.01	0.02	<0.01
OM β_E	0.12	0.07	0.23
EM Process Error	3.16	8.79	6.83
EM β_E assumption	0.18	8.96	3.81
All factors	5.32	28.39	14.71
+ All Two Way	11.80	44.67	22.70
+ All Three Way	17.83	51.54	29.46

Table S4. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to log-transformed RMSE of terminal year M with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	0.26	2.21	14.65
OM Obs. Error	1.65	0.71	4.68
$OM\sigma_e$	0.02	0.16	0.01
$OM\sigma_E$	1.06	1.23	0.84
$OM\rho_E$	0.02	0.01	<0.01
OM β_E	3.73	1.94	1.14
EM Process Error	2.04	3.85	0.87
$EM\beta_E$ assumption	0.19	0.59	0.07
EM M assumption	22.28	45.15	44.70
All factors	32.83	56.71	66.99
+ All Two Way	57.65	84.17	87.96
+ All Three Way	74.93	94.57	91.94

Table S5. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to correlation of terminal year M estimates and true values with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	0.05	0.02	0.37
OM Obs. Error	2.00	0.89	3.73
$OM\sigma_e$	2.30	2.56	0.37
$OM\sigma_E$	14.91	13.78	3.03
$OM\rho_E$	1.98	0.33	0.14
OM β_E	0.41	3.76	3.87
EM Process Error	0.43	1.52	0.62
$EM\beta_E$ assumption	12.72	12.30	1.56
EM M assumption	12.34	7.16	3.03
All factors	42.11	38.31	15.76
+ All Two Way	63.38	62.40	31.99
+ All Three Way	72.79	75.98	46.62

Table S6. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to log-transformed RMSE of terminal year SSB with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	19.88	32.67	20.92
OM Obs. Error	2.21	0.11	1.48
$OM\sigma_e$	0.05	0.16	0.03
$OM\sigma_E$	0.08	0.16	0.02
$OM\rho_E$	0.03	<0.01	0.02
OM β_E	0.05	0.03	0.14
EM Process Error	5.72	0.13	5.73
$EM\beta_E$ assumption	0.03	0.79	0.05
EM M assumption	20.05	21.11	19.43
All factors	48.09	55.17	47.82
+ All Two Way	79.09	80.98	78.40
+ All Three Way	88.30	89.49	87.67

Table S7. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to correlation of terminal year SSB estimates and true values with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	19.27	25.70	19.59
OM Obs. Error	12.53	3.60	9.46
$OM\sigma_e$	0.08	0.44	0.09
$OM\sigma_E$	0.10	0.44	0.11
$OM\rho_E$	<0.01	<0.01	0.09
OM β_E	0.09	0.05	<0.01
EM Process Error	0.07	0.92	0.71
$EM\beta_E$ assumption	<0.01	1.32	0.38
EM M assumption	28.94	19.74	25.36
All factors	61.19	52.21	56.09
+ All Two Way	81.48	72.16	74.98
+ All Three Way	85.23	83.22	80.48

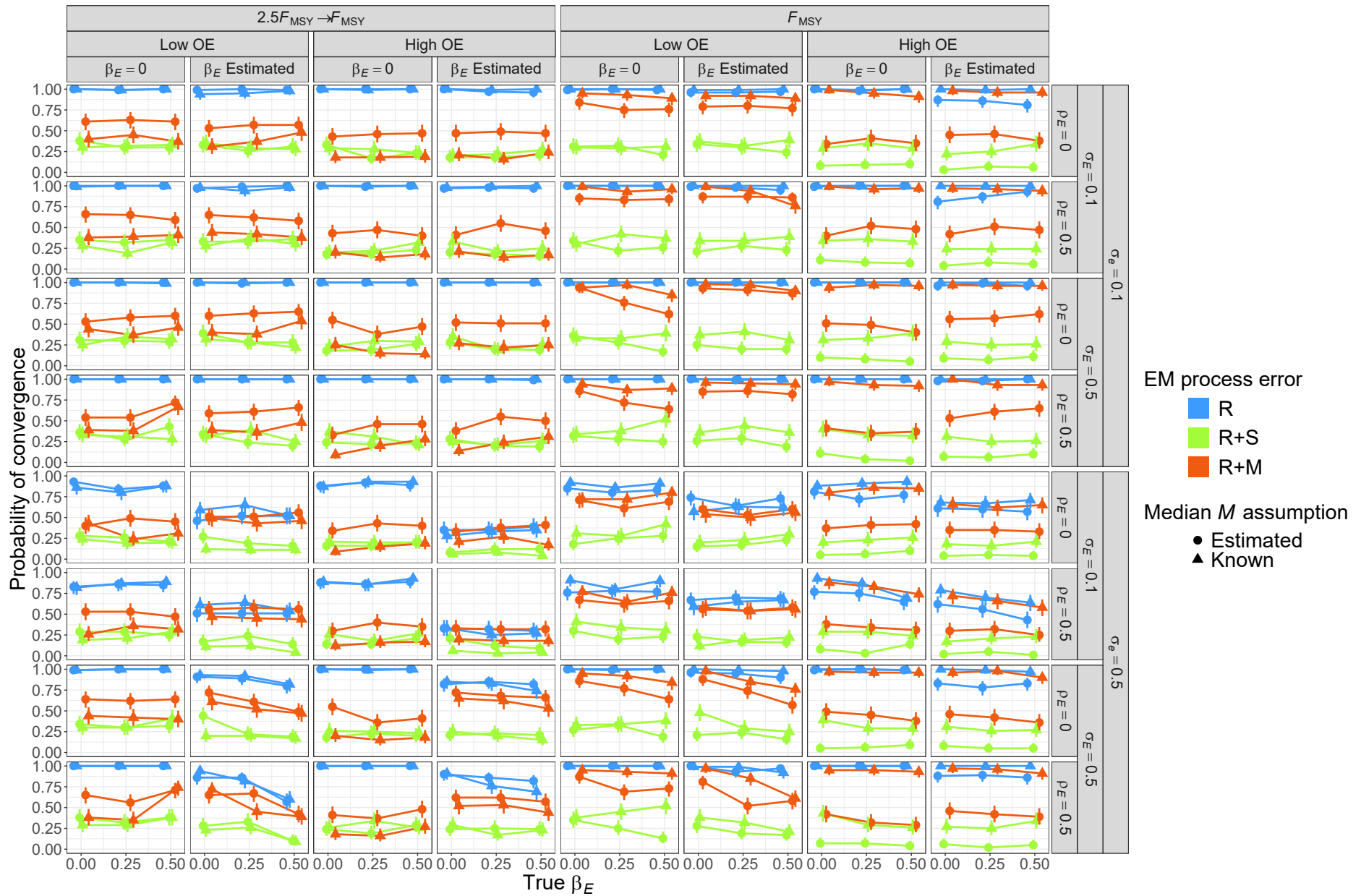


Fig. S10. Estimated probability of fits providing Hessian-based standard errors for EMs assuming alternative process error, that estimate or assume known median M , and that estimate or assume no covariate effect on median M when fitted to R OMs and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

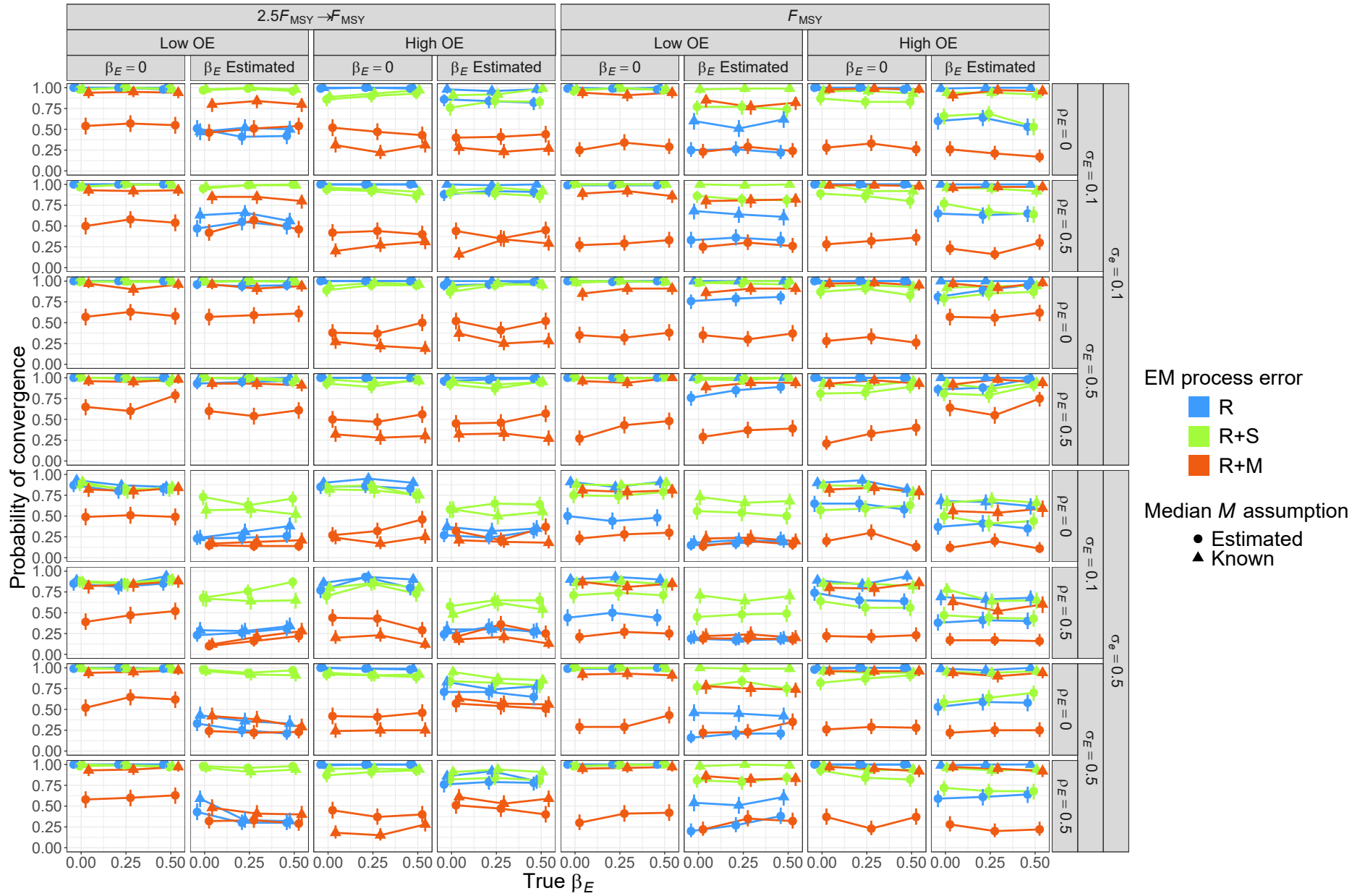


Fig. S11. Estimated probability of fits providing Hessian-based standard errors for EMs assuming alternative process error, that estimate or assume known median M , and that estimate or assume no covariate effect on median M when fitted to R+S OM and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

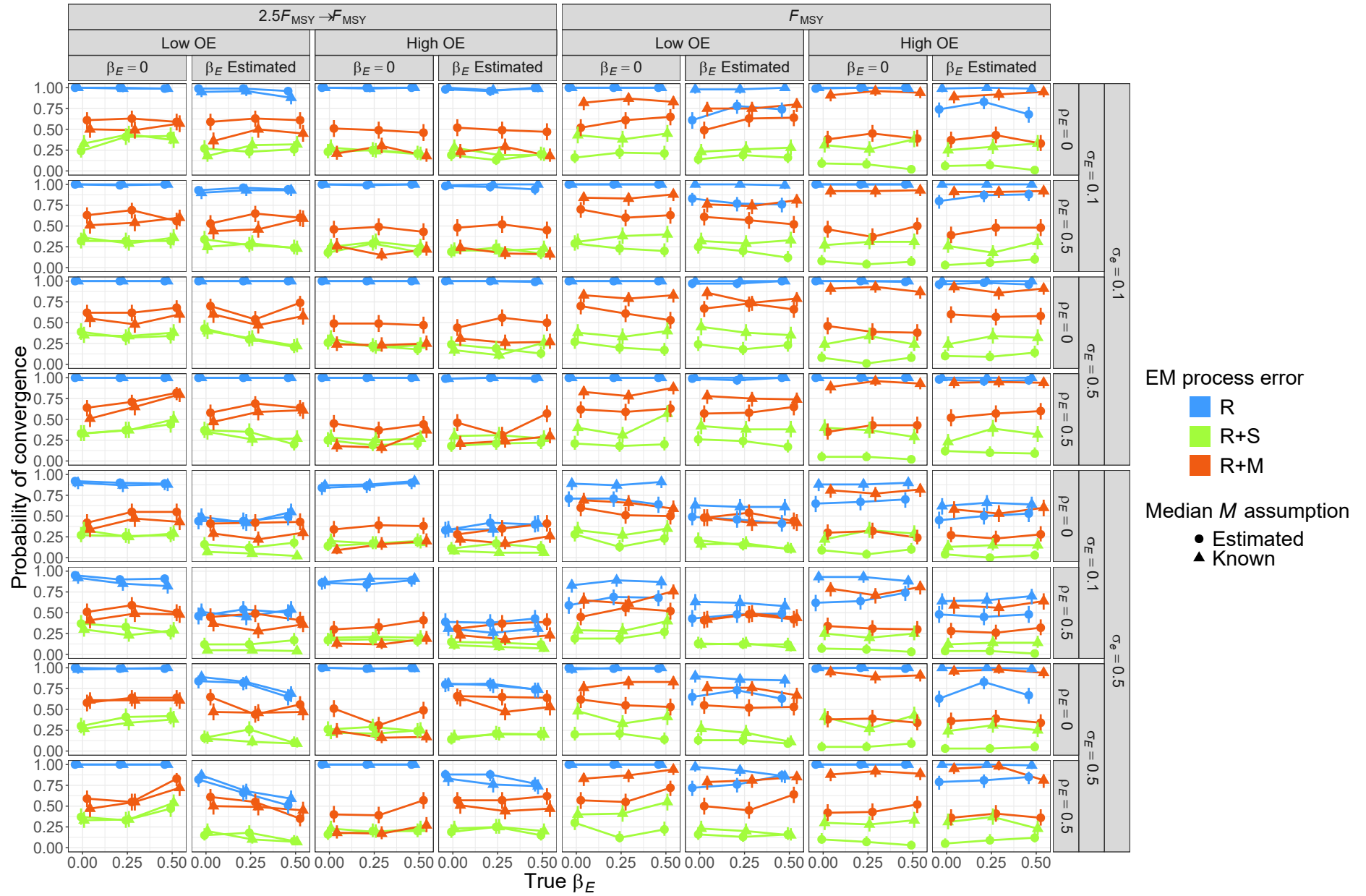


Fig. S12. Estimated probability of fits providing Hessian-based standard errors for EMs assuming alternative process error, that estimate or assume known median M , and that estimate or assume no covariate effect on median M when fitted to R+M OM and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

891 **AIC results**

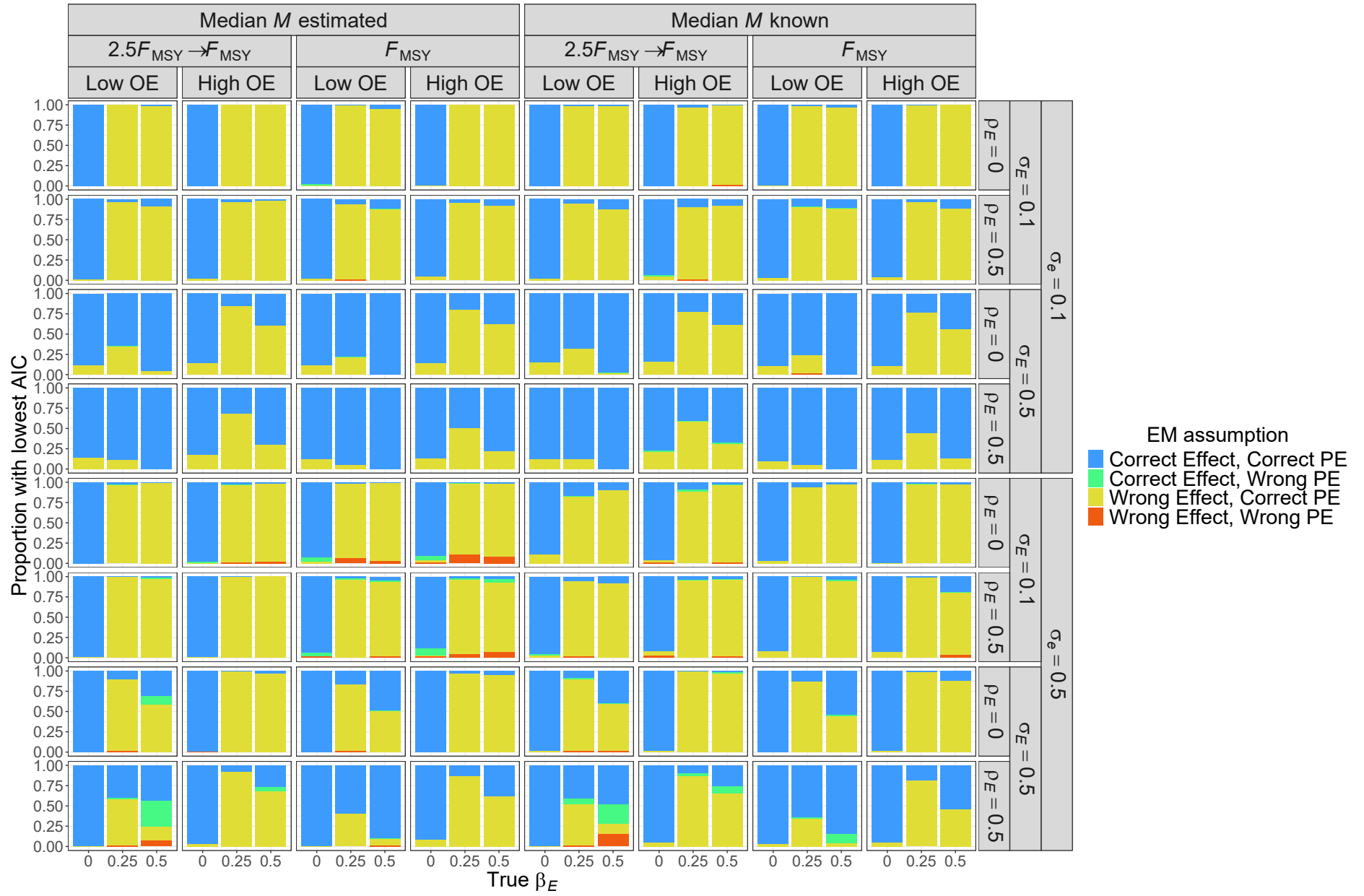


Fig. S13. Proportion of simulated data sets for R OMs where the EM type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

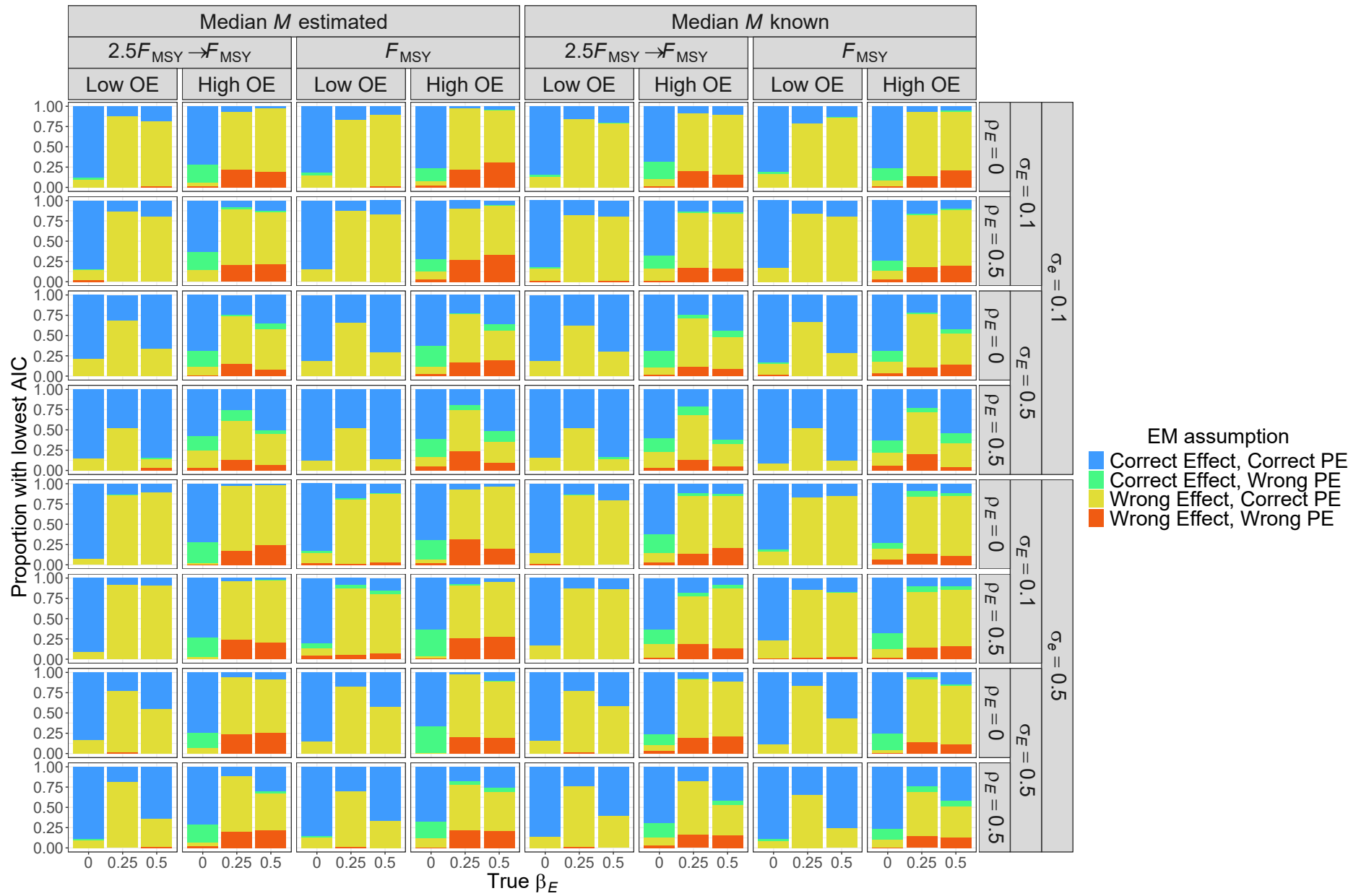


Fig. S14. Proportion of simulated data sets for R+S OMs where the EM type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

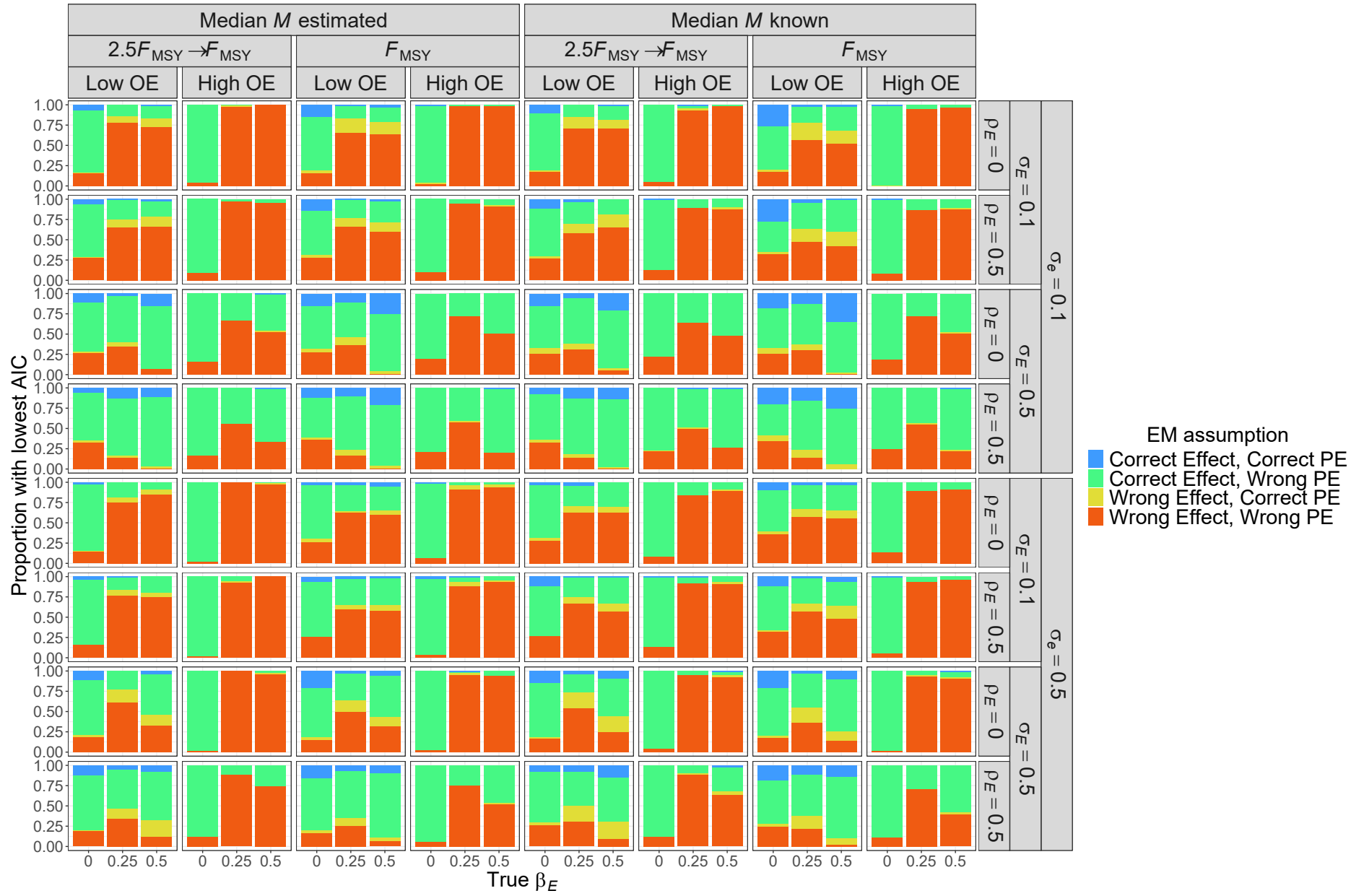


Fig. S15. Proportion of simulated data sets for R+M OMs where the EM type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

Table S8. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to errors in estimation of the covariate effect on M (β_E) with each OM and EM factor (rows) included individually, combined, and with second and third order interactions. Includes results from all unconverged and converged models.

Factor	R	R+S	R+M
OM F history	<0.01	0.01	<0.01
OM Obs. Error	<0.01	<0.01	<0.01
OM σ_e	<0.01	<0.01	<0.01
OM σ_E	<0.01	<0.01	<0.01
OM ρ_E	<0.01	<0.01	<0.01
OM β_E	<0.01	0.01	<0.01
EM Convergence	<0.01	<0.01	<0.01
EM Process Error	<0.01	0.01	0.01
EM M assumption	<0.01	<0.01	<0.01
All factors	0.02	0.03	0.03
+ All Two Way	0.16	0.20	0.18
+ All Three Way	0.58	0.73	0.59

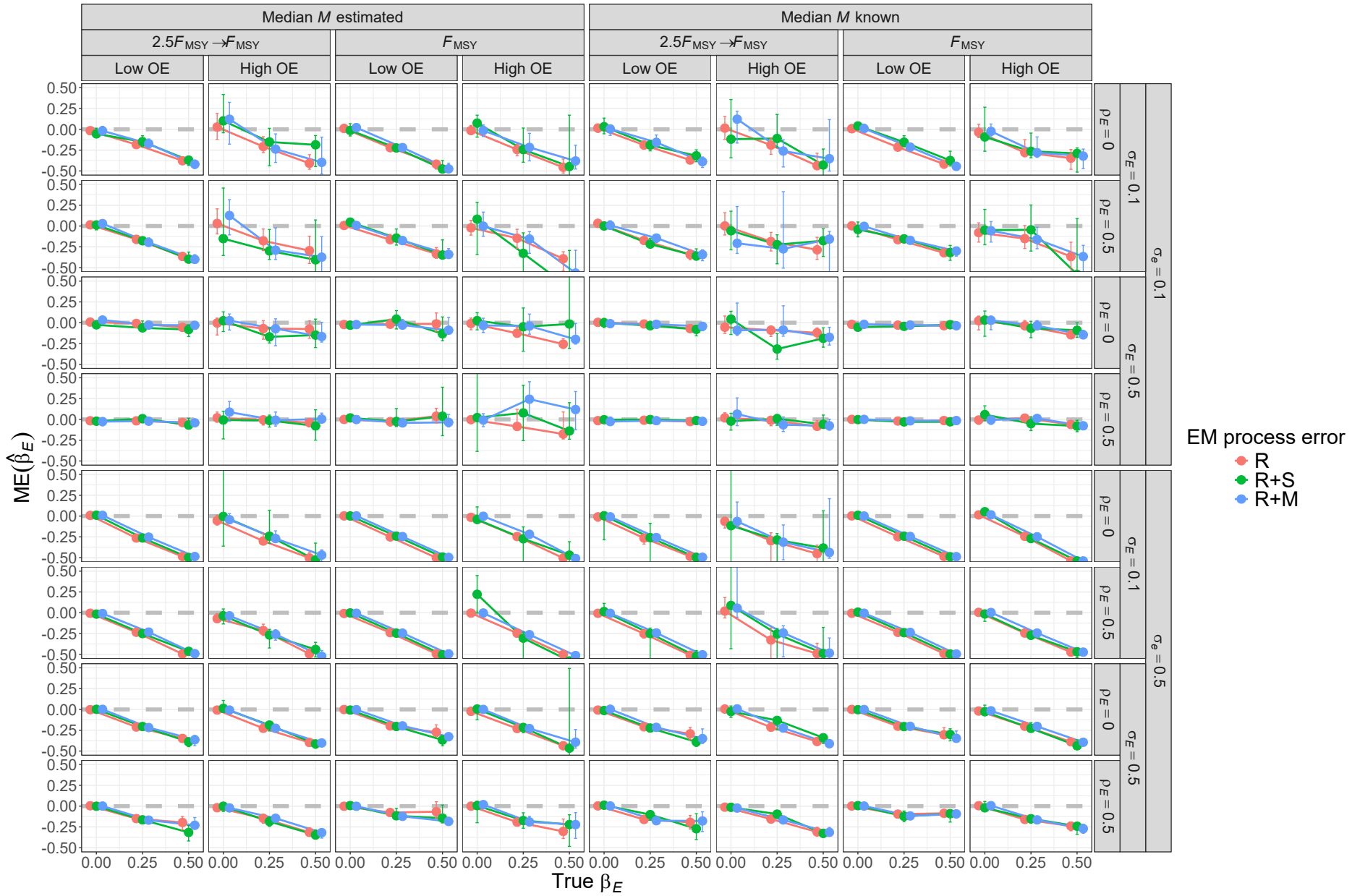


Fig. S16. For R OMs, median error (ME) of estimates of environmental effect on M (β_E) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). Vertical lines represent 95% confidence intervals.

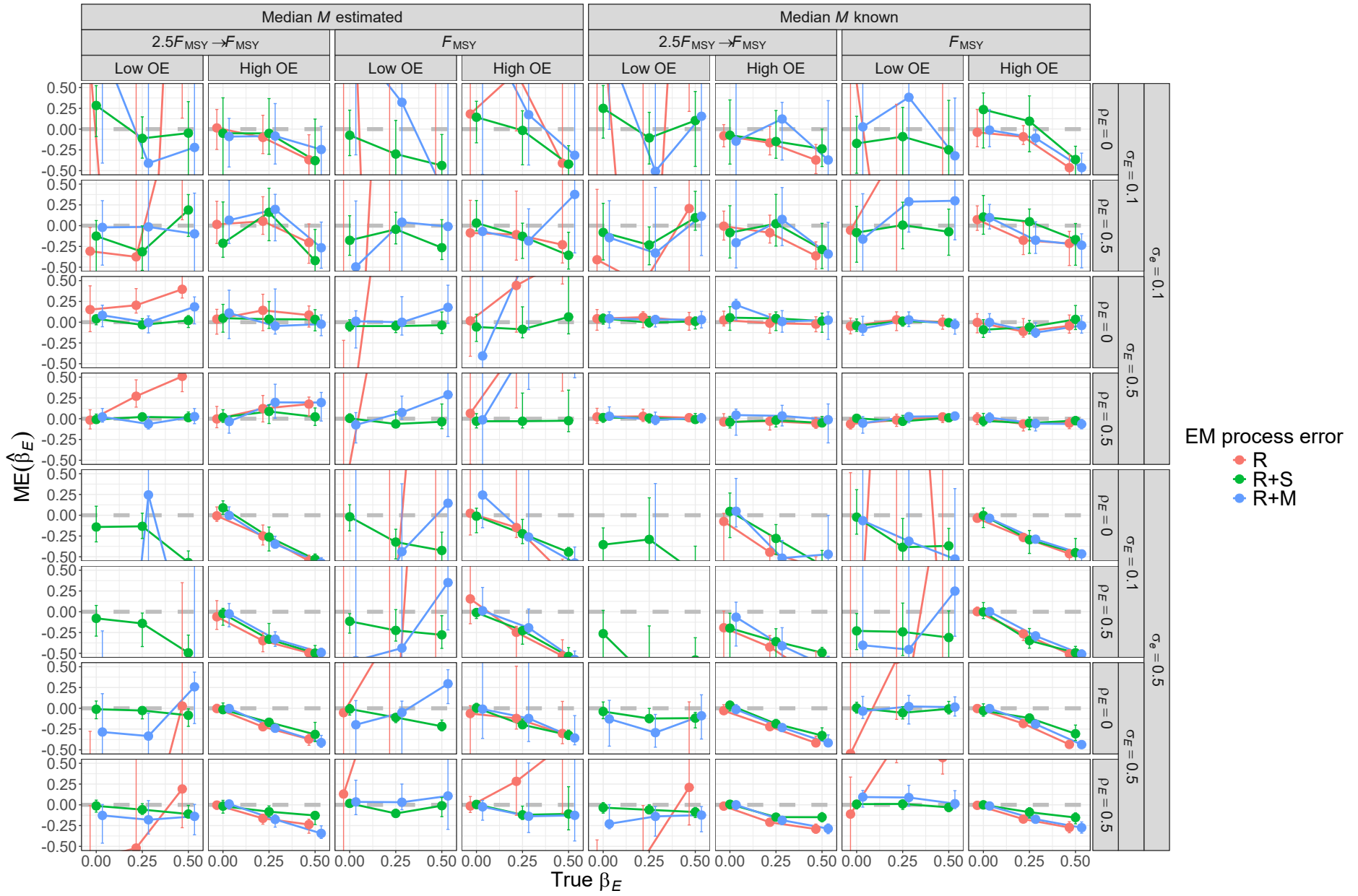


Fig. S17. For R+S OMs, median error (ME) of estimates of environmental effect on M (β_E) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). Vertical lines represent 95% confidence intervals.

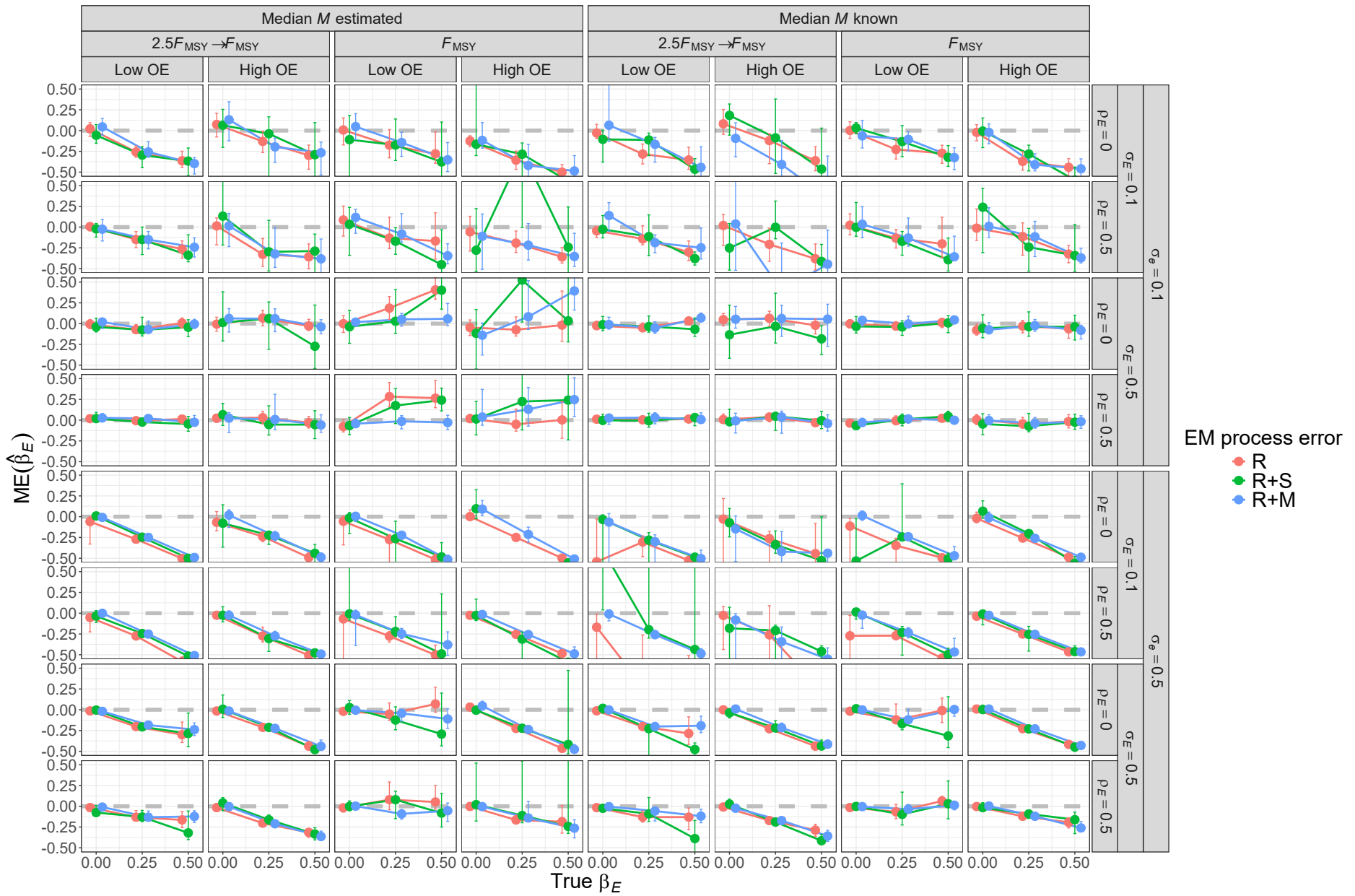


Fig. S18. For R+M OMs, median error (ME) of estimates of environmental effect on M (β_E) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). Vertical lines represent 95% confidence intervals.

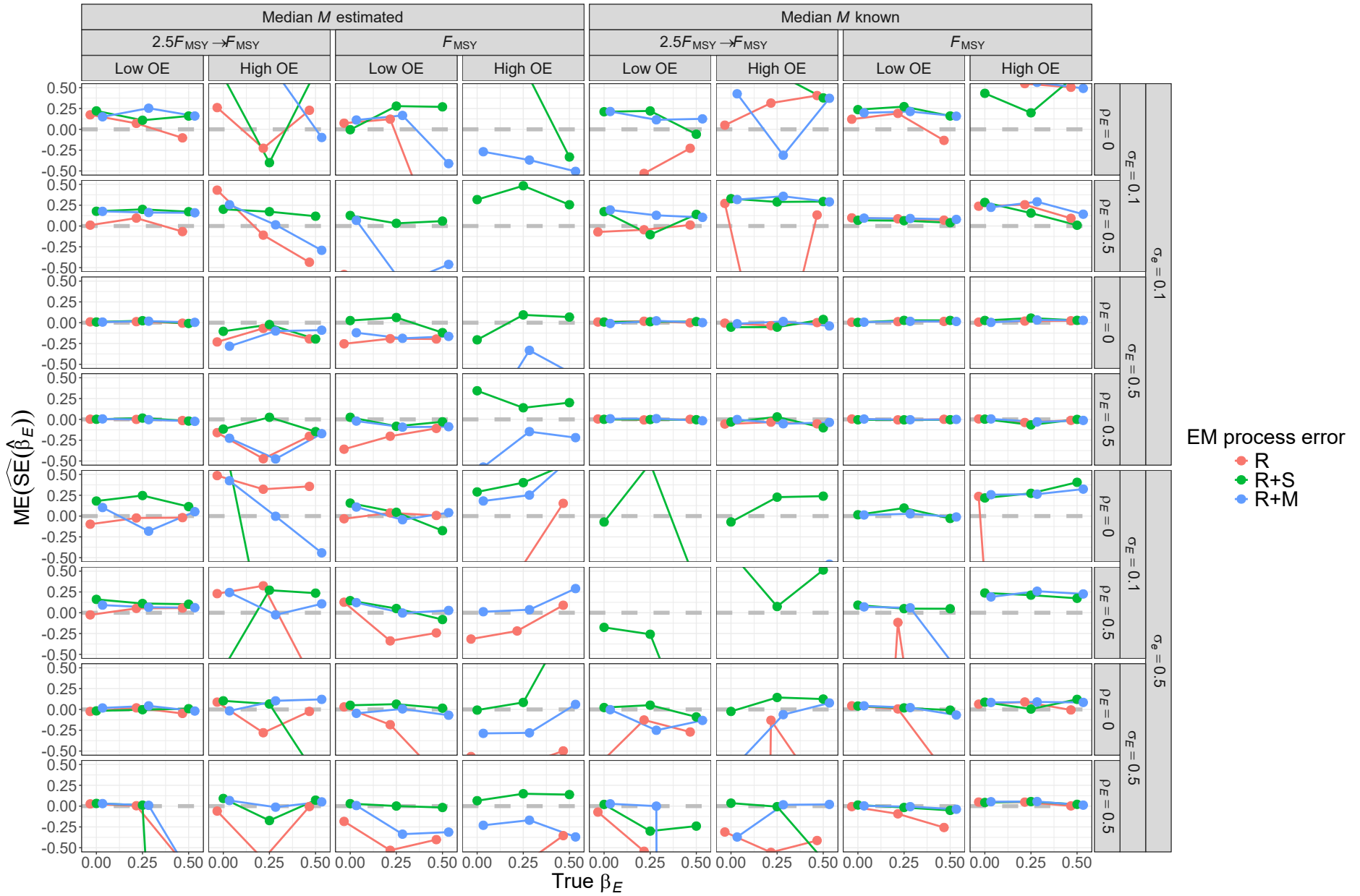


Fig. S19. For R OMs, median error (ME) of Hessian-based estimates of standard error for covariate effect on M (β_E) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given OM scenario.

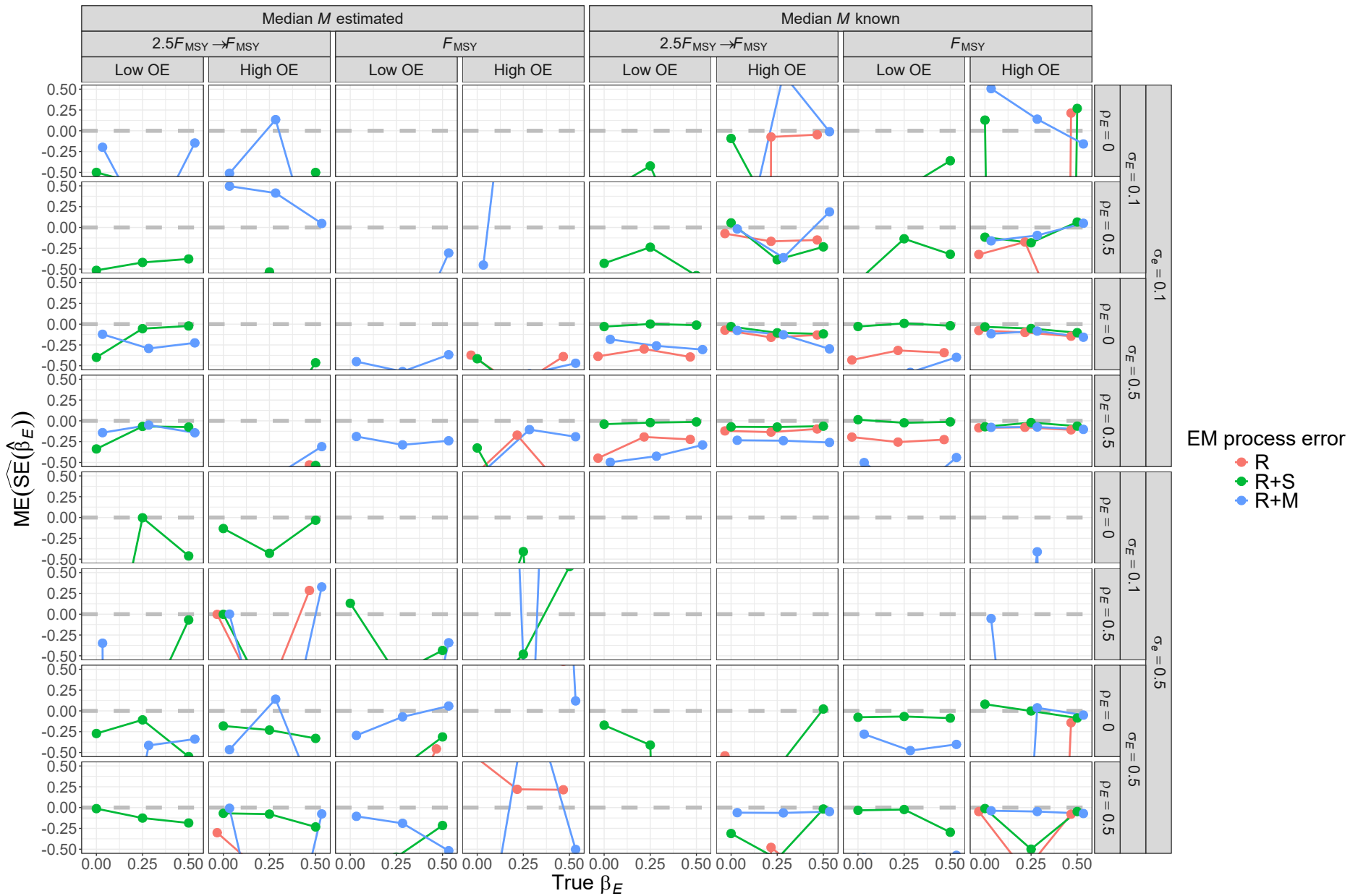


Fig. S20. For R+S OMs, median error (ME) of Hessian-based estimates of standard error for covariate effect on M (β_E) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given OM scenario.

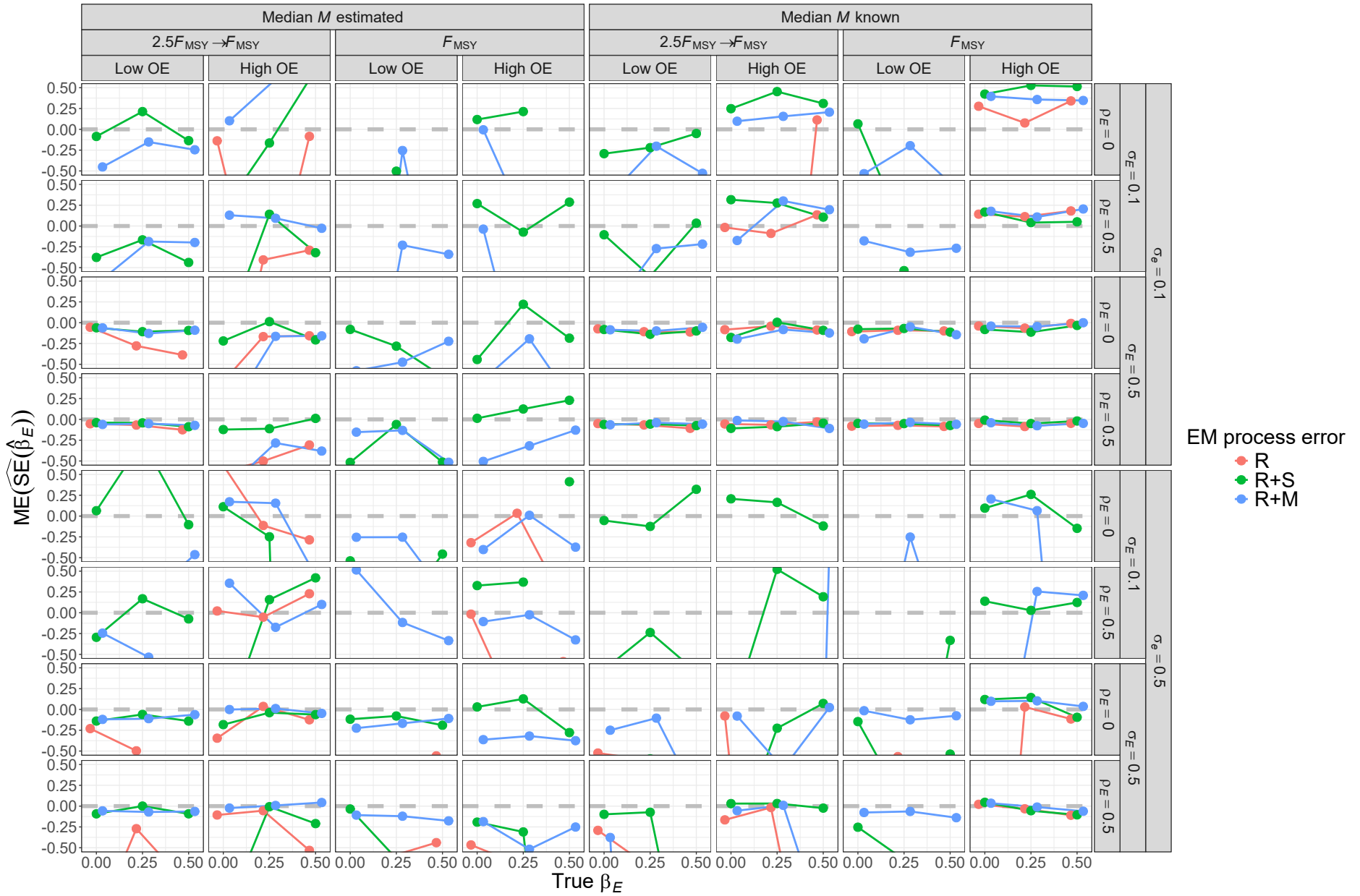


Fig. S21. For R+M OMs, median error (ME) of Hessian-based estimates of standard error for covariate effect on M (β_E) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given OM scenario.

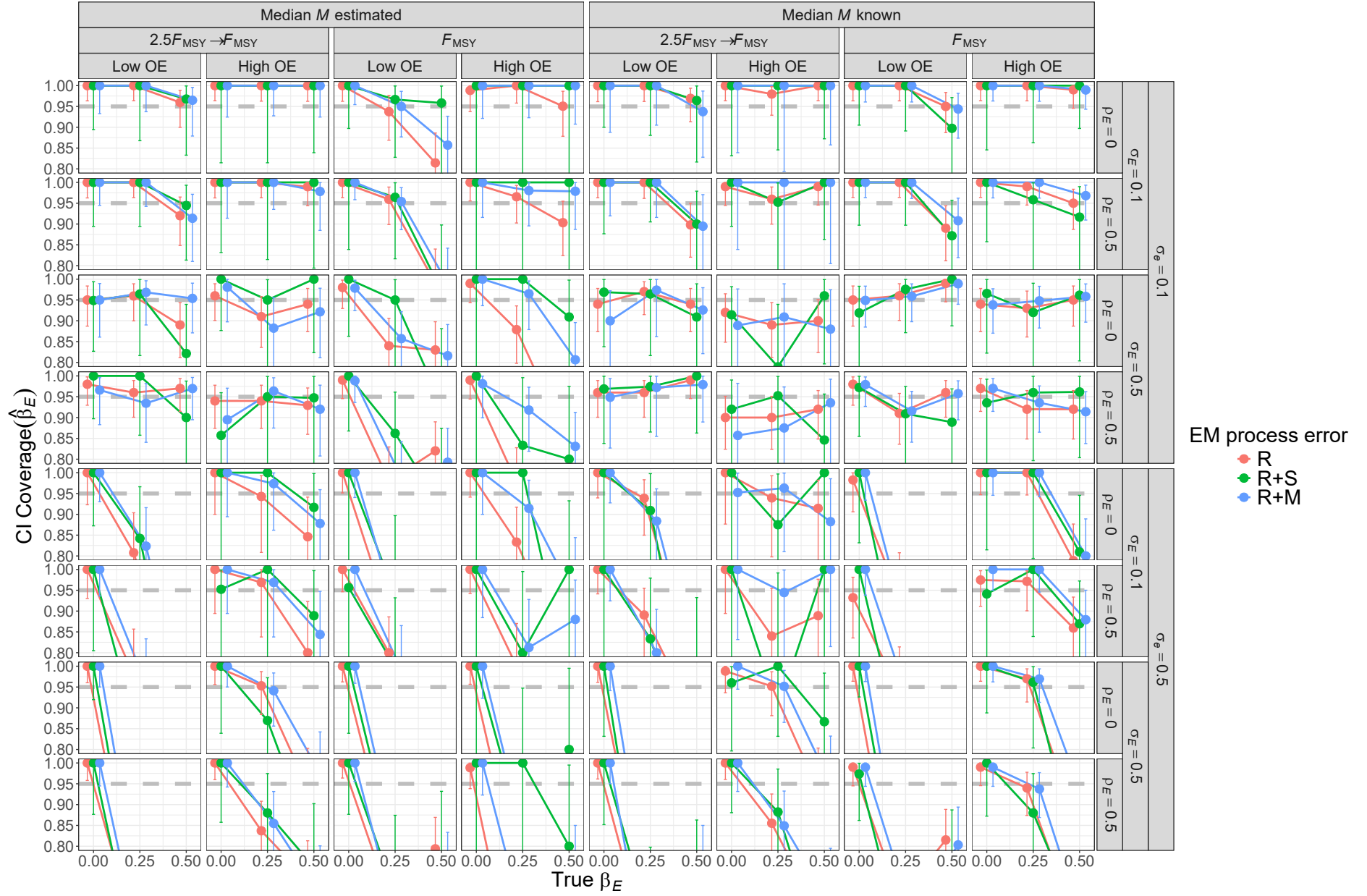


Fig. S22. For R OMs, probability of 95% confidence interval for β_E containing the true value for EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). Vertical lines represent 95% confidence intervals.

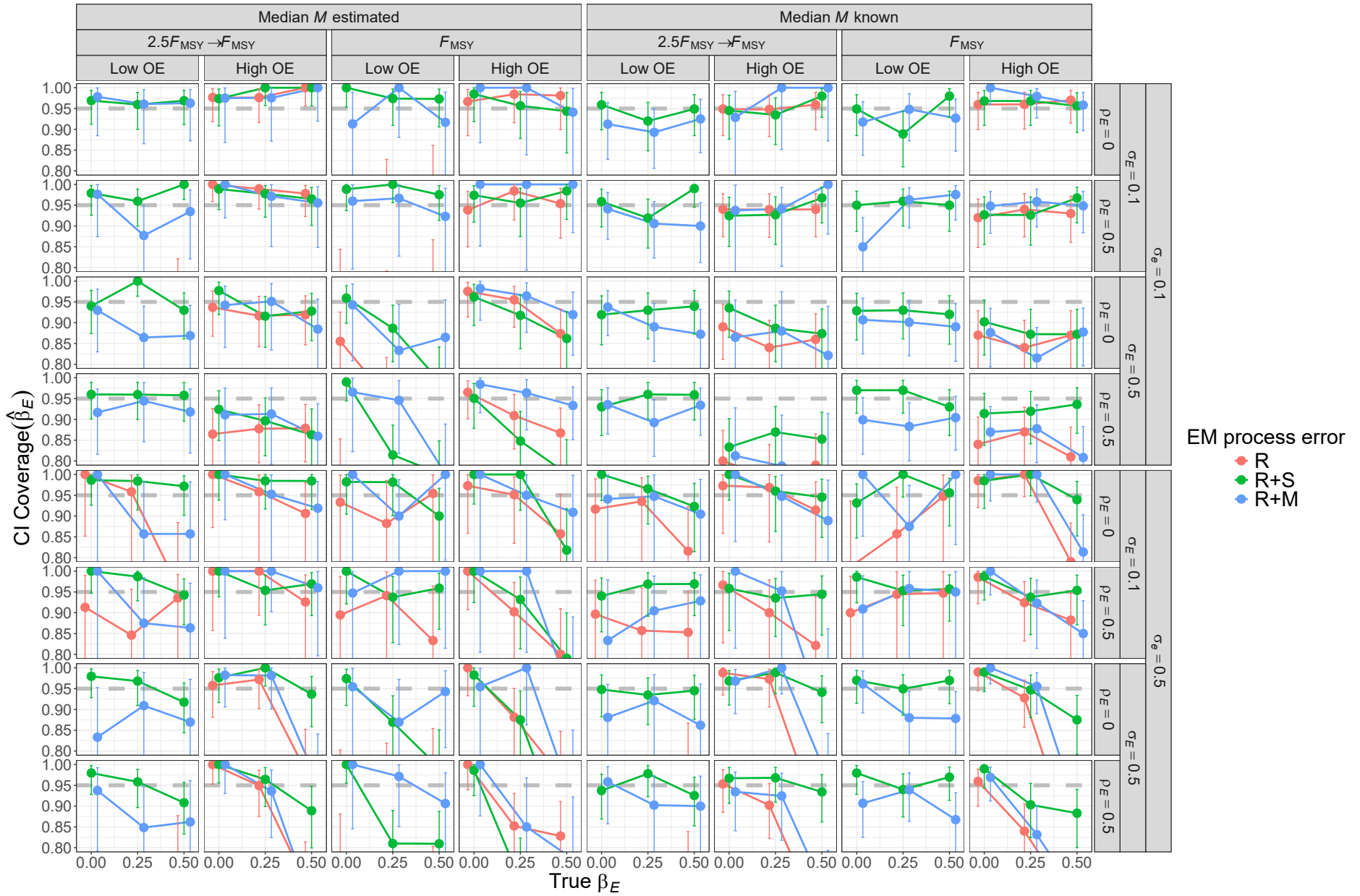


Fig. S23. For R+S OMs, probability of 95% confidence interval for β_E containing the true value for EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). Vertical lines represent 95% confidence intervals.

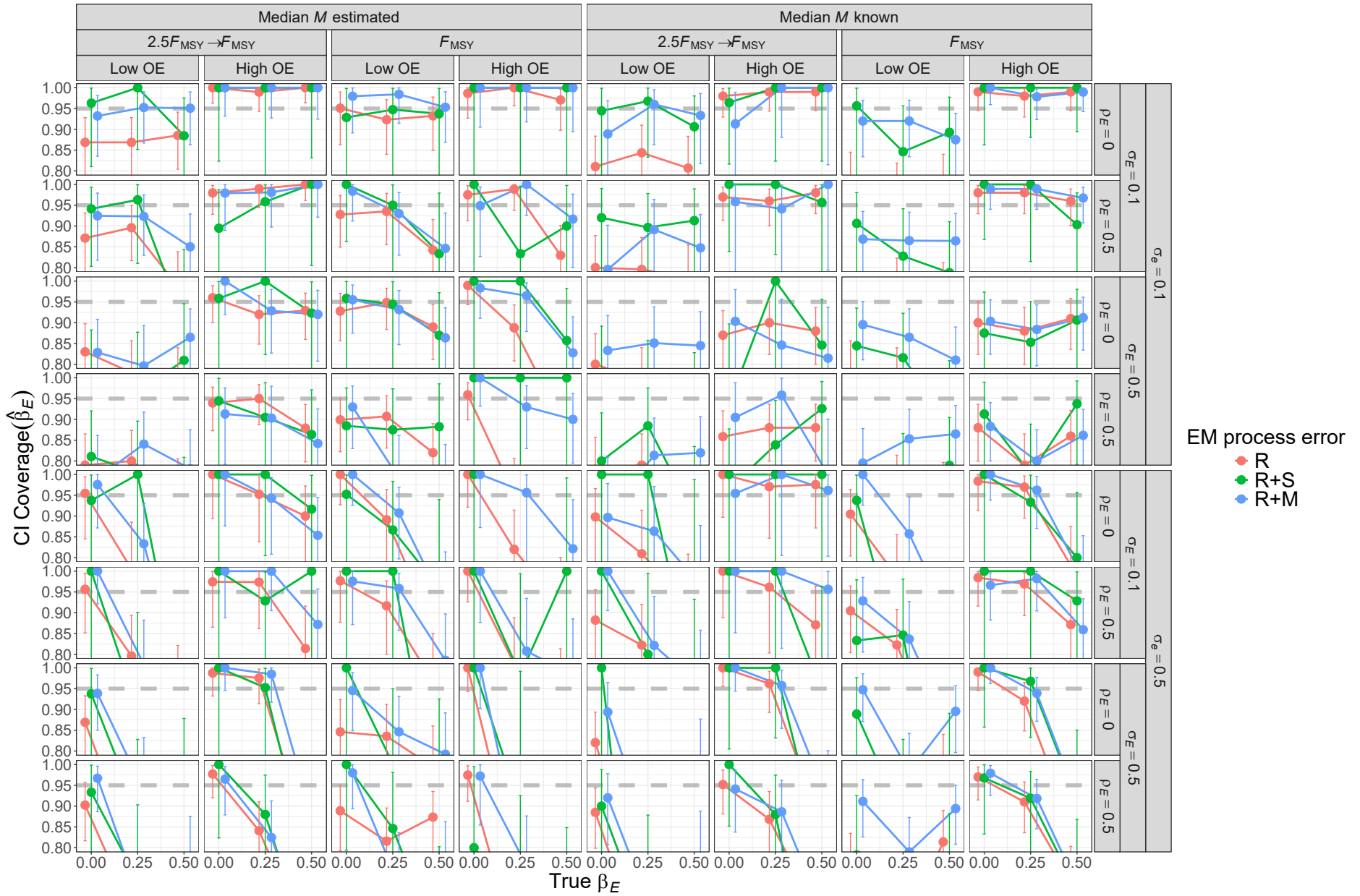


Fig. S24. For R+M OMs, probability of 95% confidence interval for β_E containing the true value for EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated). Vertical lines represent 95% confidence intervals.

895 **Covariate effect RMSE**

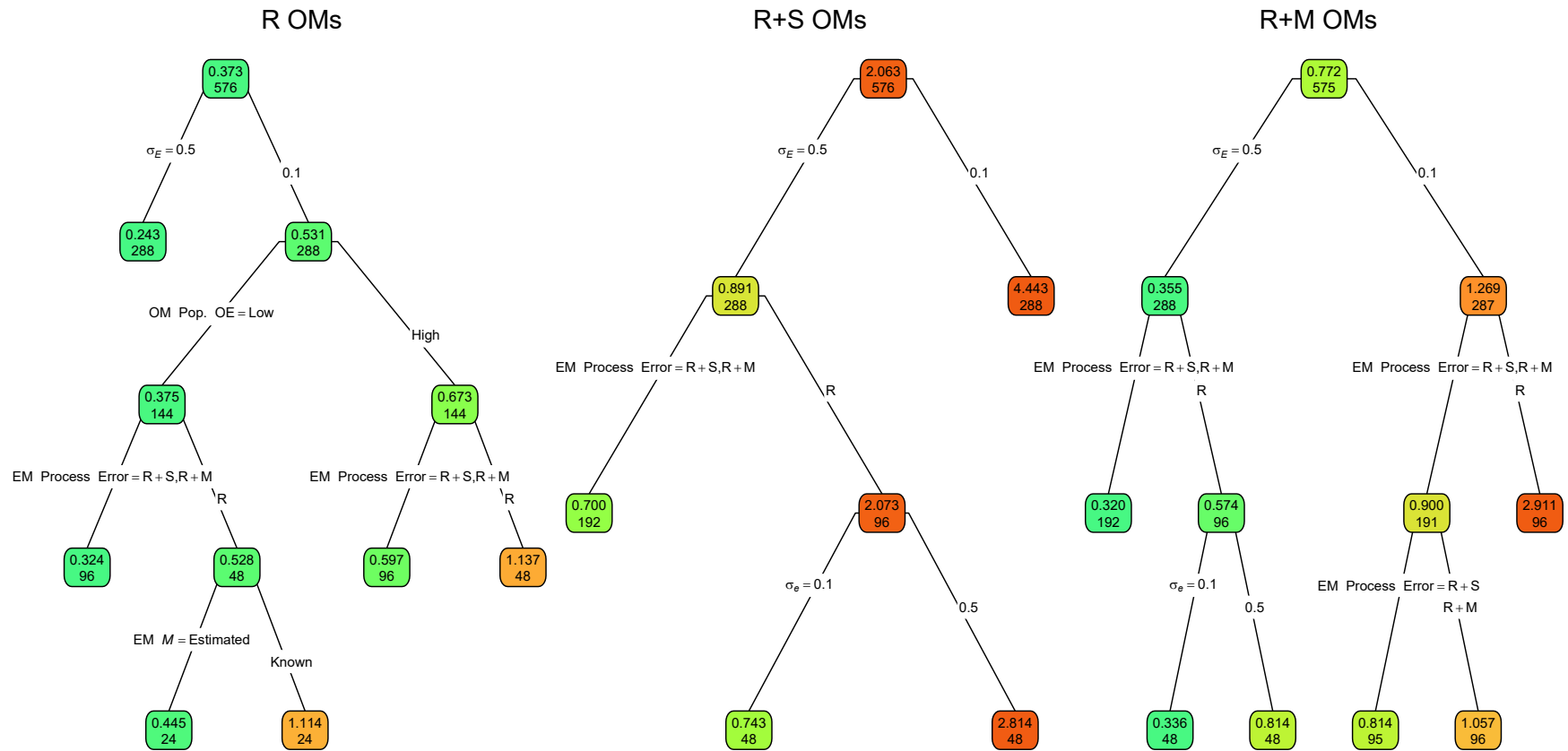


Fig. S25. Regression trees indicating primary factors determining reductions in sums of squares for the log-transformed RMSE of covariate effect on M ($\text{RMSE}(\hat{\beta}_E)$) for operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more green or red polygons, respectively.

Table S9. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to the log-transformed RMSE for estimates of the covariate effect on M ($\text{RMSE}(\hat{\beta}_E)$) with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	0.50	0.11	0.12
OM Obs. Error	12.04	5.66	0.36
$OM\sigma_e$	0.87	5.78	2.60
$OM\sigma_E$	15.85	24.18	24.26
$OM\rho_E$	0.36	0.82	0.34
OM β_E	6.72	0.07	0.96
EM Process Error	6.49	11.31	11.98
EM M assumption	0.04	0.64	0.07
All factors	42.87	48.58	40.65
+ All Two Way	67.34	74.09	61.74
+ All Three Way	79.42	82.75	71.92

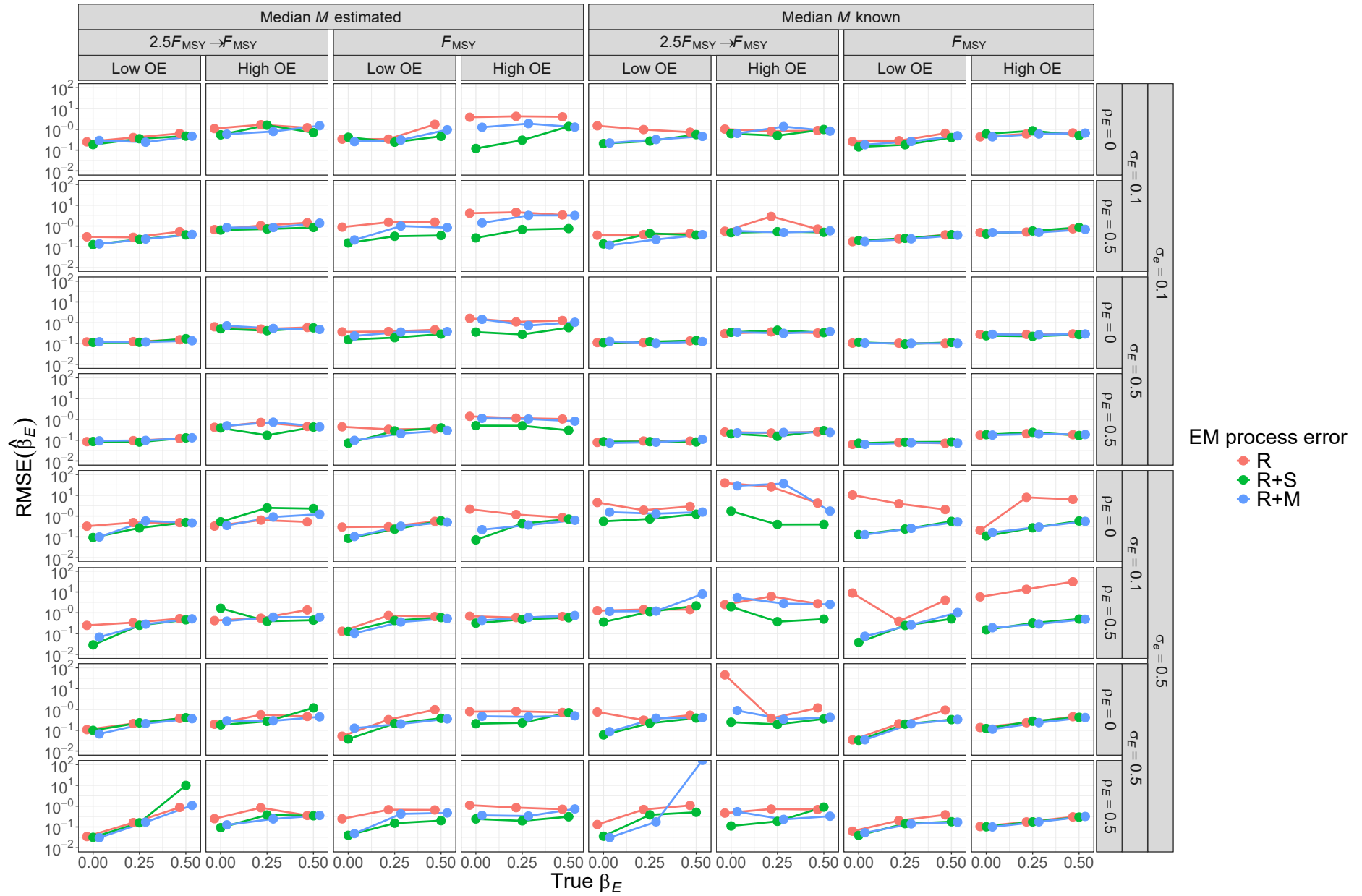


Fig. S26. For R OMs, RMSE of estimates of covariate effect on M (β_E) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated).

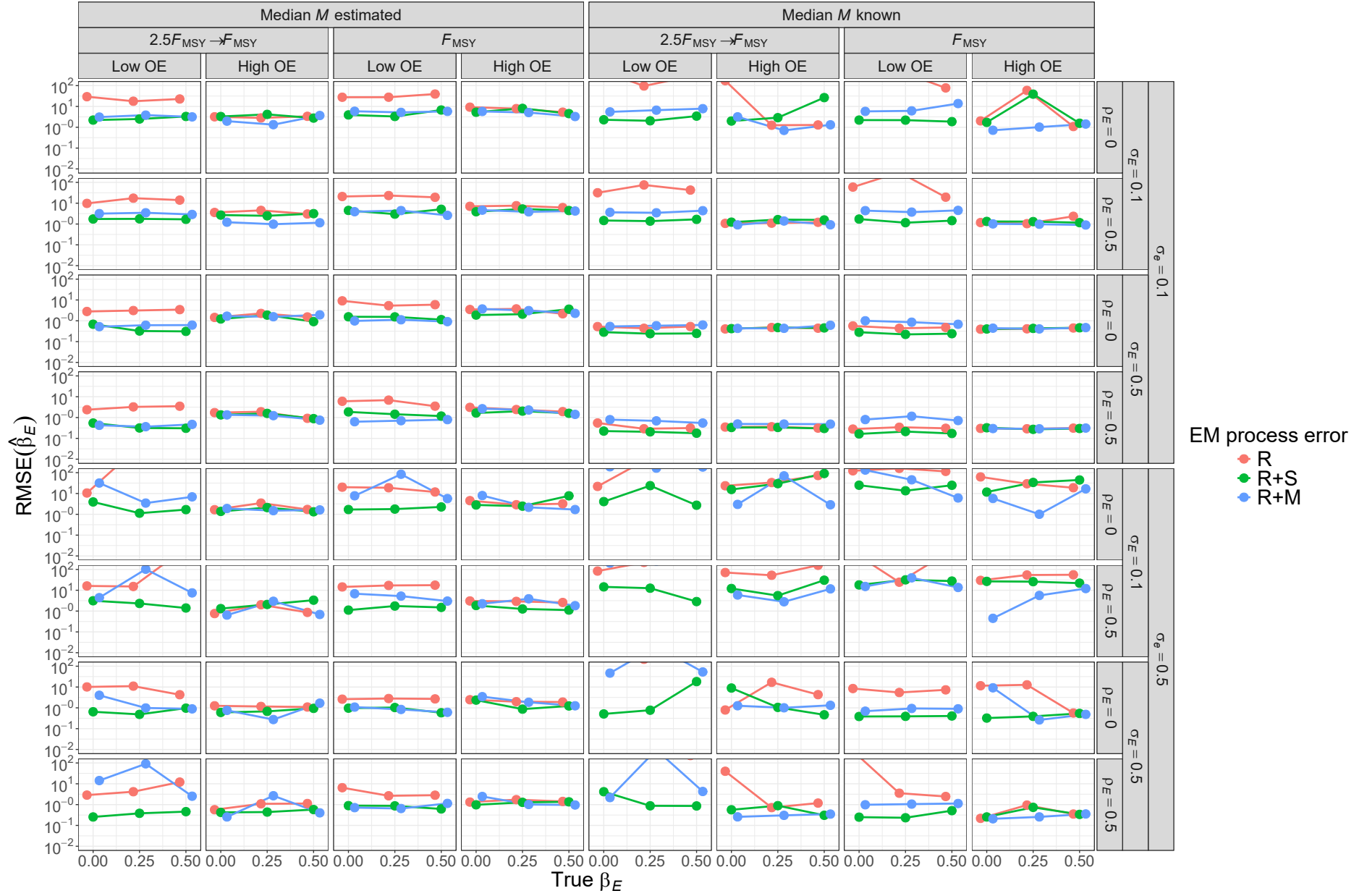


Fig. S27. For R+S OMs, RMSE of estimates of covariate effect on M ($\hat{\beta}_E$) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated).

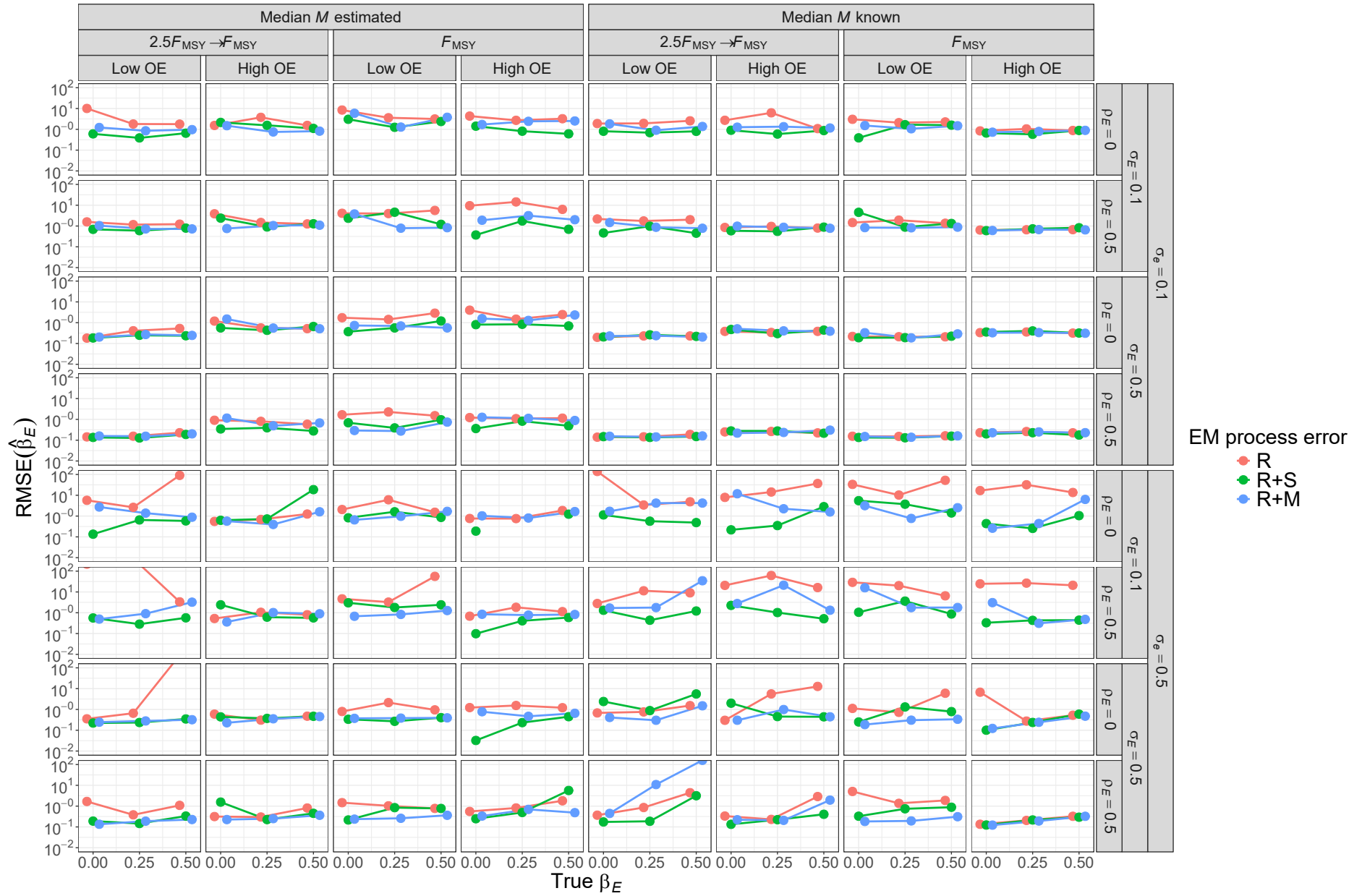


Fig. S28. For R+M OM, RMSE of estimates of covariate effect on M (β_E) from fitting EMs with alternative process error assumptions and treatment of median M (e_M^β known or estimated).

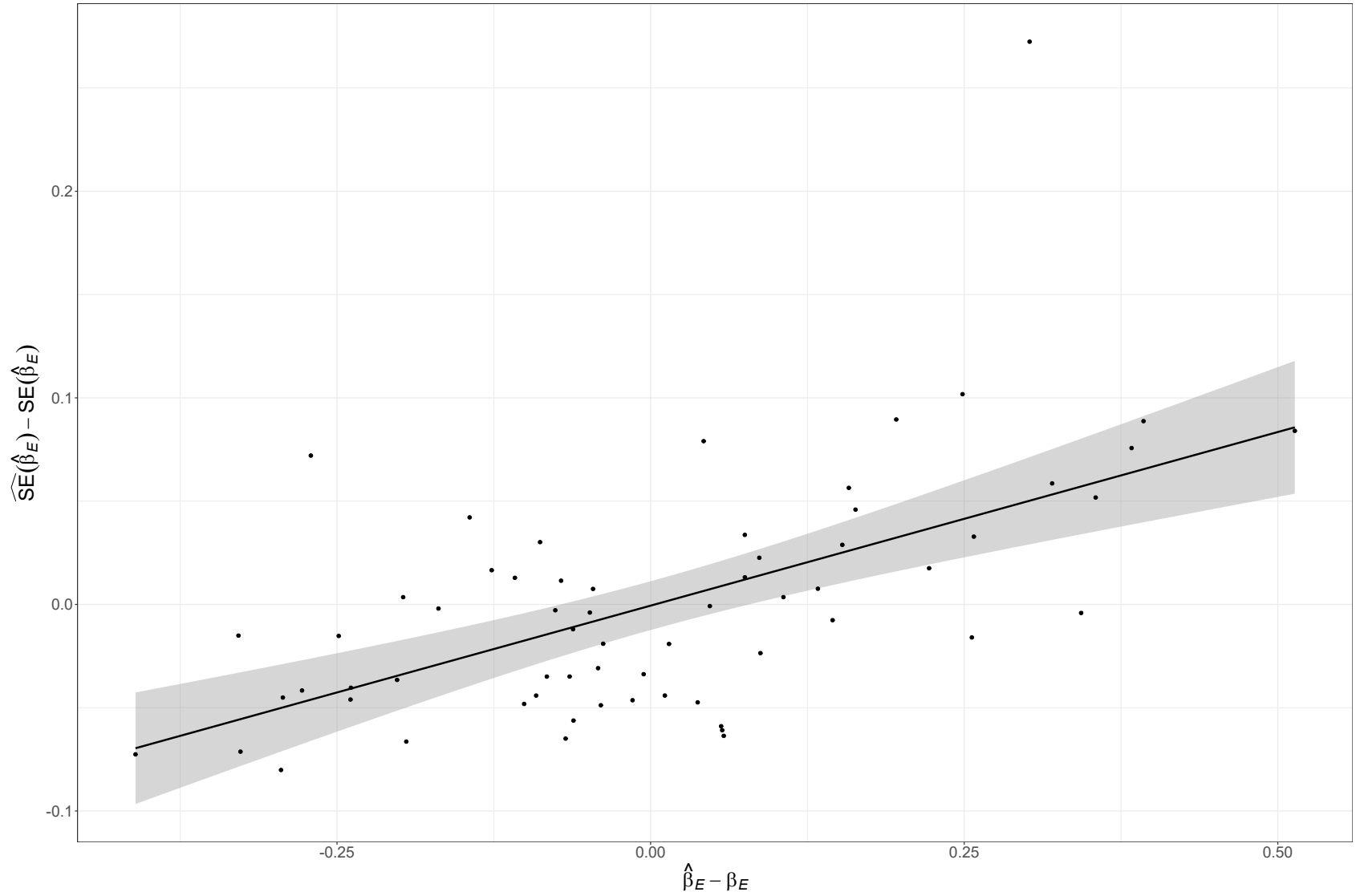


Fig. S29. Positive correlation of covariate effect estimates and Hessian-based standard error estimates for EM that also estimates the median M parameter (β_M) and has correct R+M process error assumption fitted to simulated data from the OM with R+M process errors, temporal contrast in fishing pressure, low observation uncertainty for both population (*LowOE*) and covariate observations ($\sigma_e = 0.1$), high and uncorrelated temporal variability in the true covariate ($\sigma_E = 0.5$ and $\rho_E = 0$), and the strongest covariate effect on M ($\beta_E = 0.5$).

Table S10. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to errors in estimation of the median M parameter (β_M) with each OM and EM factor (rows) included individually, combined, and with second and third order interactions. Includes results from all unconverged and converged models.

Factor	R	R+S	R+M
OM F history	<0.01	<0.01	<0.01
OM Obs. Error	<0.01	<0.01	<0.01
OM σ_e	<0.01	<0.01	<0.01
OM σ_E	<0.01	<0.01	<0.01
OM ρ_E	<0.01	<0.01	<0.01
OM β_E	<0.01	<0.01	0.01
EM Convergence	<0.01	0.01	0.01
EM Process Error	<0.01	0.01	0.01
EM β_E assumption	<0.01	<0.01	<0.01
All factors	0.02	0.02	0.05
+ All Two Way	0.15	0.15	0.24
+ All Three Way	0.57	0.54	0.73

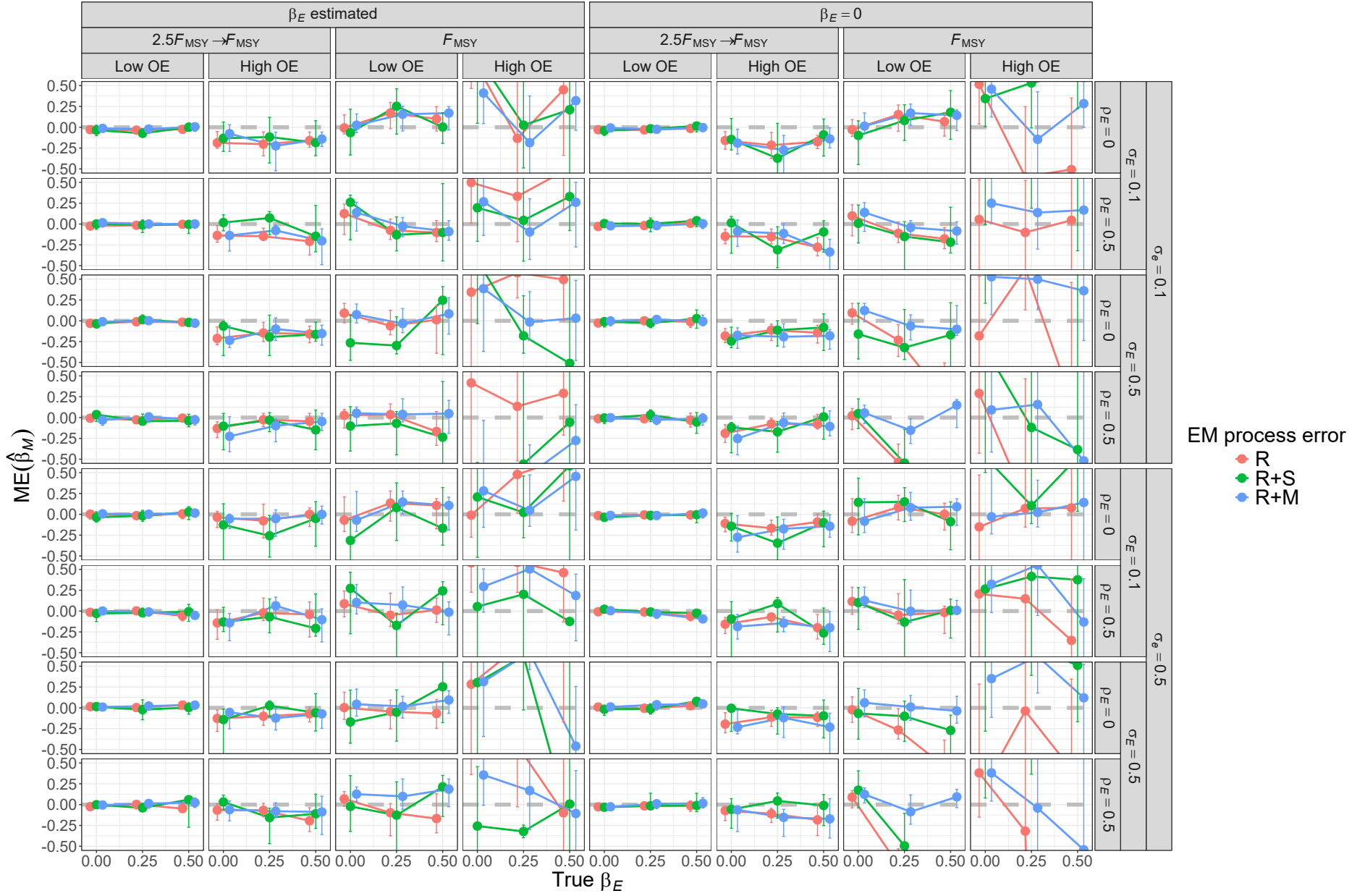


Fig. S30. For R OMs, median error (ME) of estimates of β_M from fitting EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

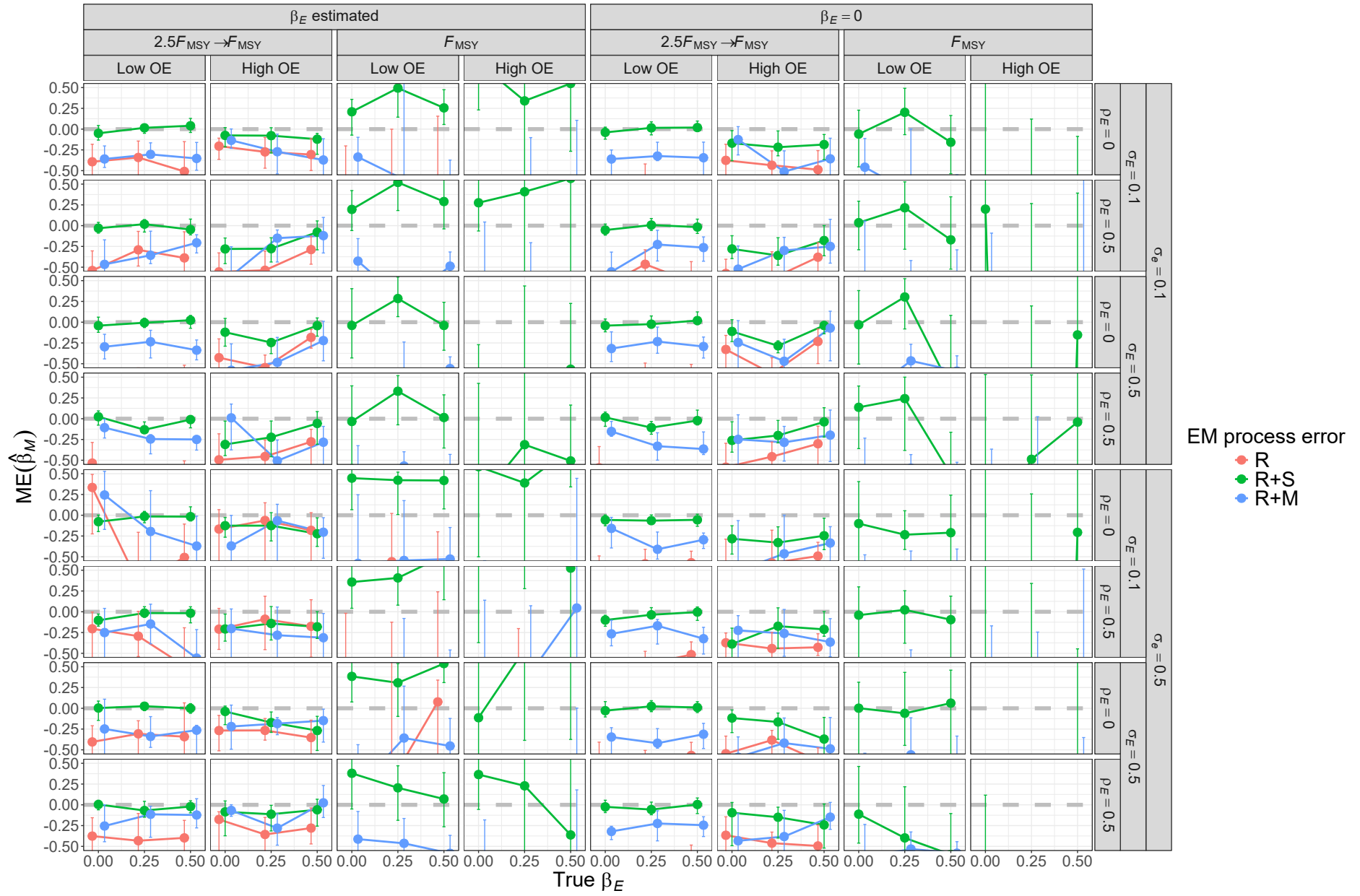


Fig. S31. For R+S OMs, median error (ME) of estimates of β_M from fitting EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

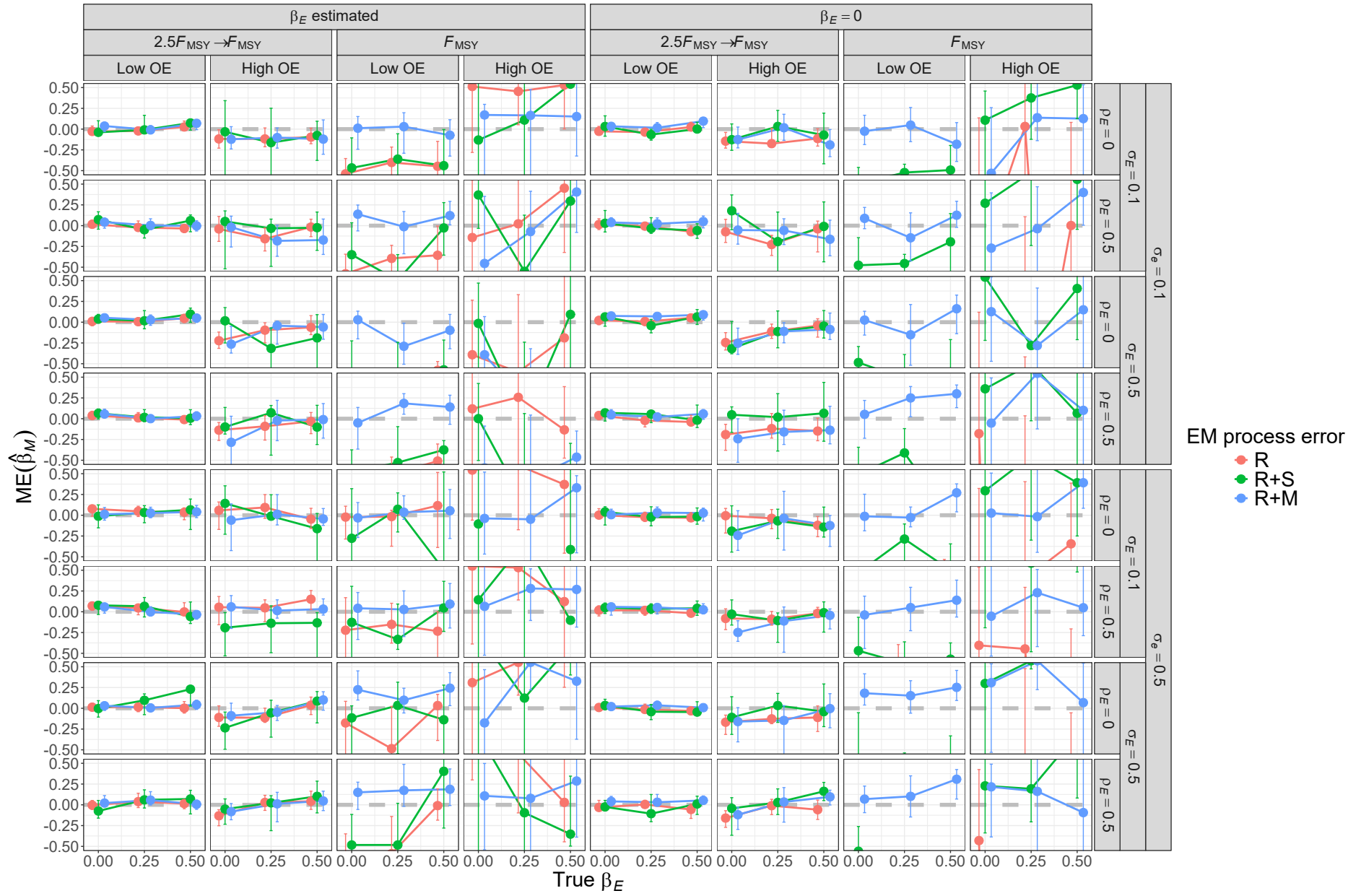


Fig. S32. For R+M OMs, median error (ME) of estimates of β_M from fitting EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

898 **Median natural mortality parameter standard error estimation bias**

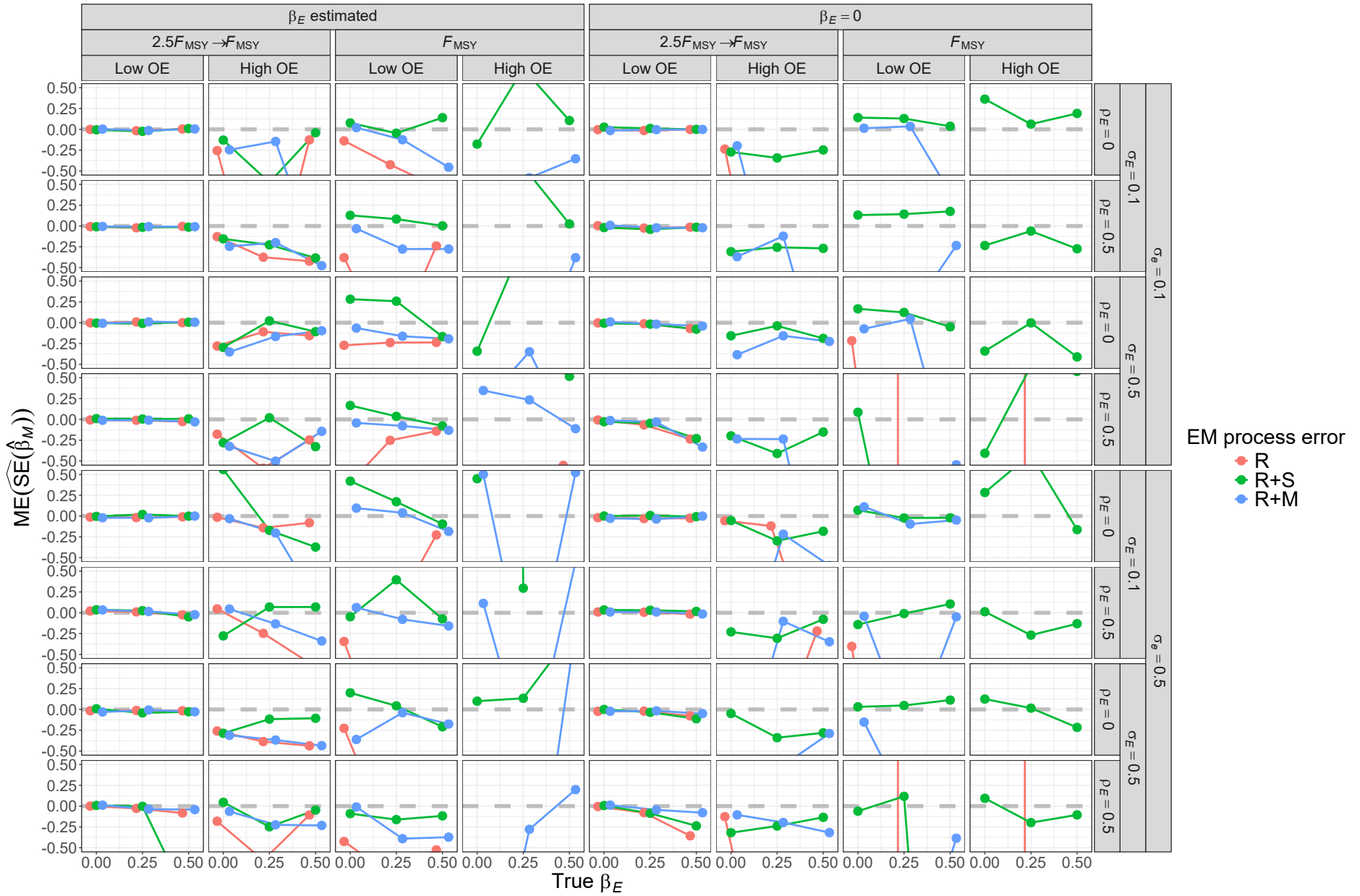


Fig. S33. For R OMs, median error (ME) of Hessian-based estimates of standard error for median M parameter (β_M) from fitting EMs with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given OM scenario.

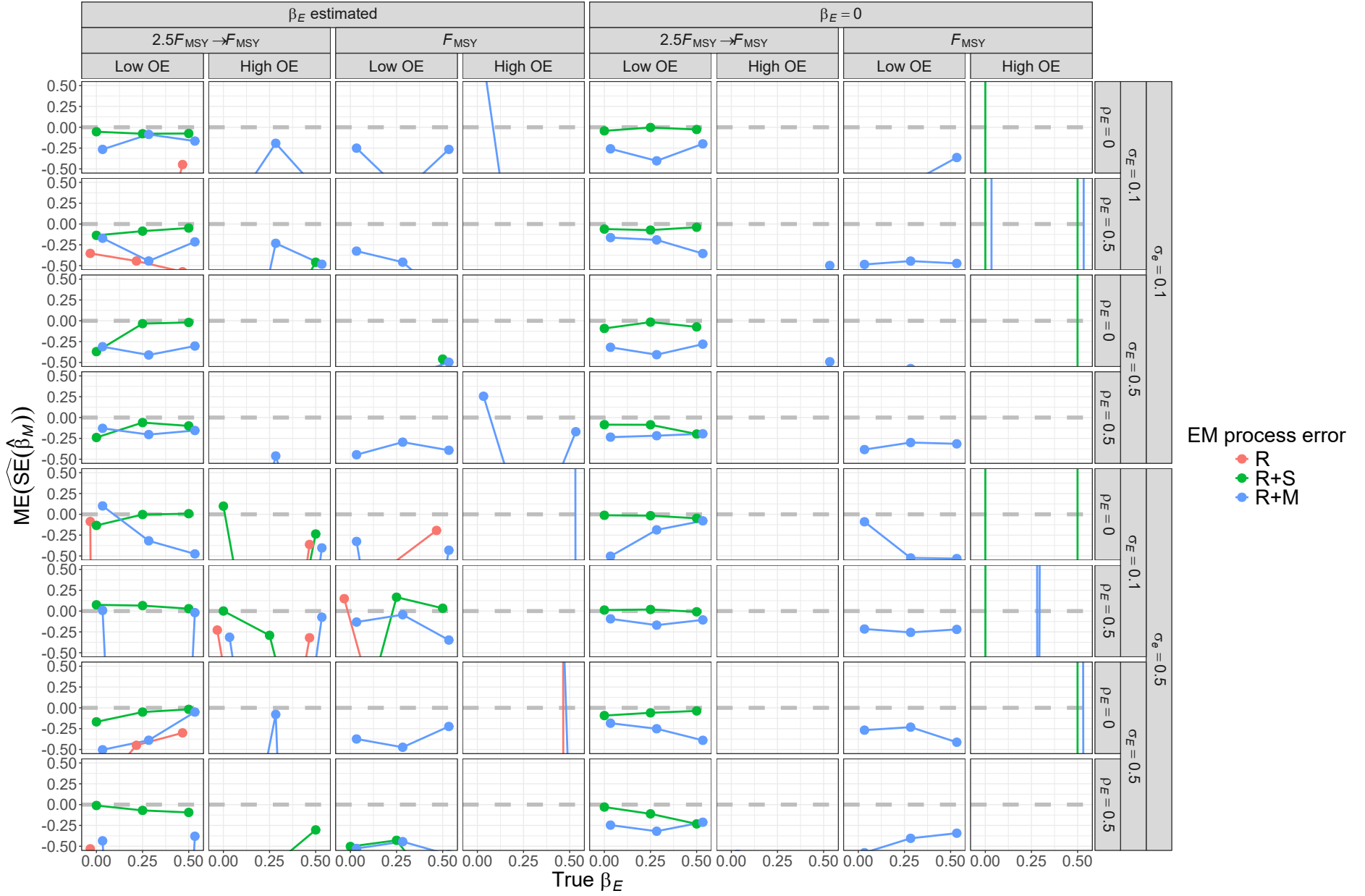


Fig. S34. For R+S OMs, median error (ME) of Hessian-based estimates of standard error for median M parameter (β_M) from fitting EMs with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given OM scenario.

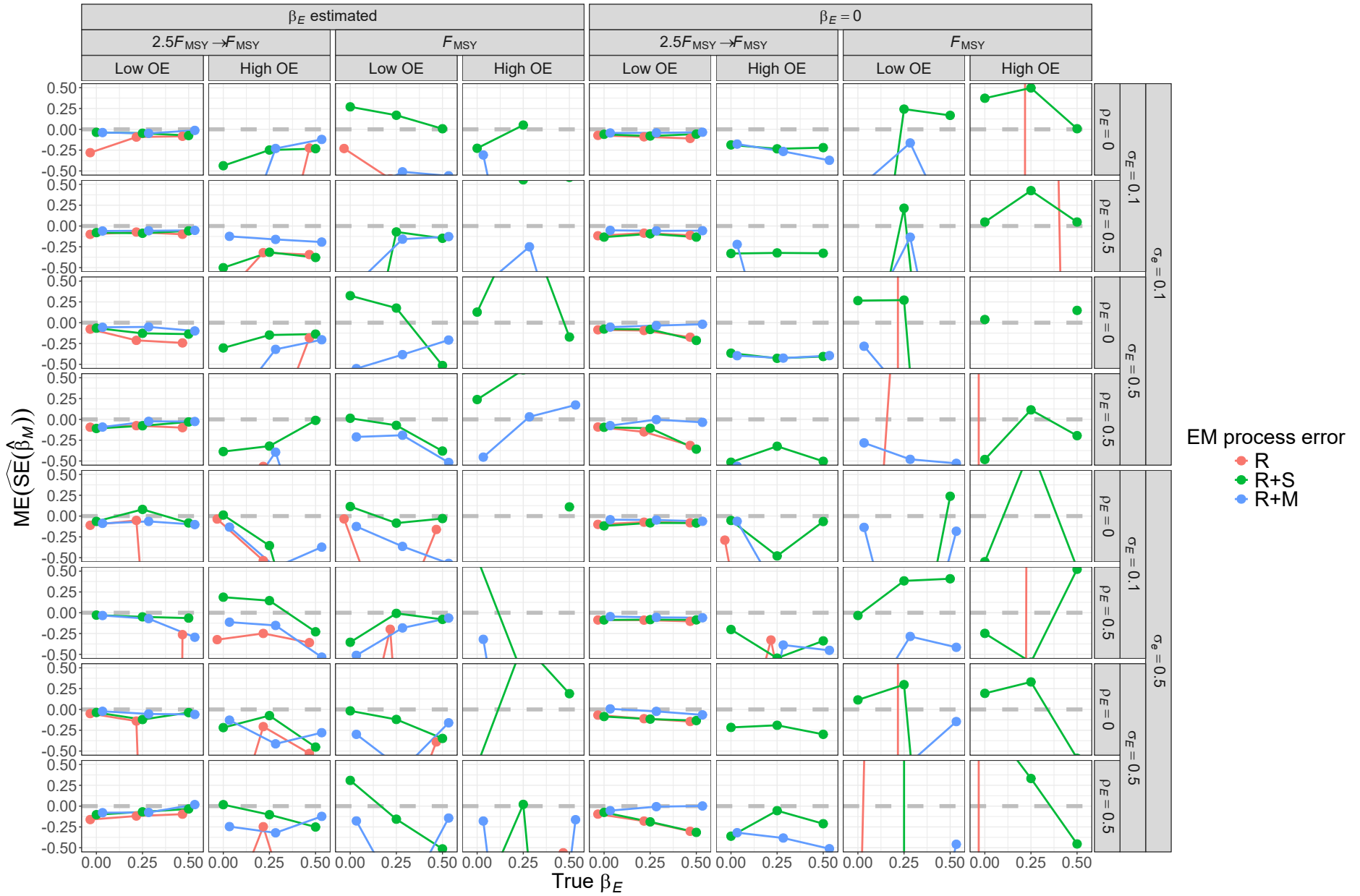


Fig. S35. For R+M OMs, median error (ME) of Hessian-based estimates of standard error for median M parameter ($\hat{\beta}_M$) from fitting EMs with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given OM scenario.

899 **Median Natural mortality parameter confidence interval coverage**

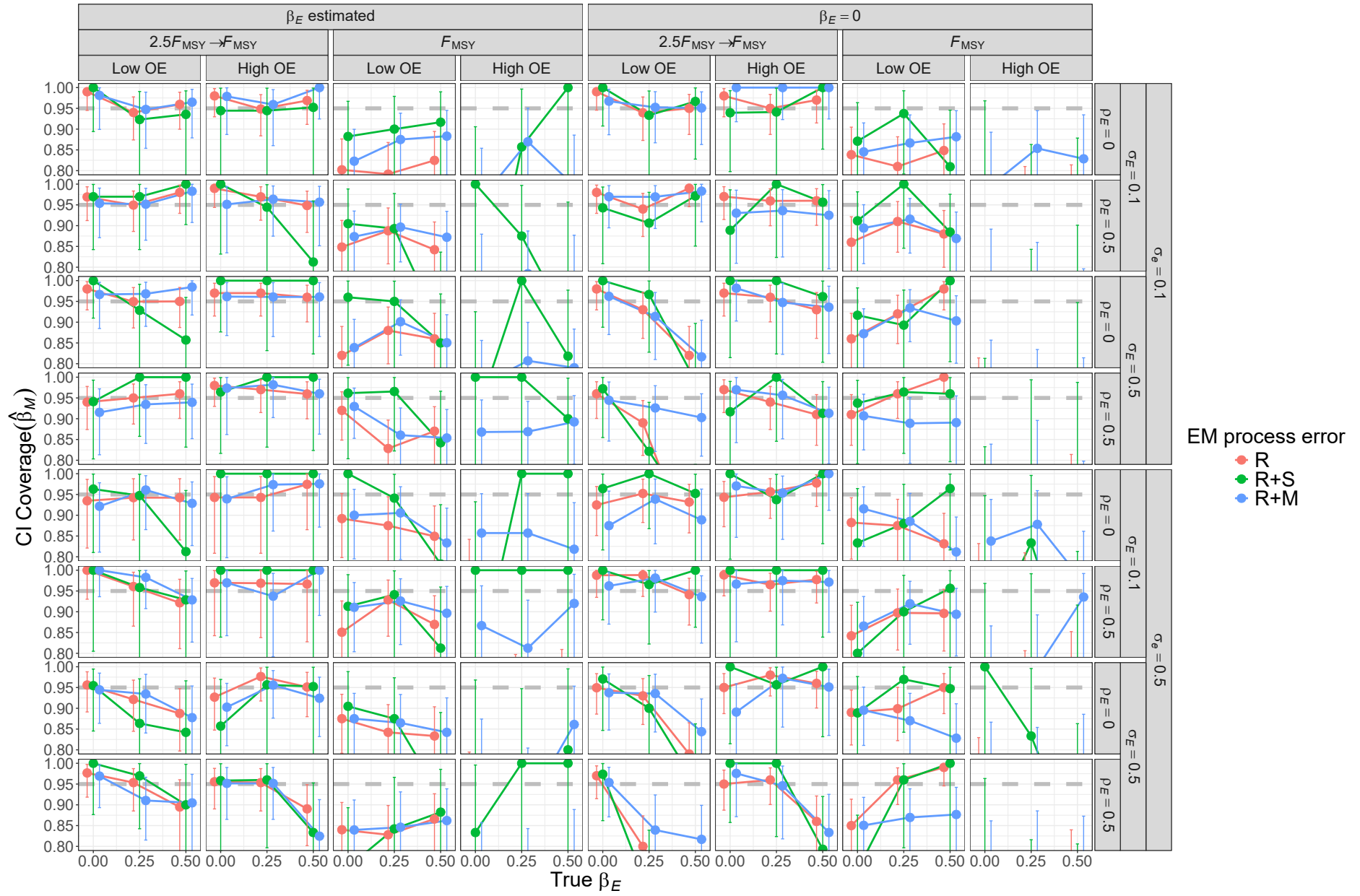


Fig. S36. For R OMs, probability of 95% confidence interval for β_M containing the true value for EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

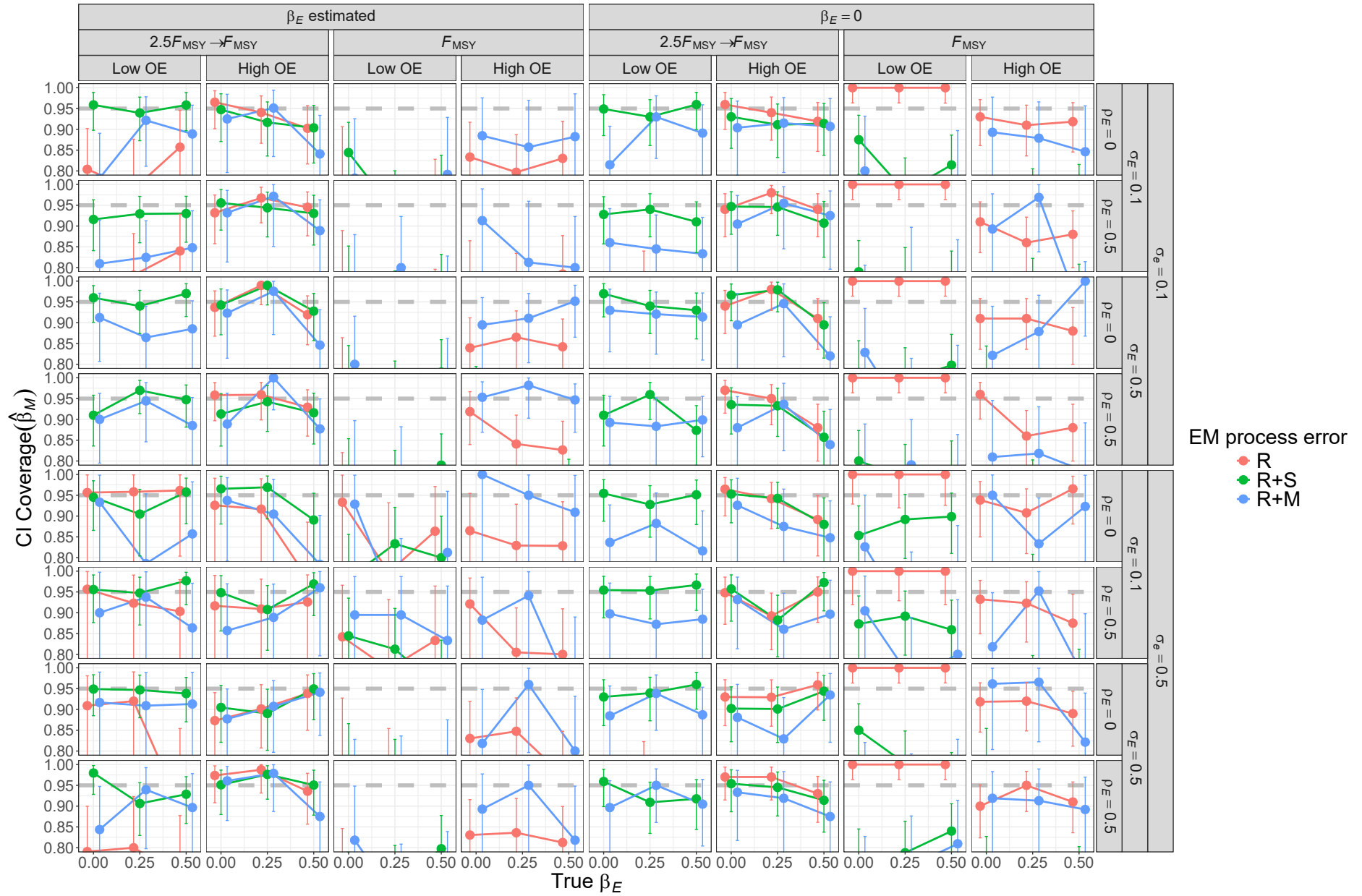


Fig. S37. For R+S OM, probability of 95% confidence interval for β_M containing the true value for EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

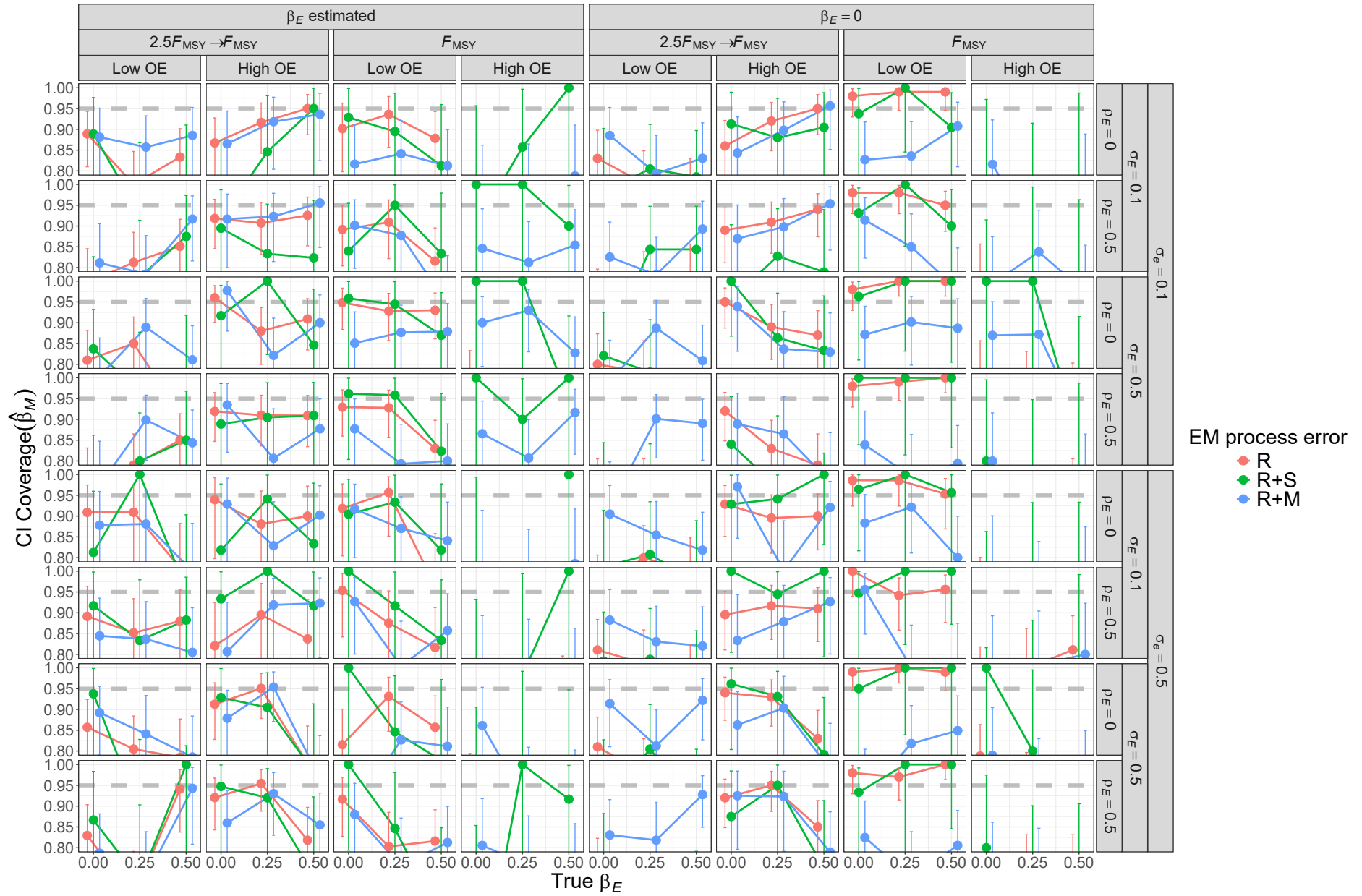


Fig. S38. For R+M OMs, probability of 95% confidence interval for β_M containing the true value for EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

900 Median natural mortality parameter estimate and standard error
901 example

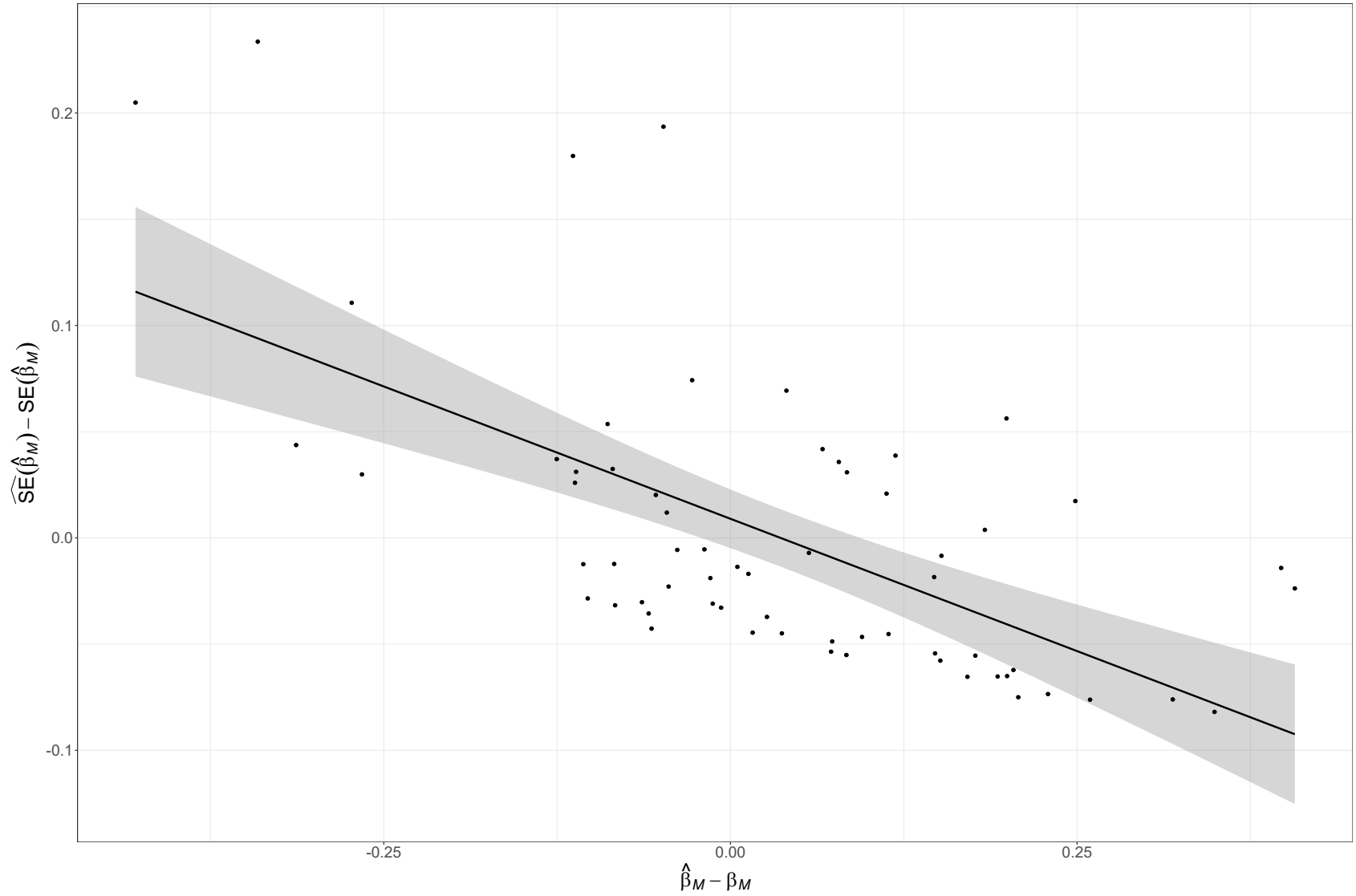


Fig. S39. Negative correlation of β_M estimates and Hessian-based standard error estimates for EM that also estimates the covariate effect and has correct R+M process error assumption fitted to simulated data from the OM with R+M process errors, temporal contrast in fishing pressure, low observation uncertainty for both population (*LowOE*) and covariate observations ($\sigma_e = 0.1$), high and uncorrelated temporal variability in the true covariate ($\sigma_E = 0.5$ and $\rho_E = 0$), and the strongest covariate effect on M ($\beta_E = 0.5$).

902 **Median Natural mortality parameter accuracy**

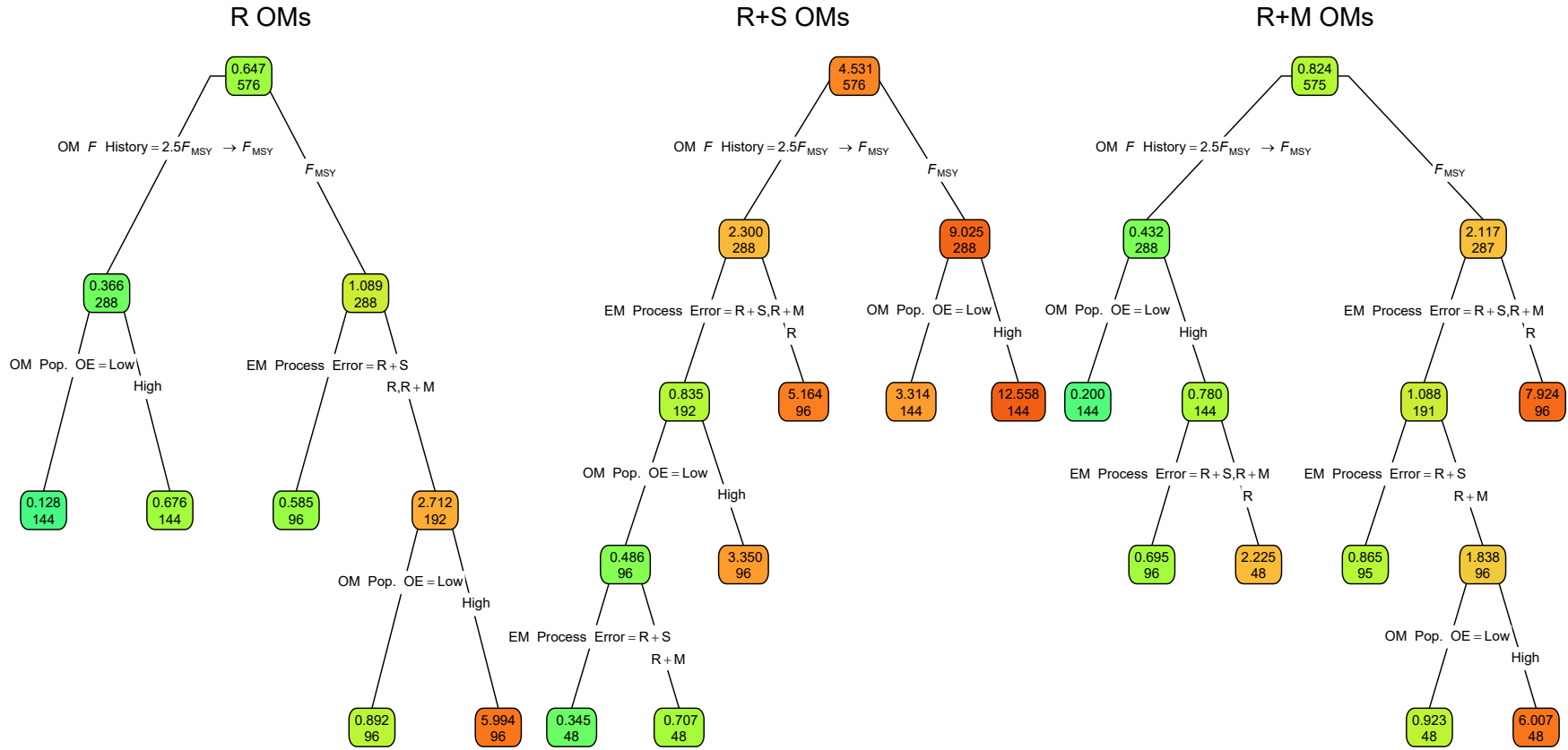


Fig. S40. Regression trees indicating primary factors determining reductions in sums of squares of errors for the log-transformed RMSE of estimates for the median M parameter ($\widehat{\beta}_M$) in EMs fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more green or red polygons, respectively.

Table S11. For each OM process error source (columns), percent reduction in deviance for linear regression models fit to the log-transformed RMSE for estimates of the median M parameter ($\text{RMSE}(\hat{\beta}_M)$) with each OM and EM factor (rows) included individually, combined, and with second and third order interactions.

Factor	R	R+S	R+M
OM F history	34.70	24.62	34.91
OM Obs. Error	25.55	17.82	12.77
$OM\sigma_e$	0.01	<0.01	<0.01
$OM\sigma_E$	0.19	0.02	0.07
$OM\rho_E$	0.05	0.26	0.03
OM β_E	0.58	0.23	0.16
EM Process Error	10.75	18.12	14.50
$EM\beta_E$ assumption	3.96	6.39	2.93
All factors	75.79	67.47	65.26
+ All Two Way	90.21	80.65	77.59
+ All Three Way	93.72	88.68	86.44

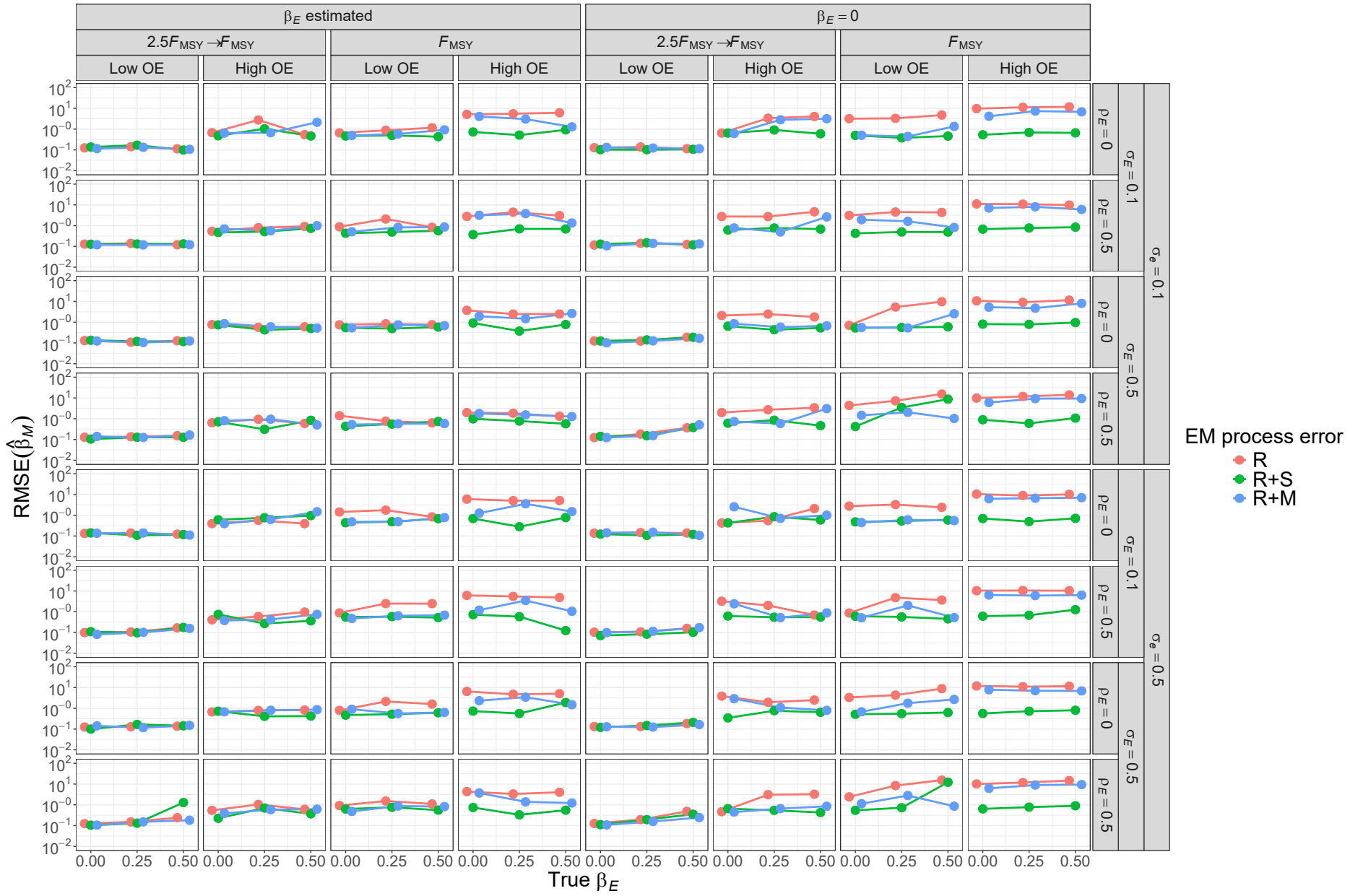


Fig. S41. For R OMs, RMSE of estimates of β_M from fitting EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated).

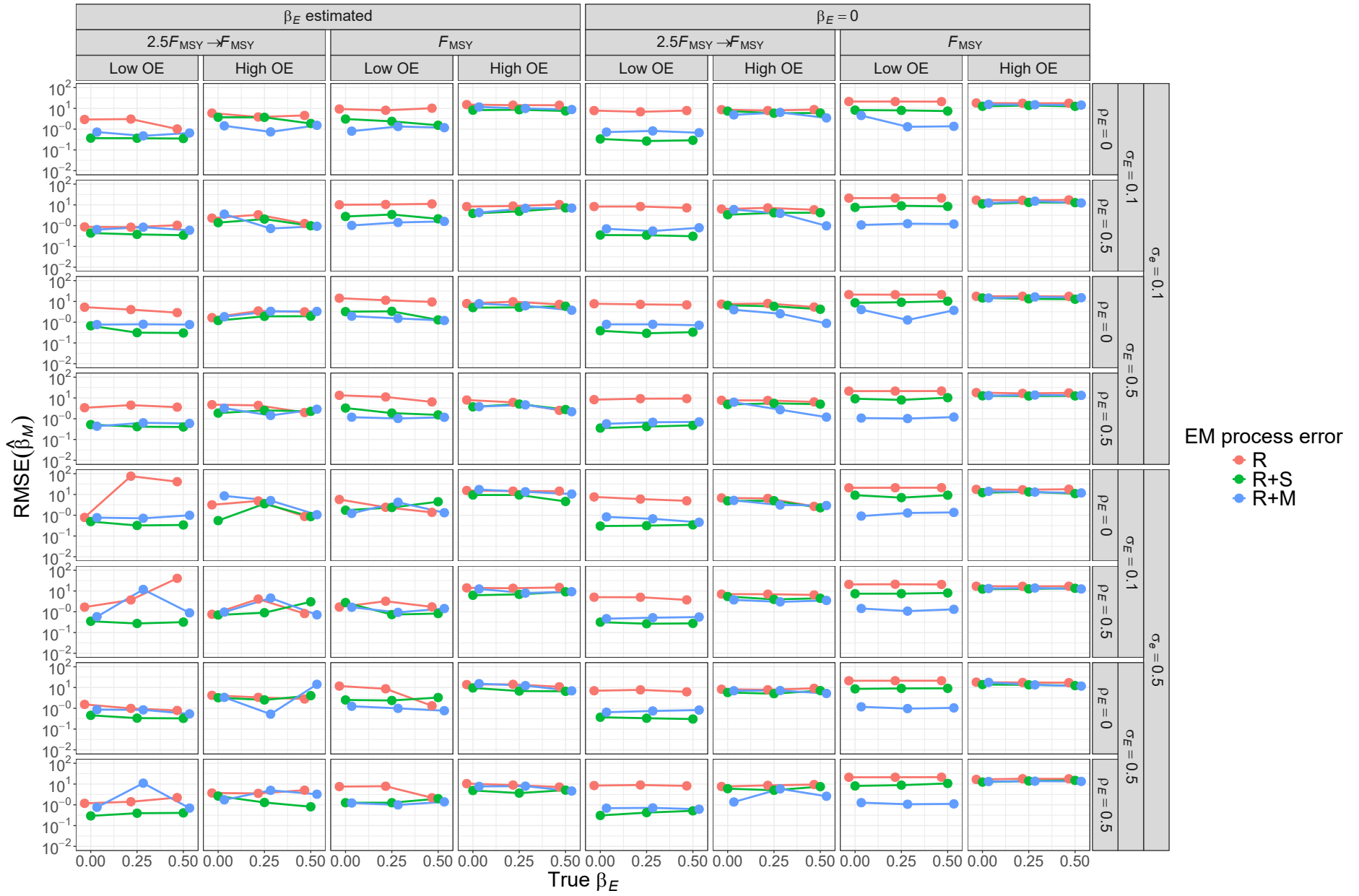


Fig. S42. For R+S OMs, RMSE of estimates of β_M from fitting EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated).

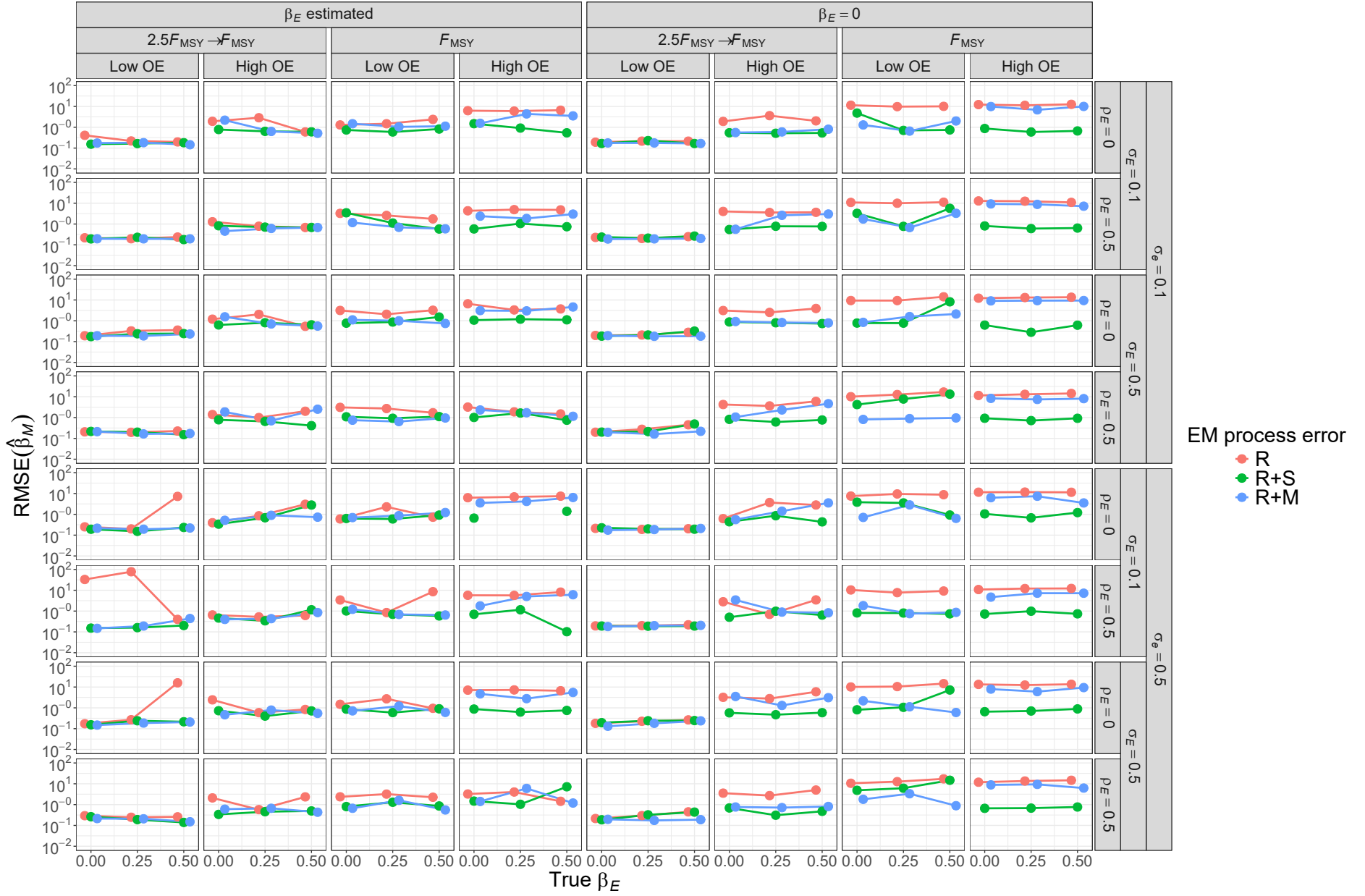


Fig. S43. For R+M OMs, RMSE of estimates of β_M from fitting EMs with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated).

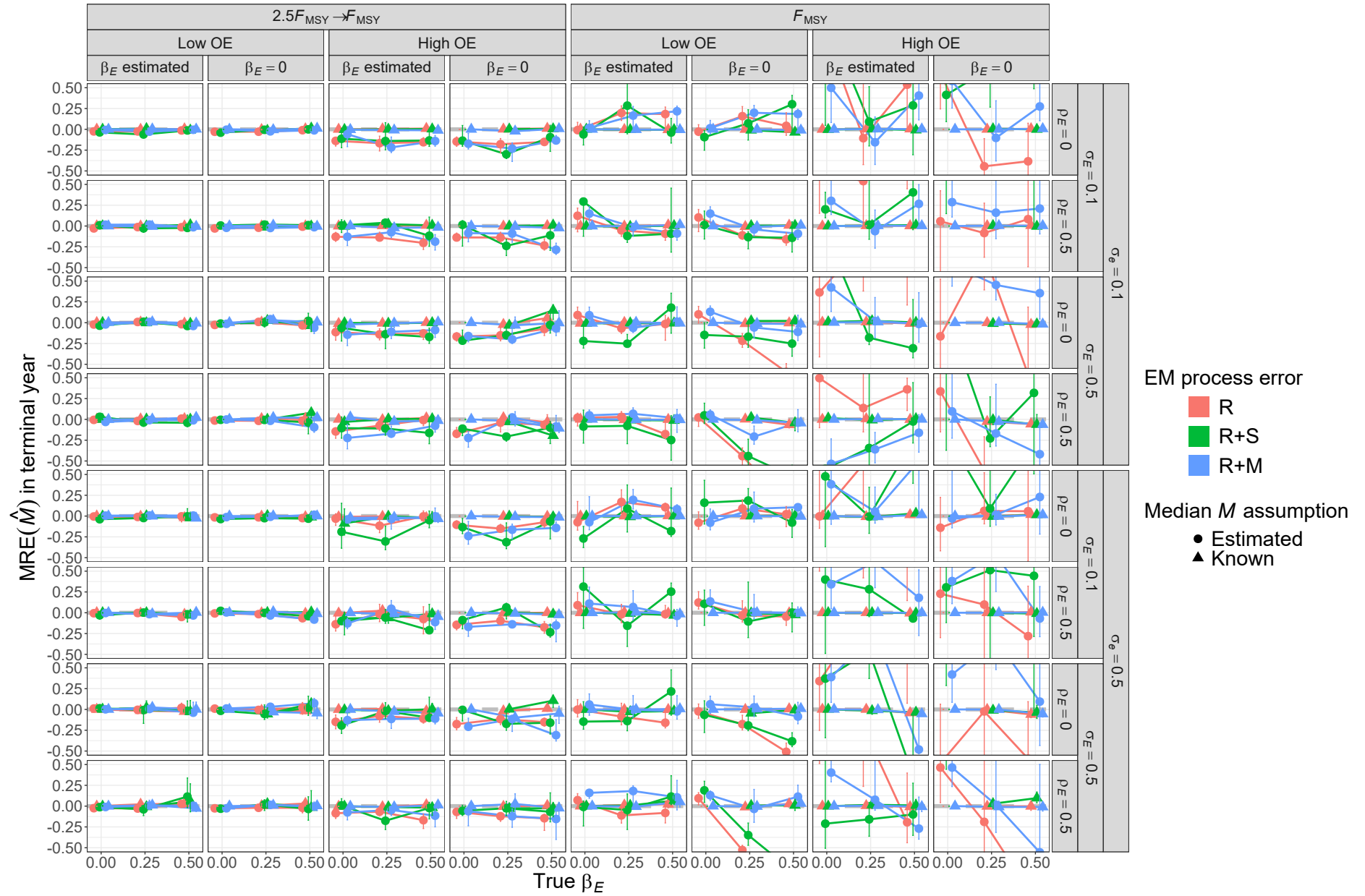


Fig. S44. For R OMs, median relative error (MRE) of estimates of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

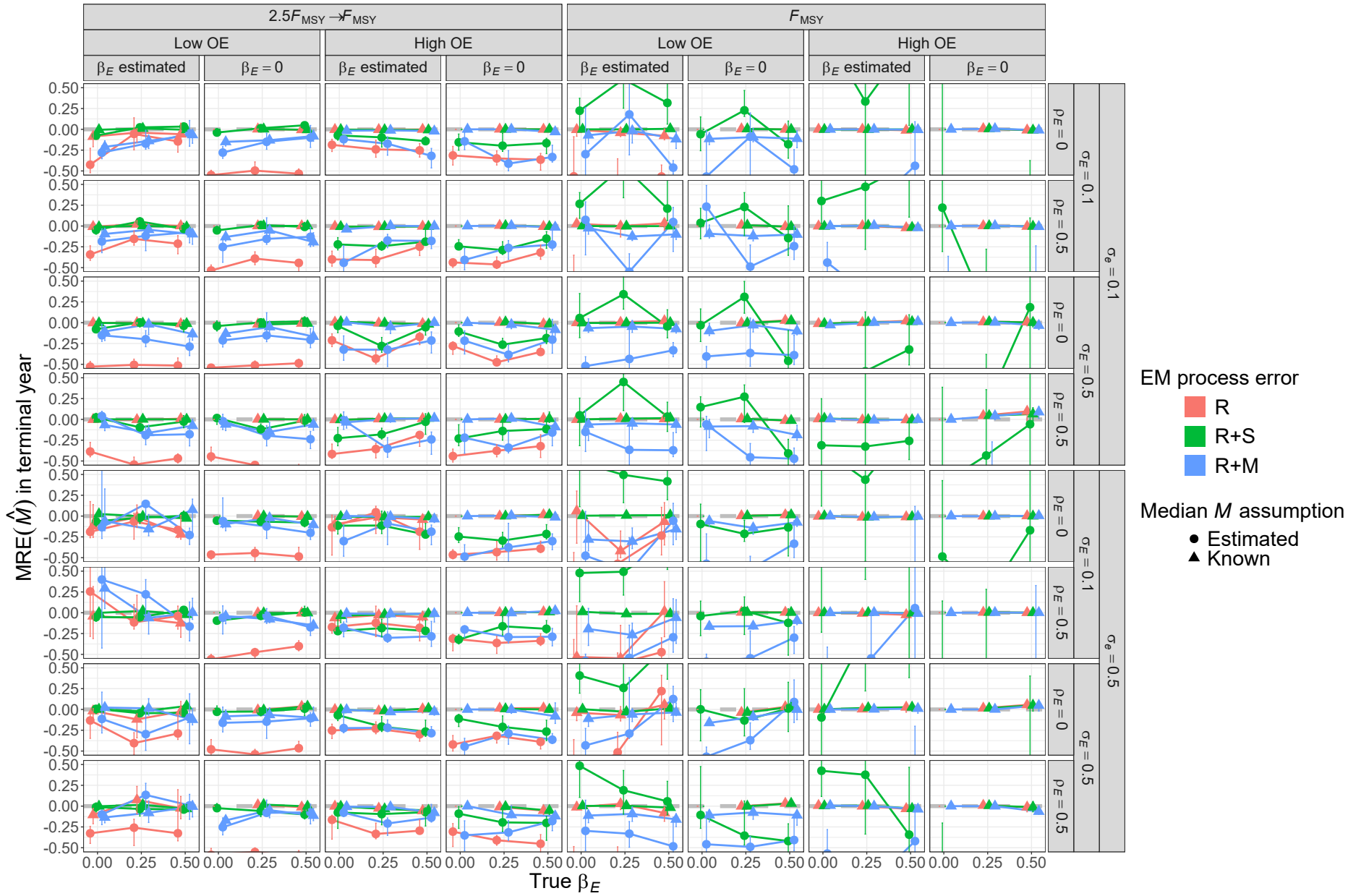


Fig. S45. For R+S OM, median relative error (MRE) of estimates of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

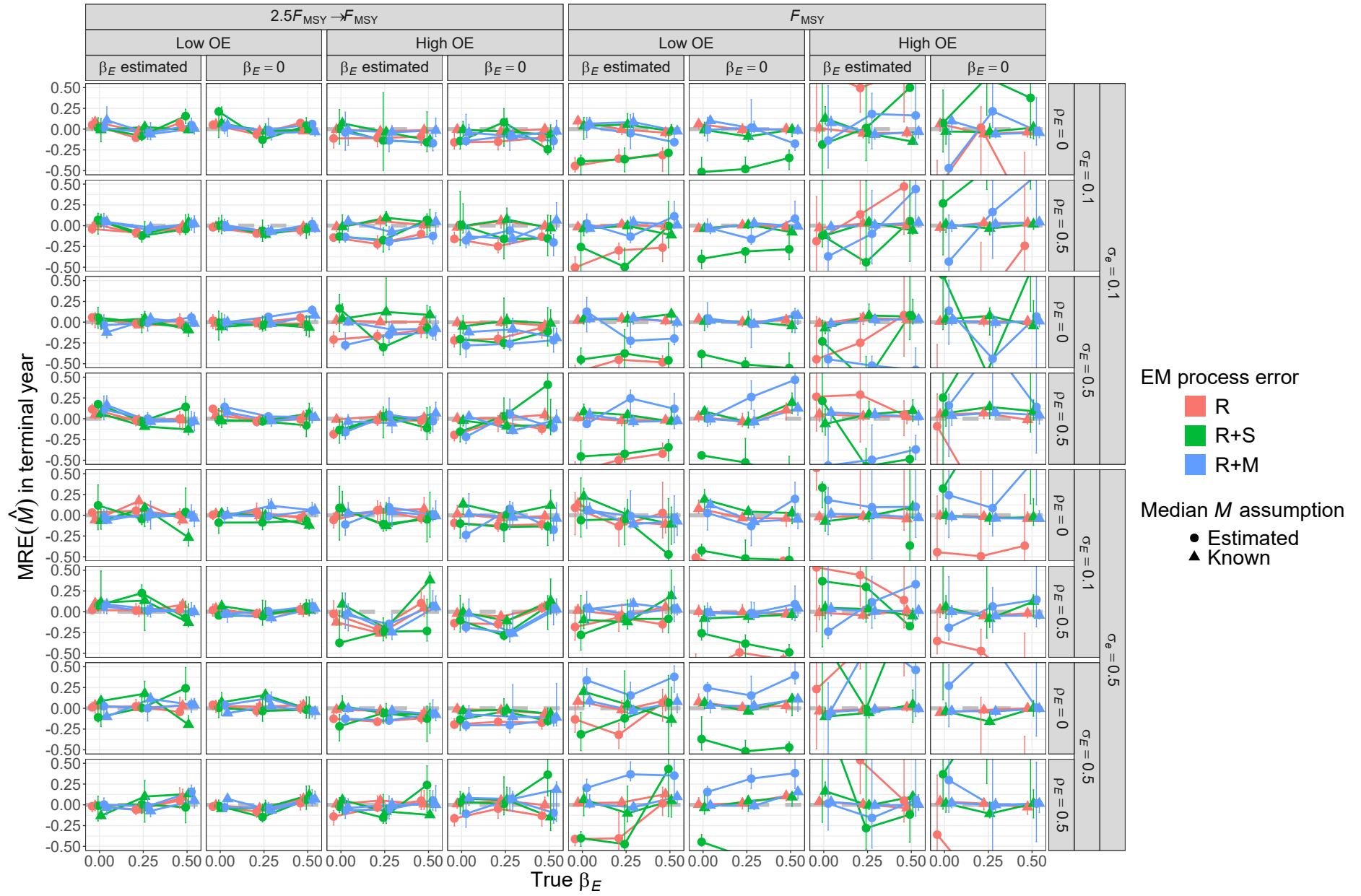


Fig. S46. For R+M OM, median relative error (MRE) of estimates of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

904 **Terminal year natural mortality accuracy**

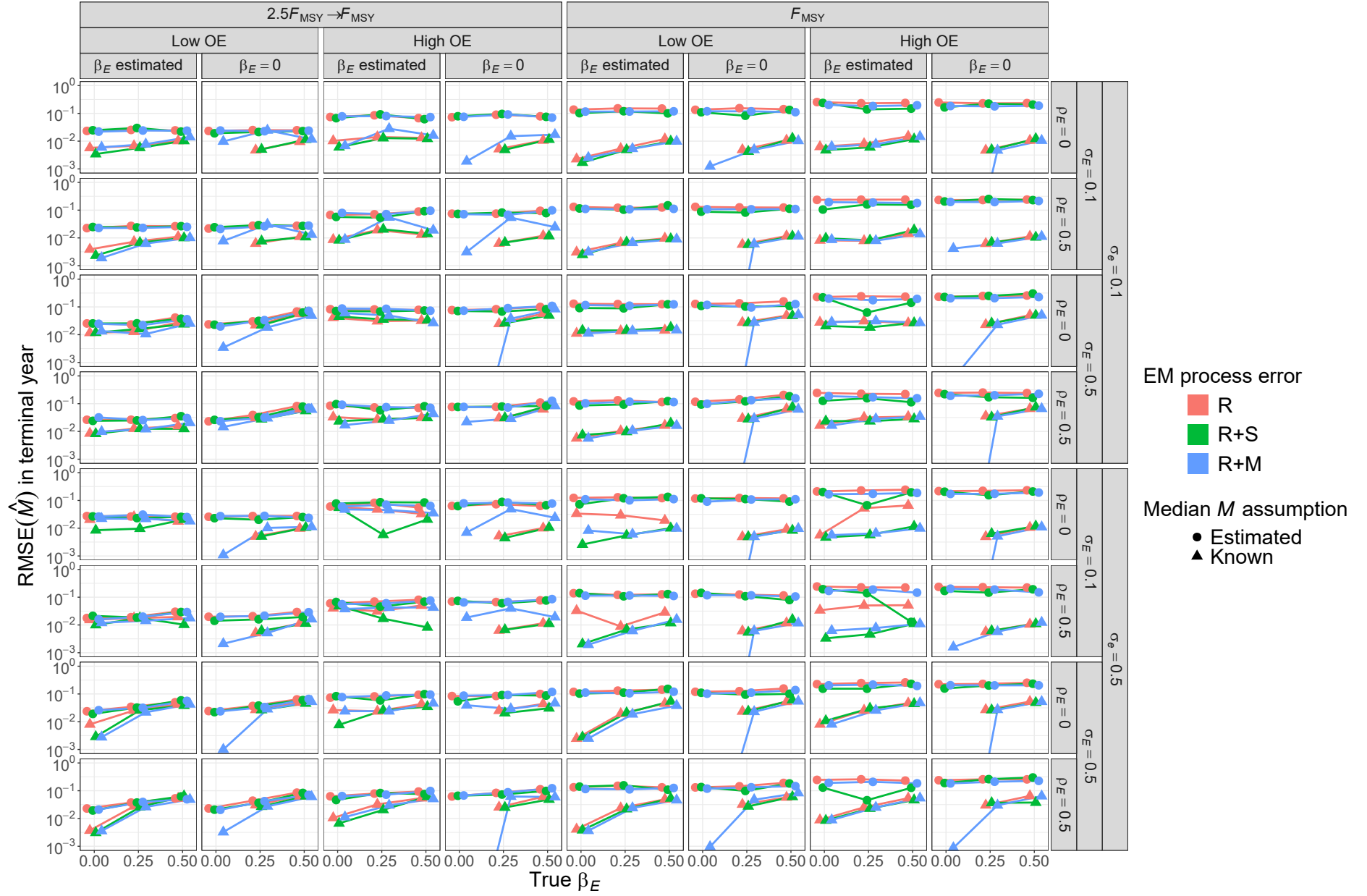


Fig. S47. For R OMs, RMSE of estimates of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

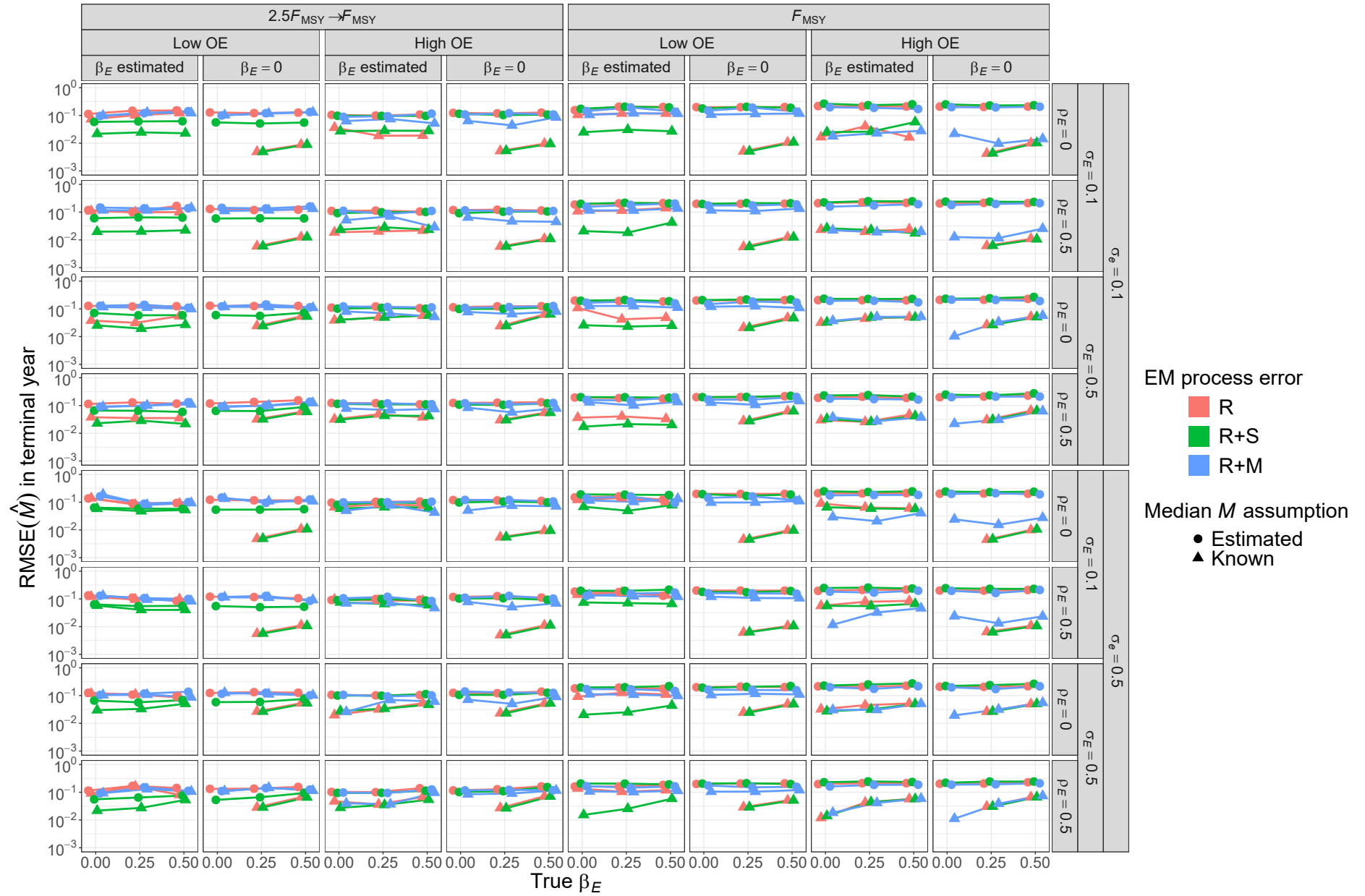


Fig. S48. For R+S OM, RMSE of estimates of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

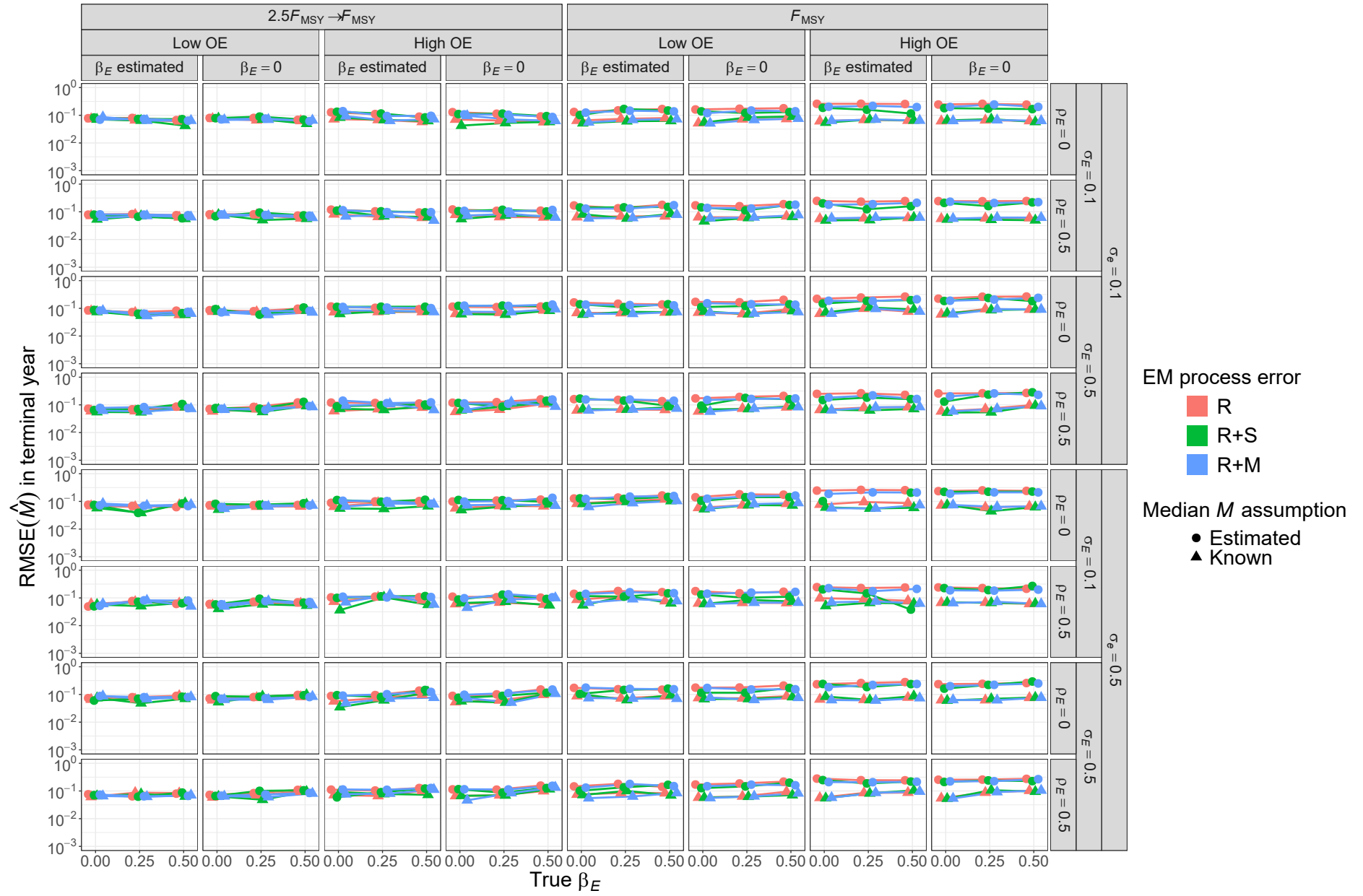


Fig. S49. For R+M OMs, RMSE of estimates of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

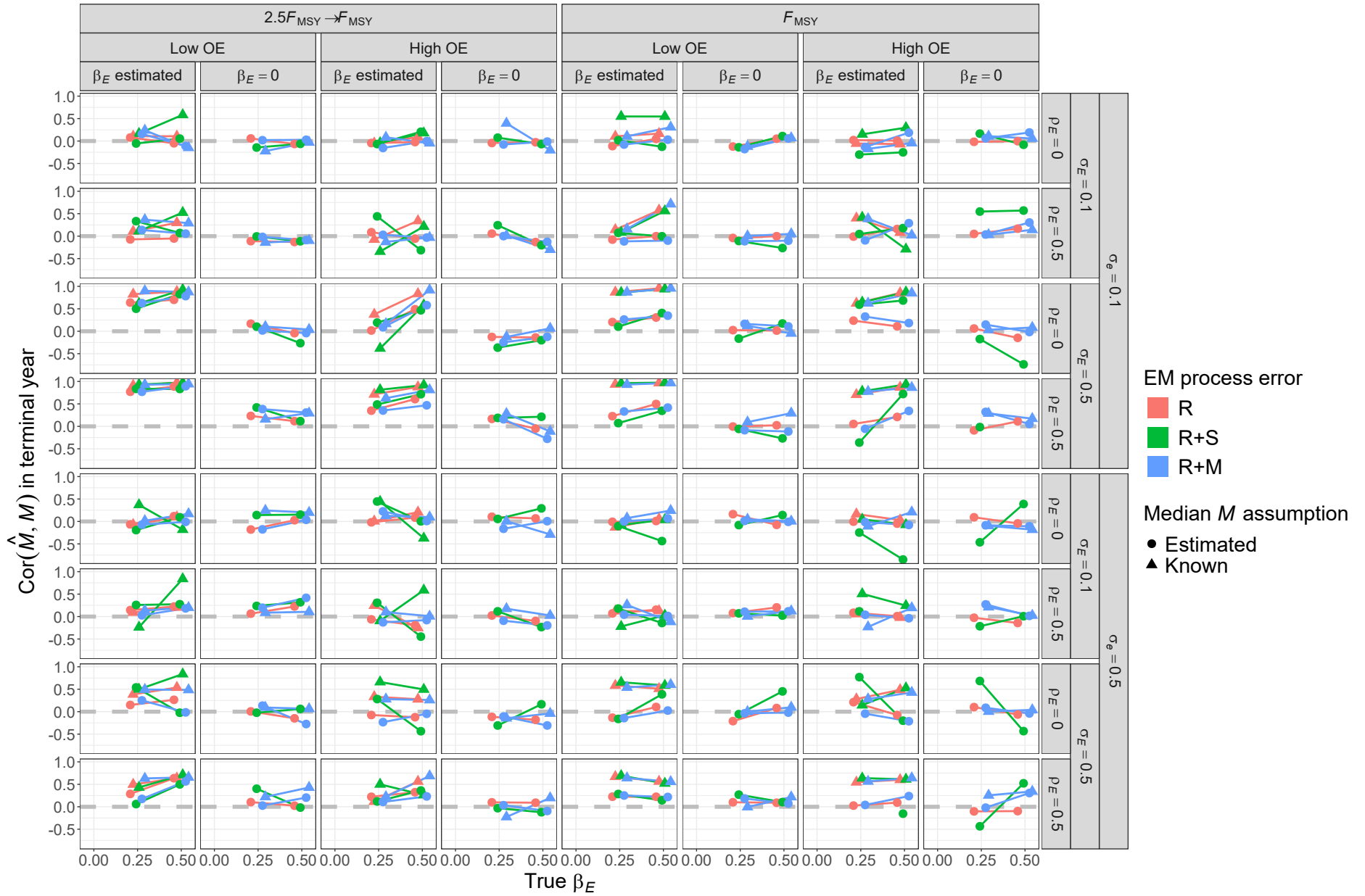


Fig. S50. For R OMs, correlation of estimates and true values of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

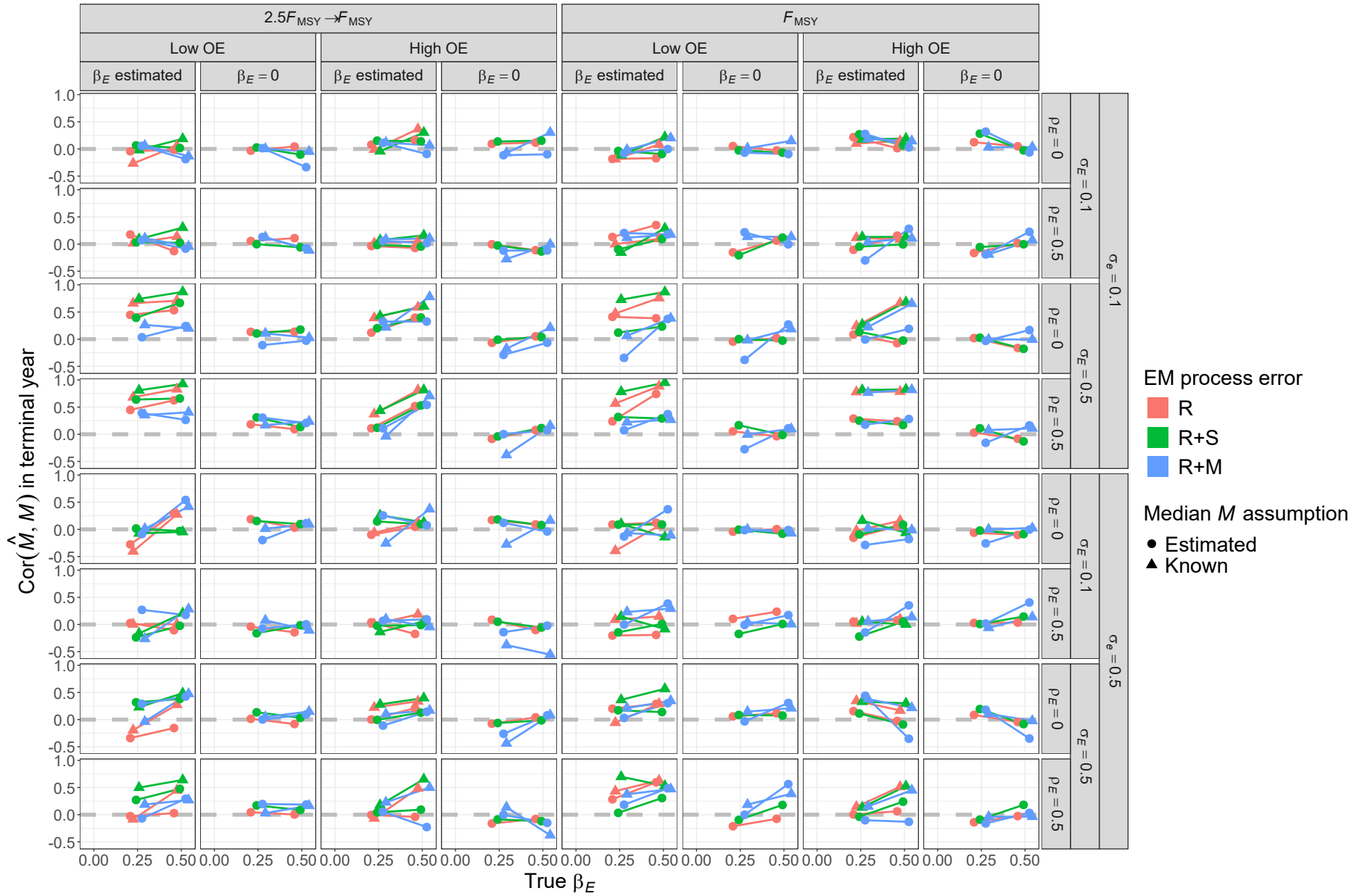


Fig. S51. For R+S OM, correlation of estimates and true values of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

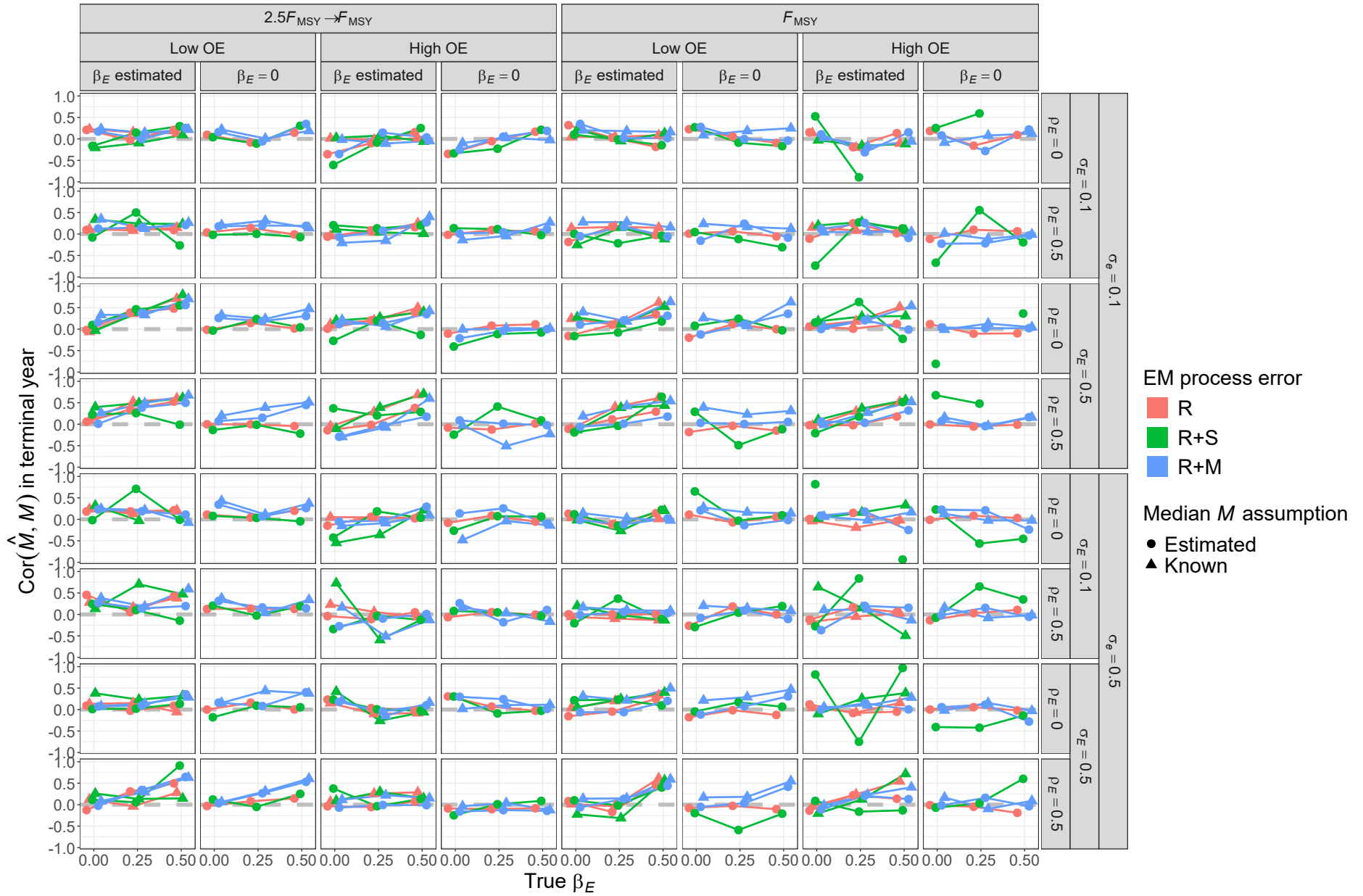


Fig. S52. For R+M OM, correlation of estimates and true values of terminal year M for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

905 **Terminal year spawning stock biomass bias**

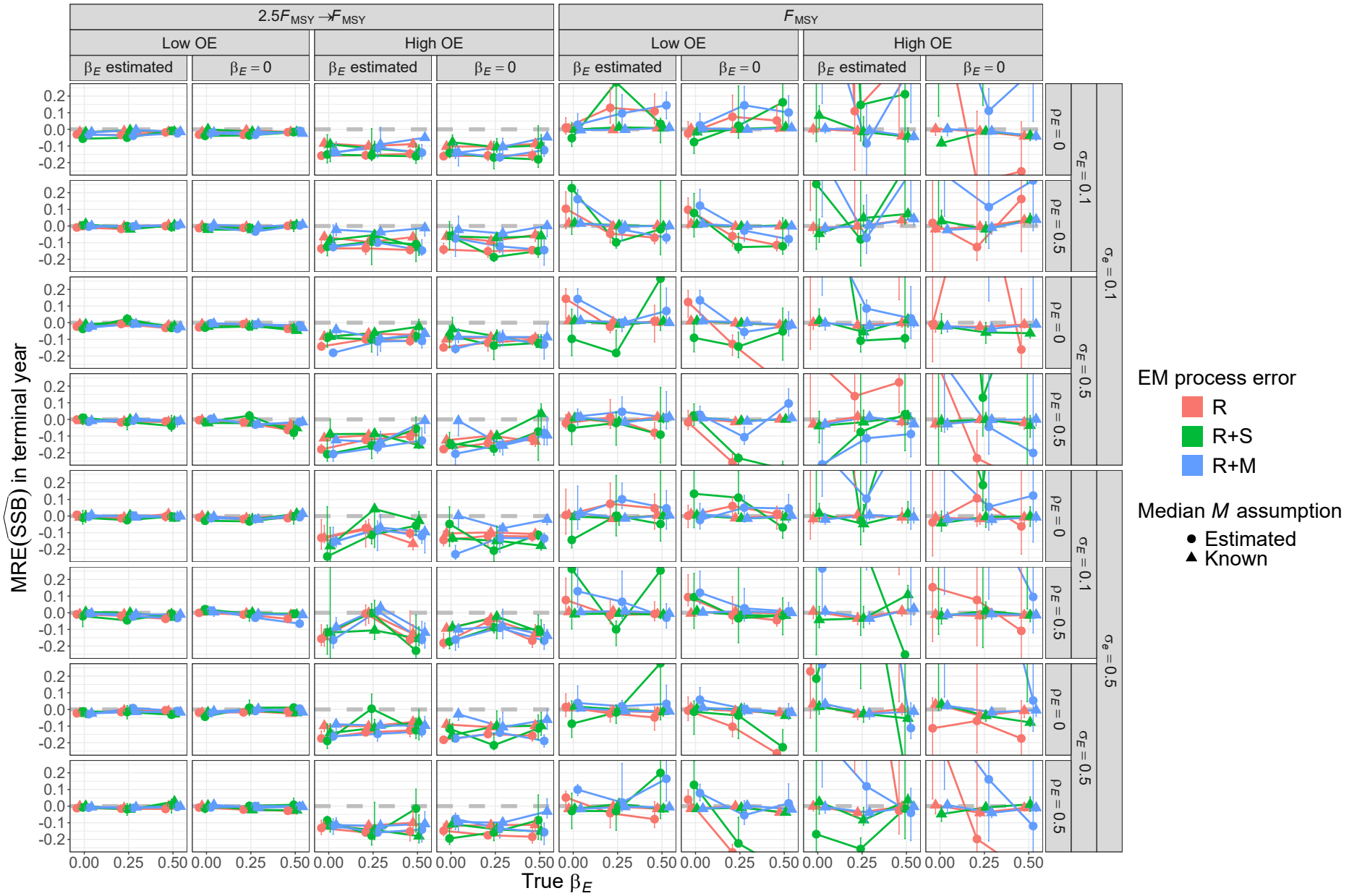


Fig. S53. For R OM, median relative error (MRE) of estimates of SSB in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

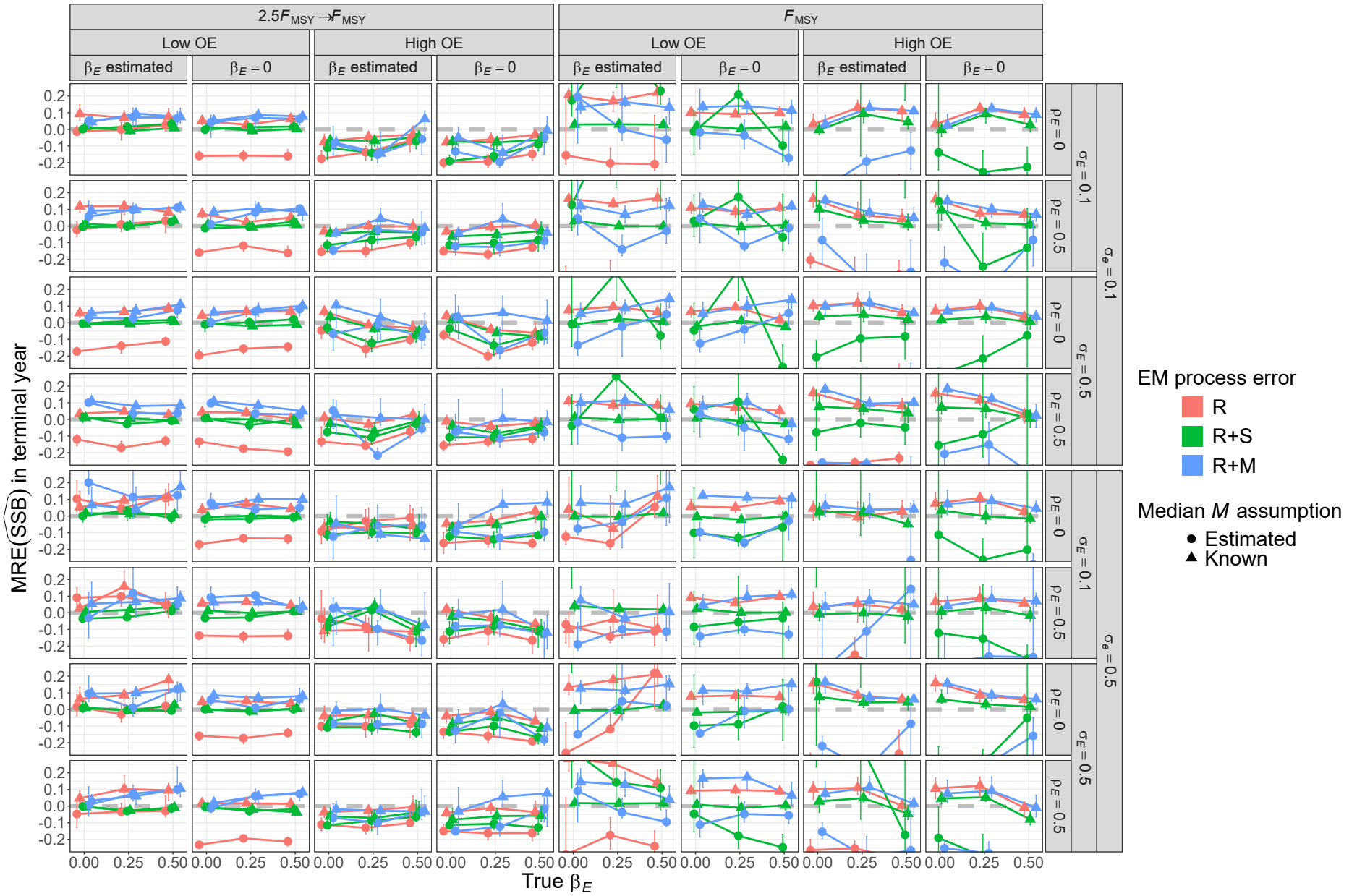


Fig. S54. For R+S OMs, median relative error (MRE) of SSB in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

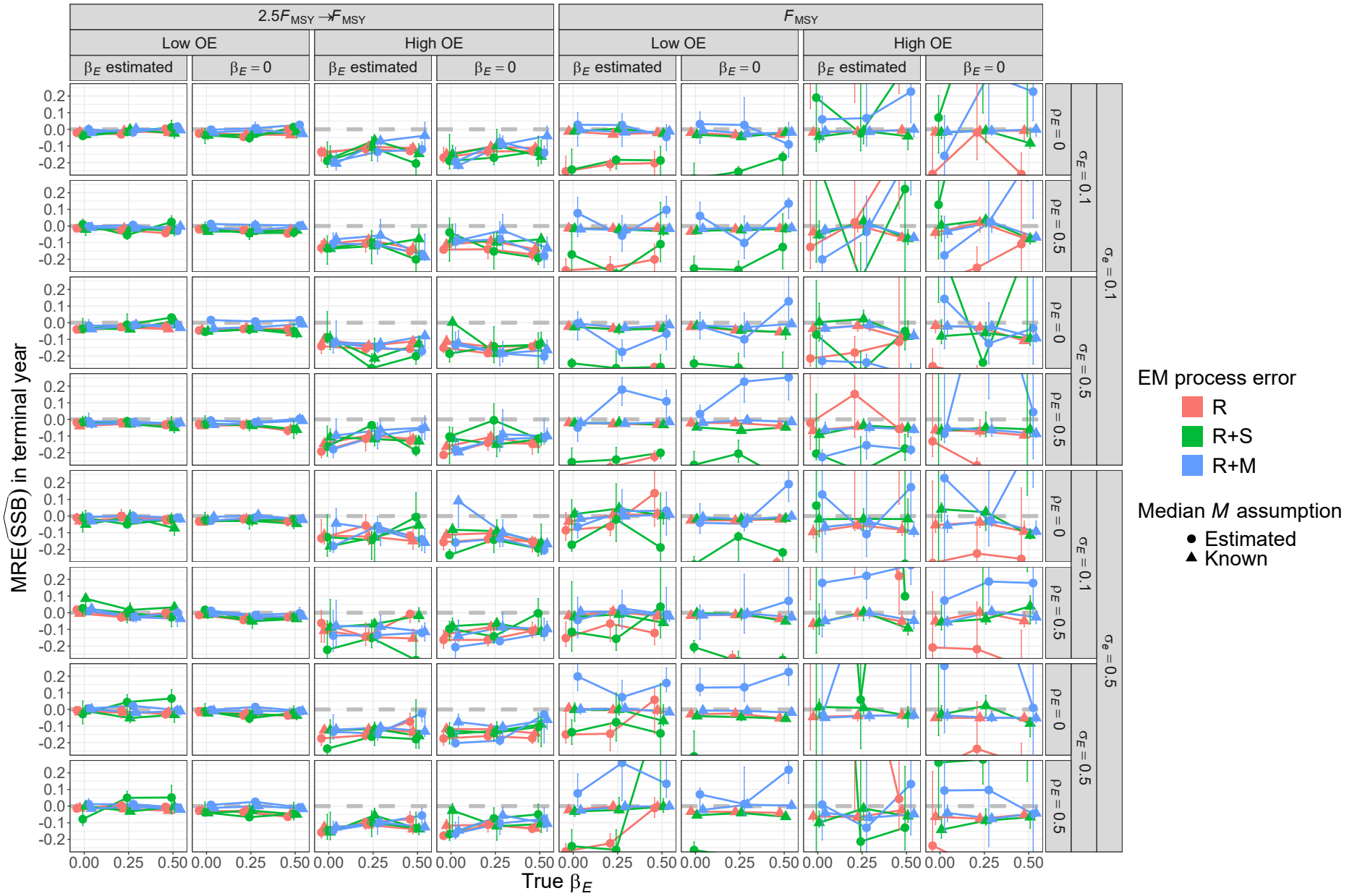


Fig. S55. For R+M OM, median relative error (MRE) of estimates of SSB in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

906 **Terminal year spawning stock biomass accuracy**

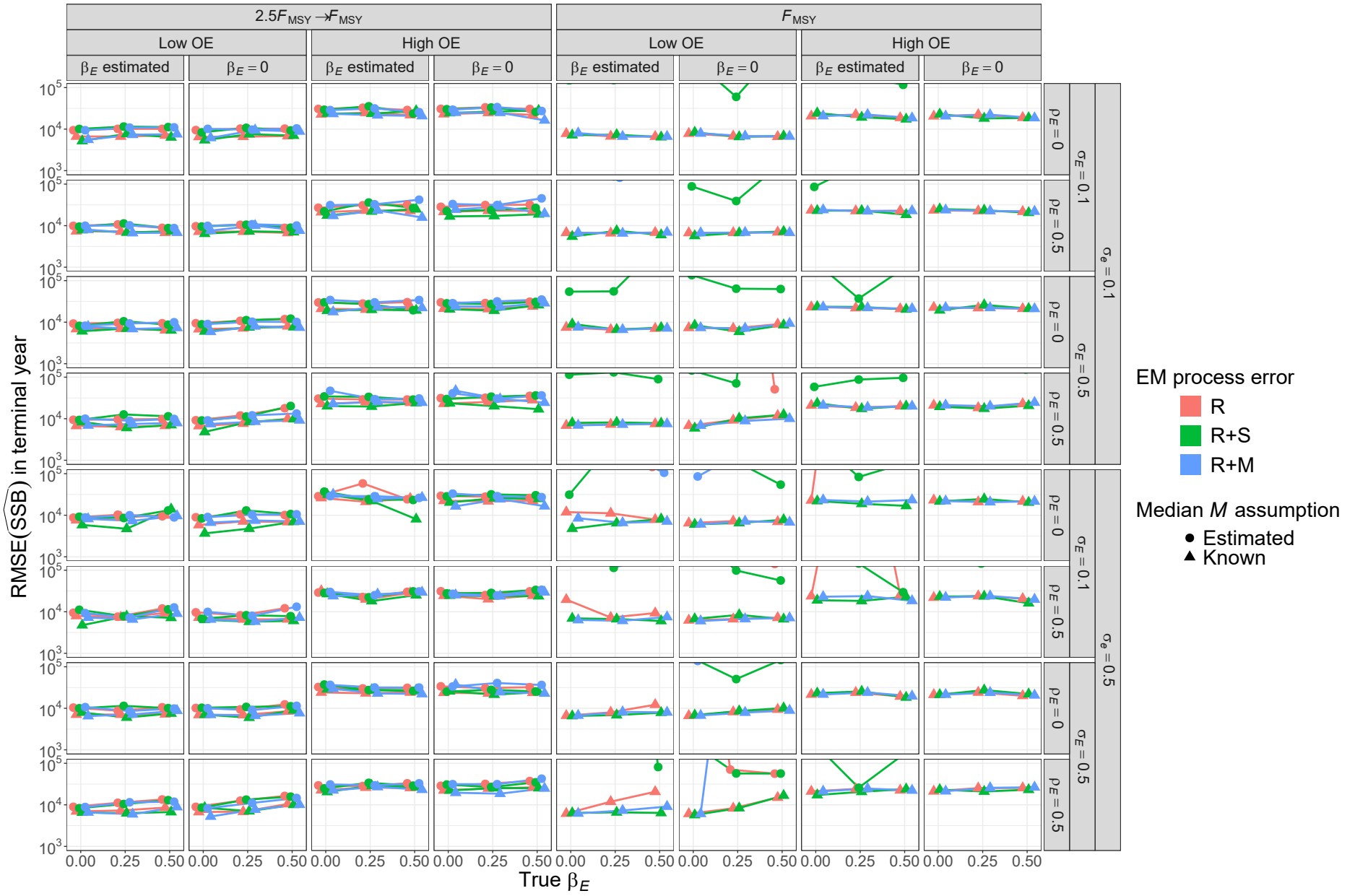


Fig. S56. For R OMs, RMSE of estimates of SSB in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

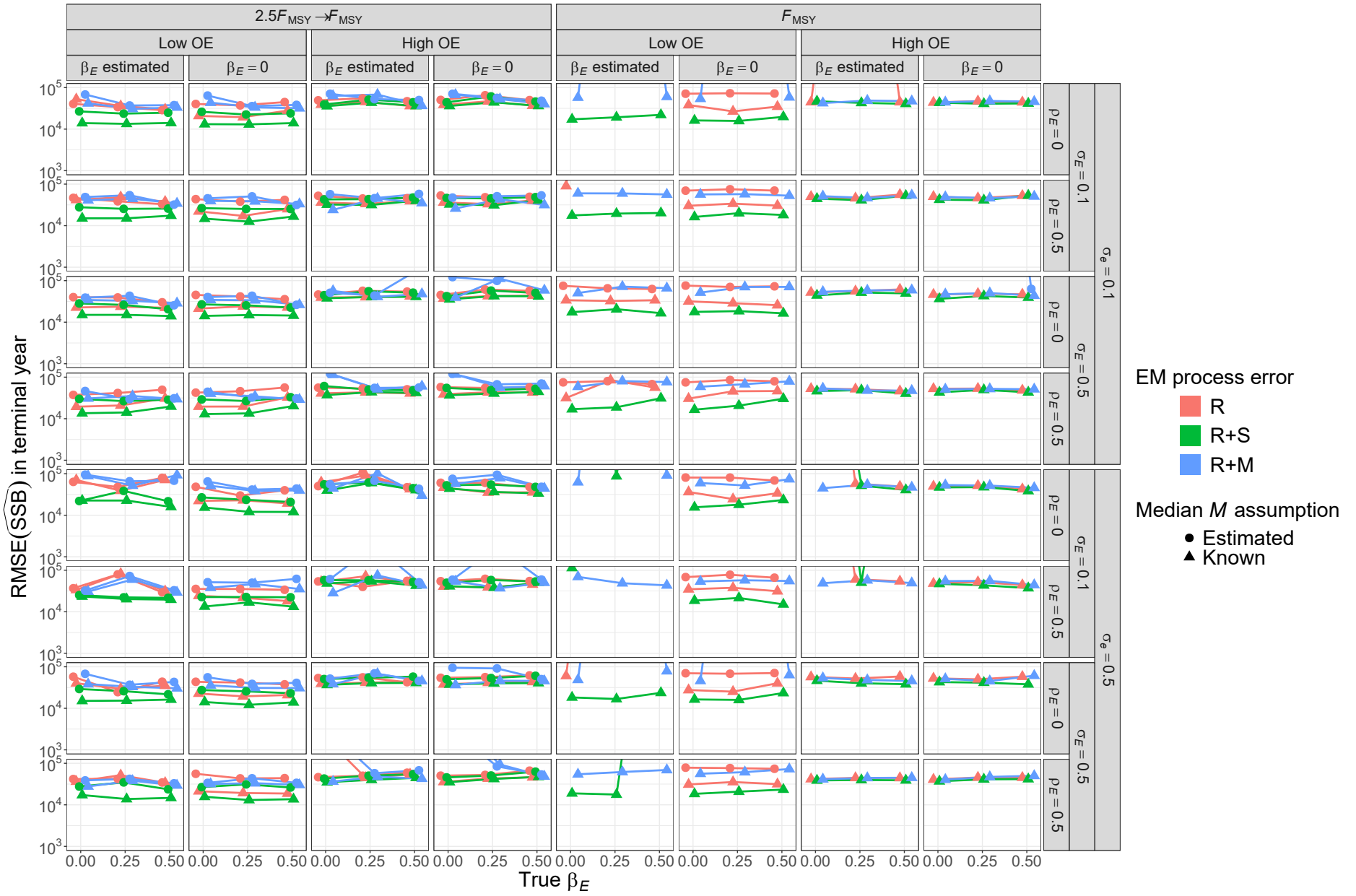


Fig. S57. For R+S OMs, RMSE of estimates of SSB in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

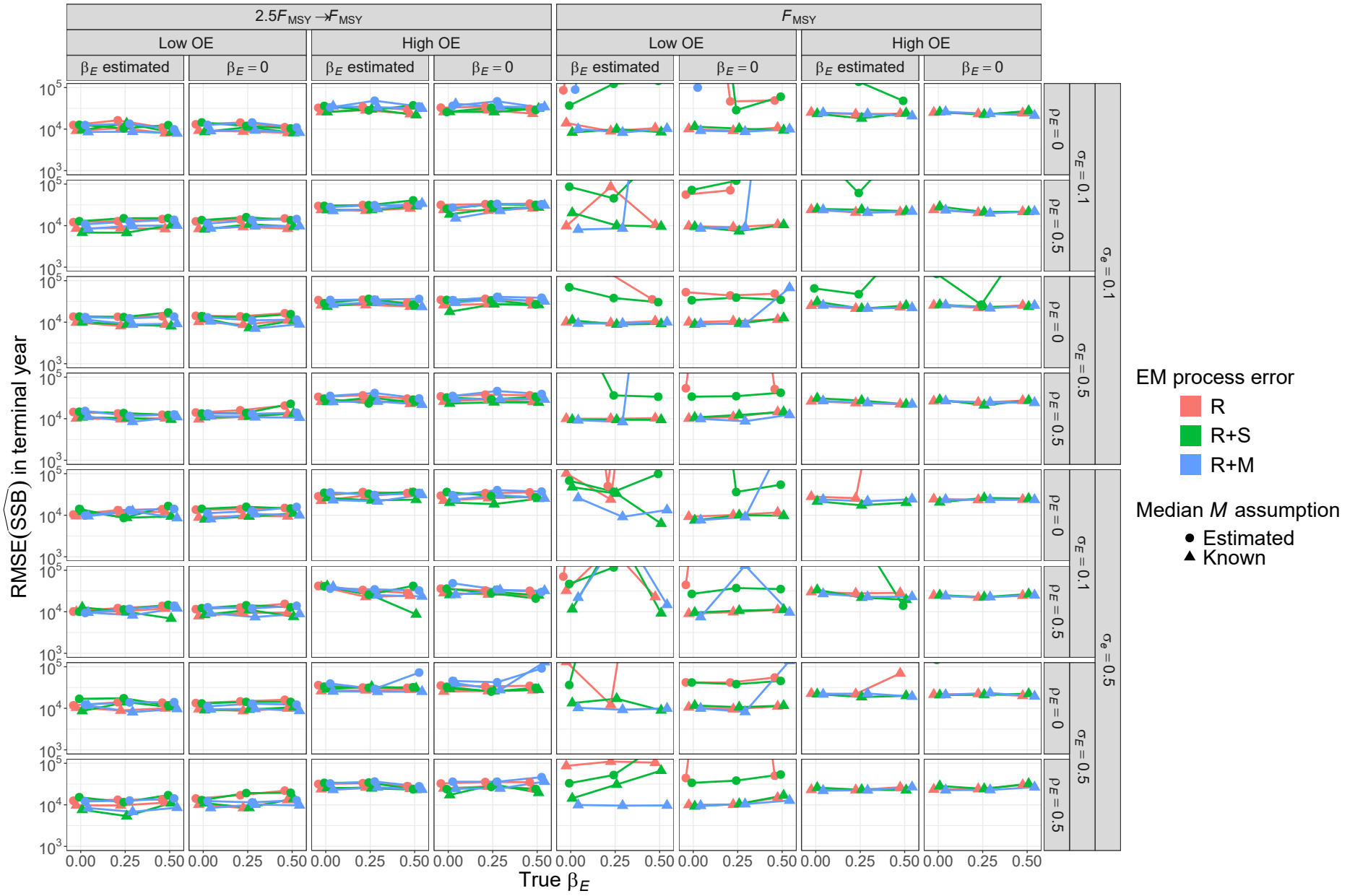


Fig. S58. For R+M OM, RMSE of estimates of SSB in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

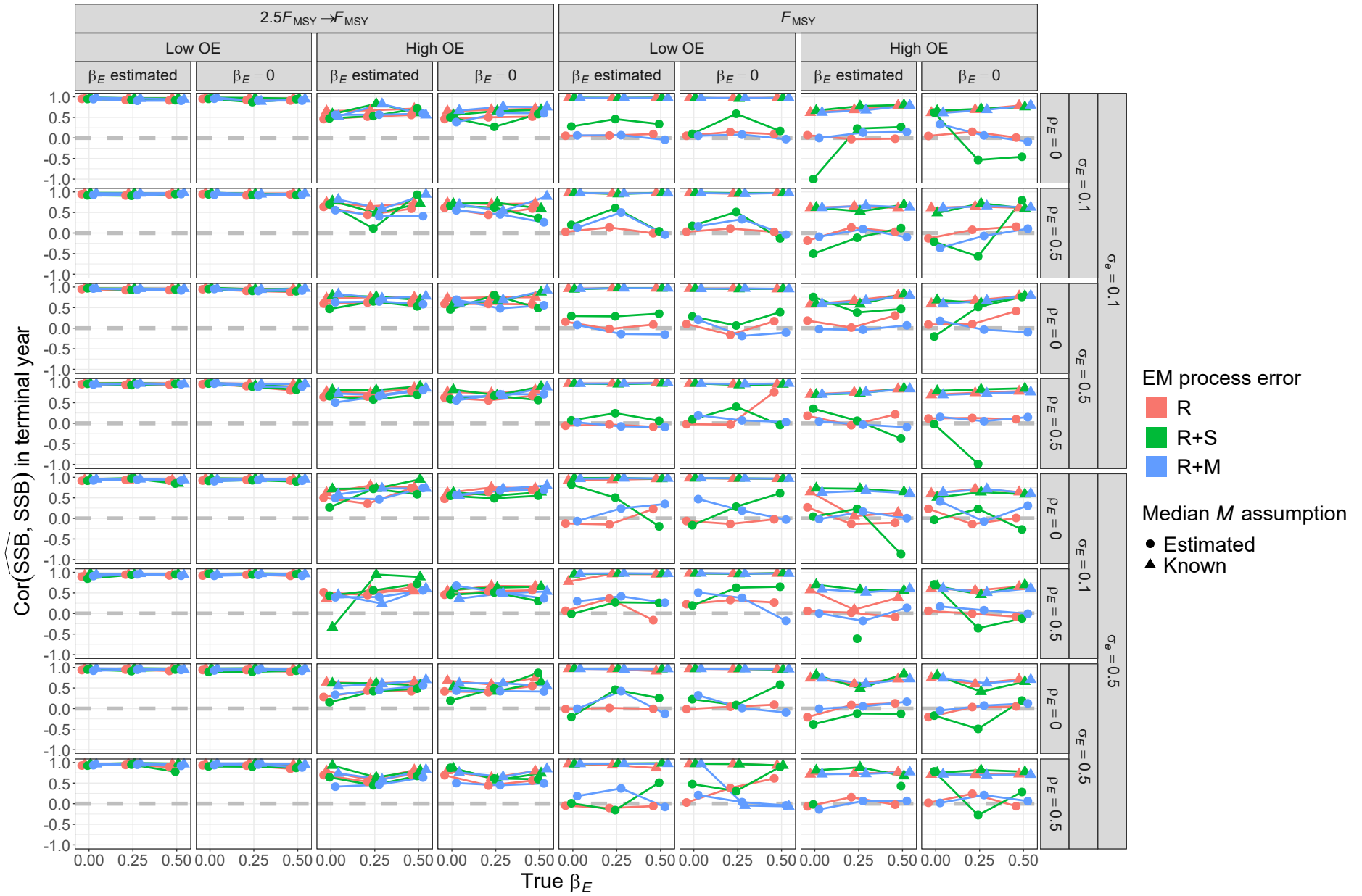


Fig. S59. For R OMs, correlation of estimates and true values of terminal year SSB for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

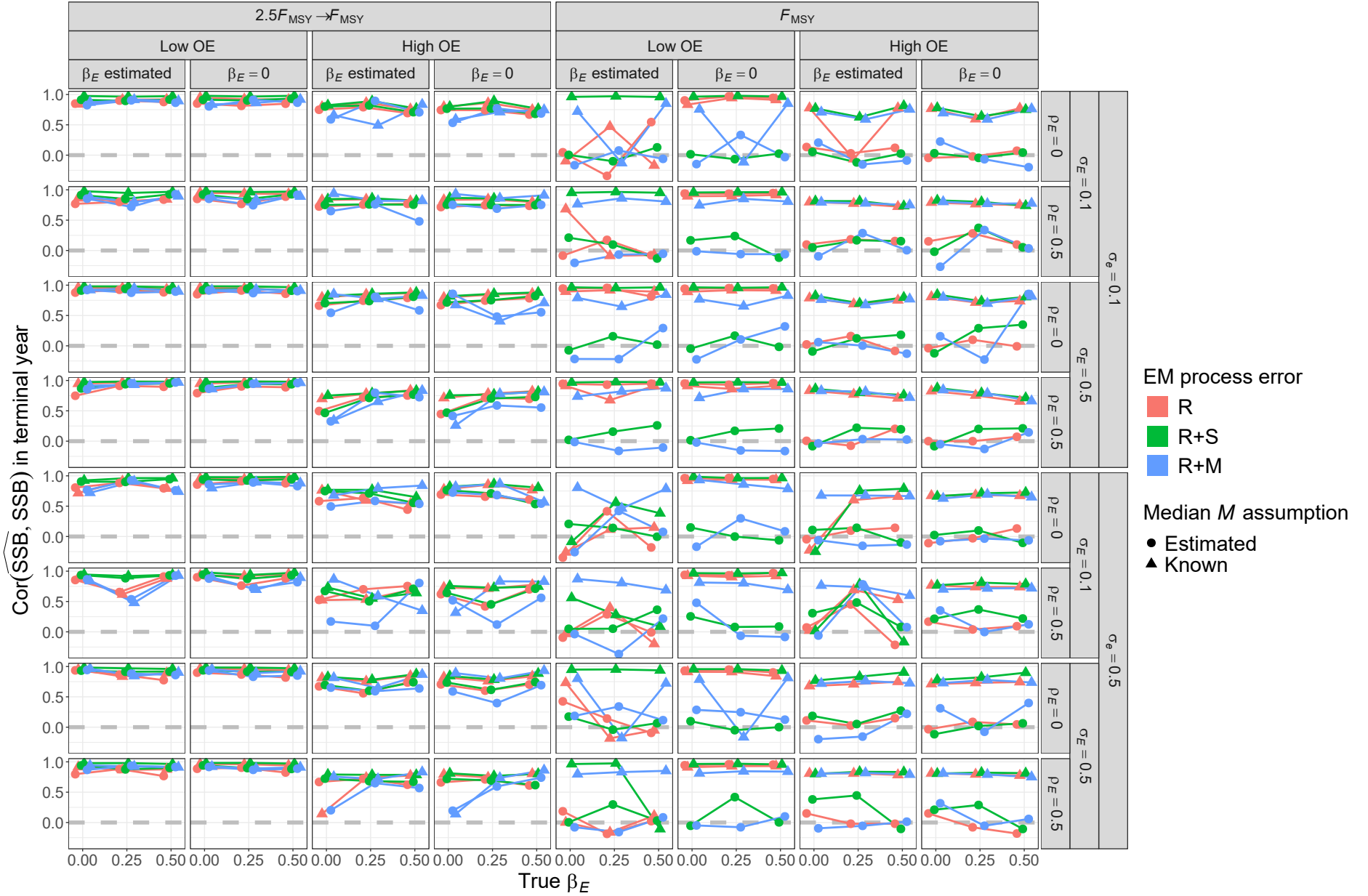


Fig. S60. For R+S OM, correlation of estimates and true values of terminal year SSB for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

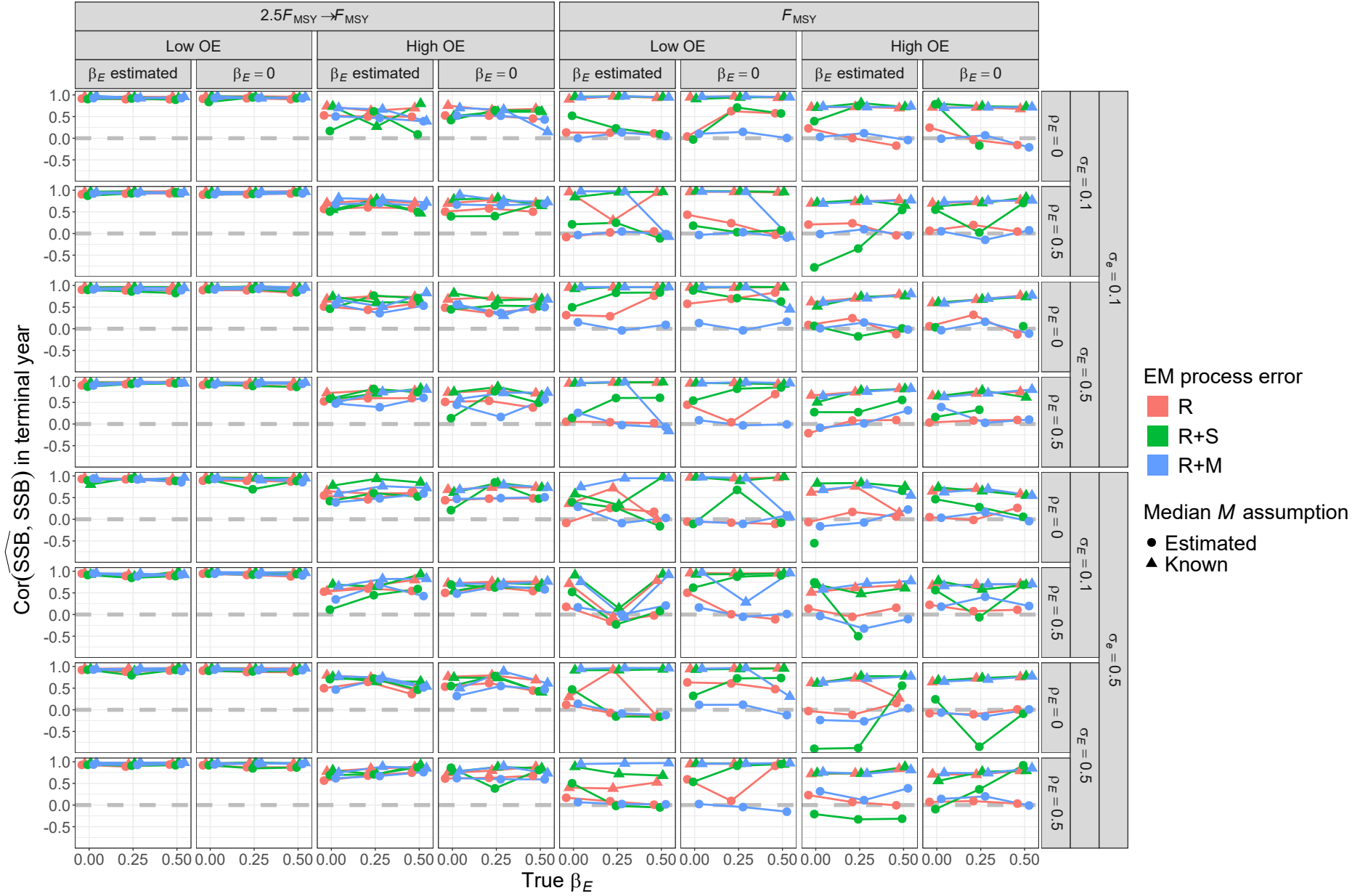


Fig. S61. For R+M OM, correlation of estimates and true values of terminal year SSB for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

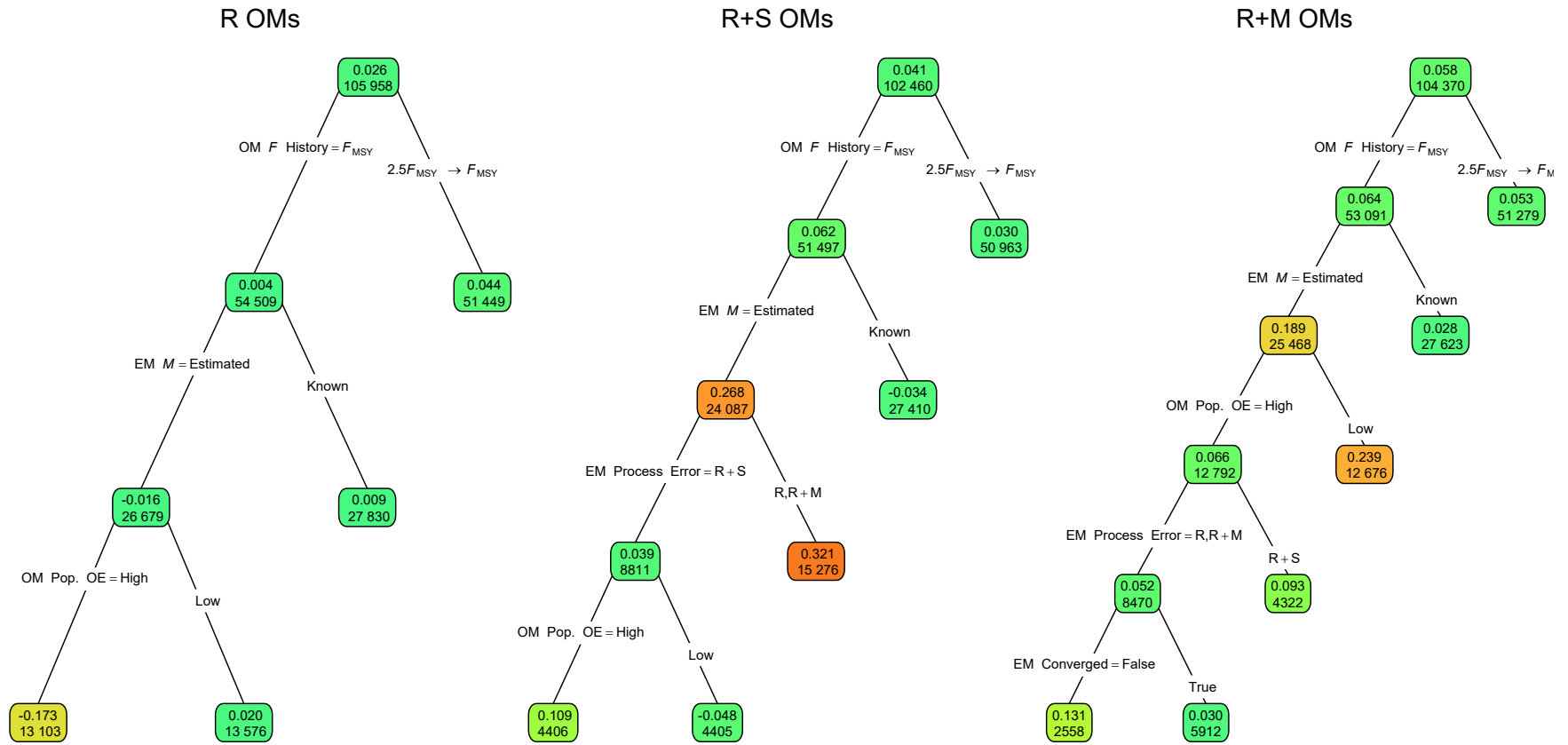


Fig. S62. Regression trees indicating primary factors determining reductions in sums of squares of errors in estimation measured by Eq. 2 for the terminal year fishing mortality rate in EMs fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median absolute errors of the process error assumption are indicated by more green or red polygons, respectively.

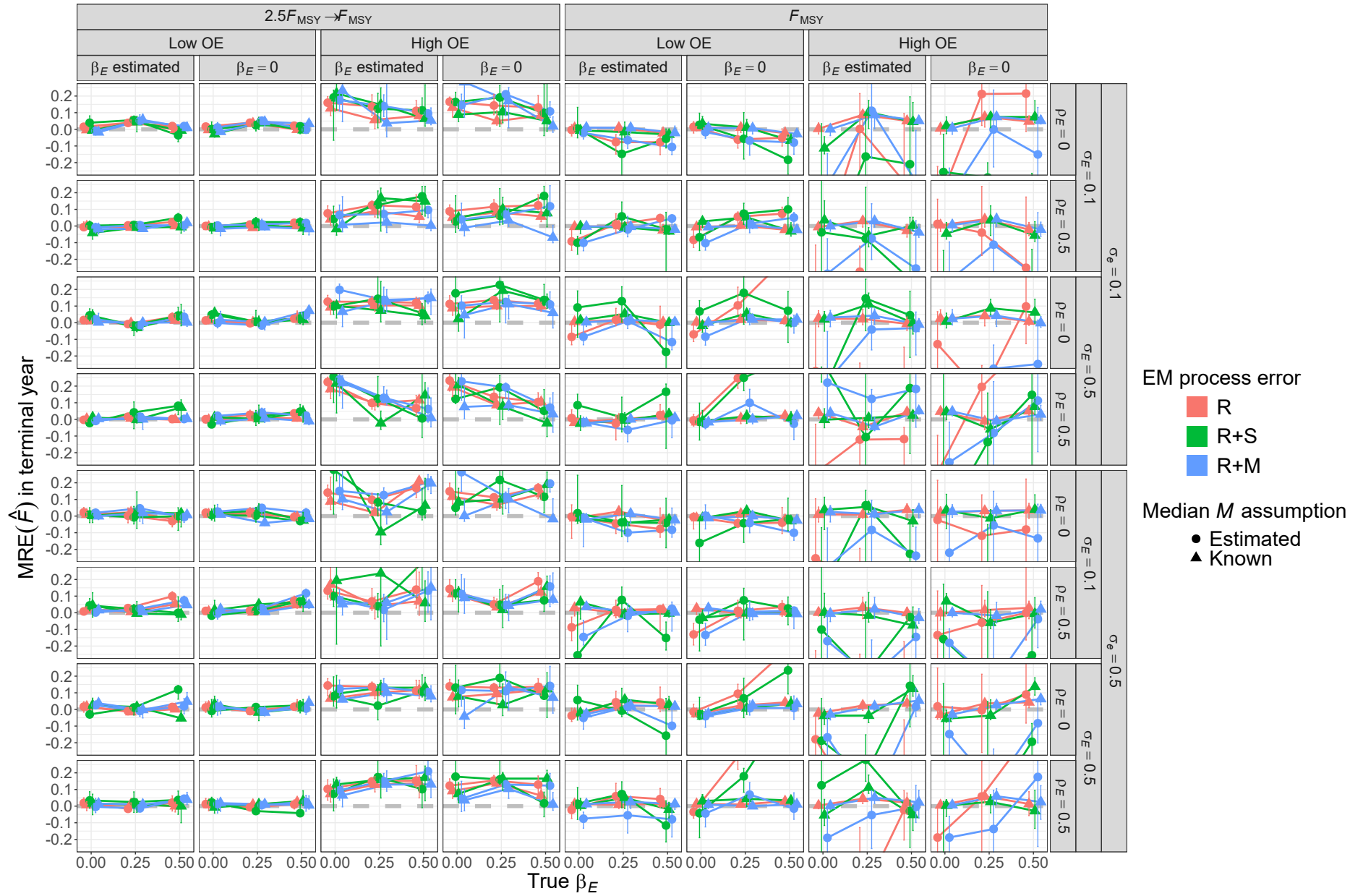


Fig. S63. For R OMs, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known). All OMs had low observation error and contrast in fishing mortality.

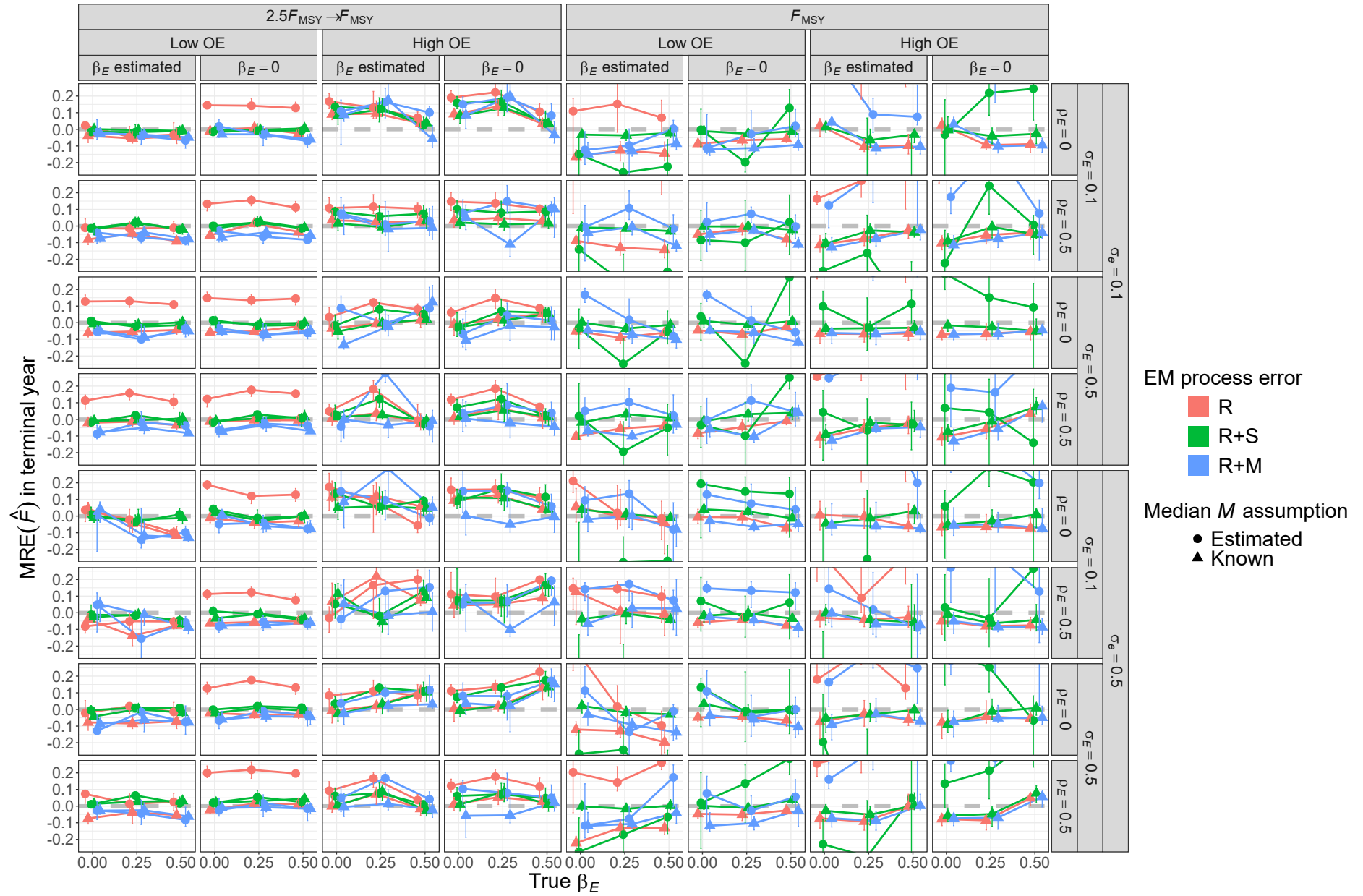


Fig. S64. For R+S OMs, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known). All OMs had low observation error and contrast in fishing mortality.

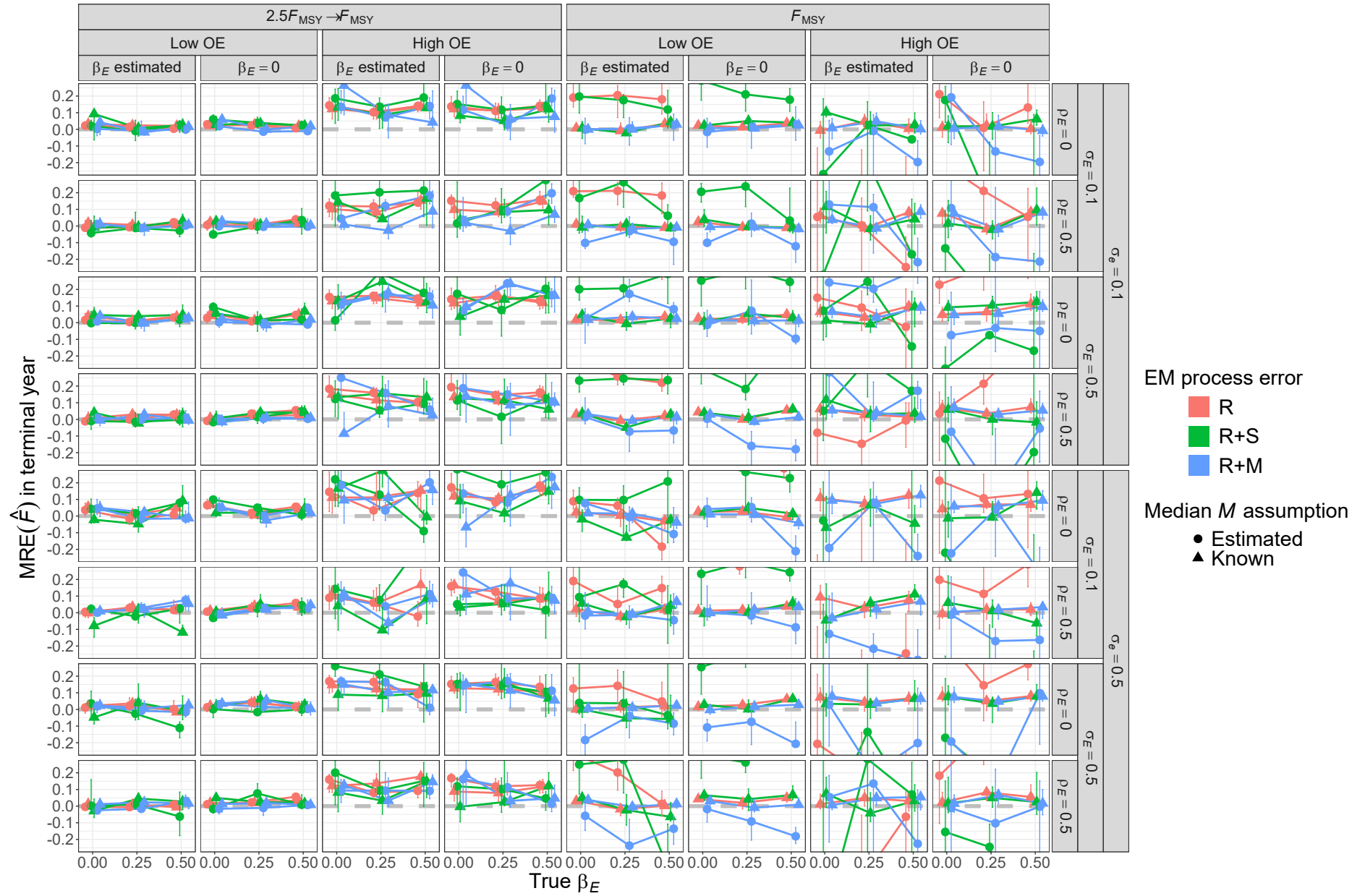


Fig. S65. For R+M OMs, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known). All OMs had low observation error and contrast in fishing mortality.

908 **Terminal year fishing mortality accuracy**

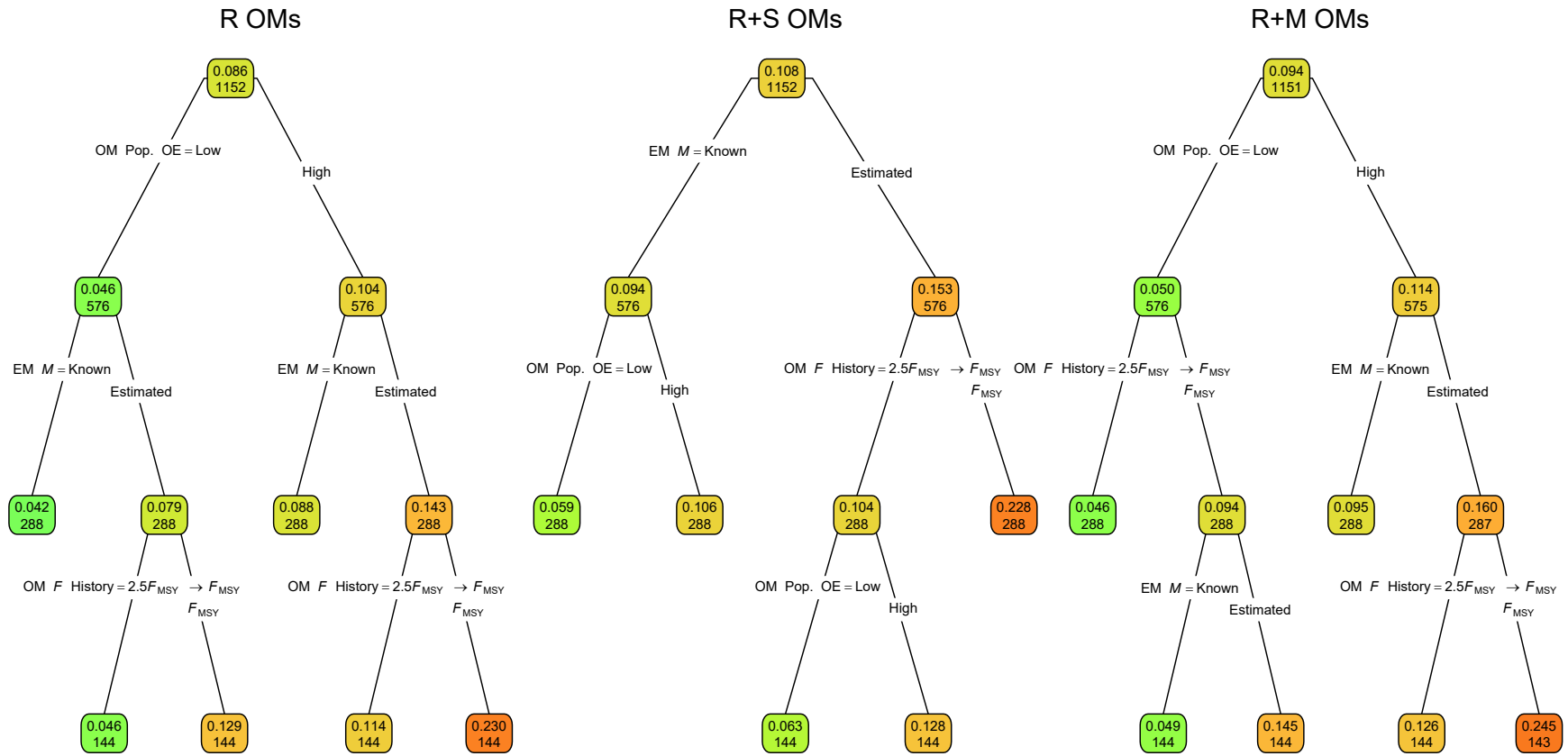


Fig. S66. Regression trees indicating primary factors determining reductions in sums of squares of errors for the log-transformed RMSE of terminal year fishing mortality rate in EMs fitted to operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Each node shows the median error (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more green or red polygons, respectively.

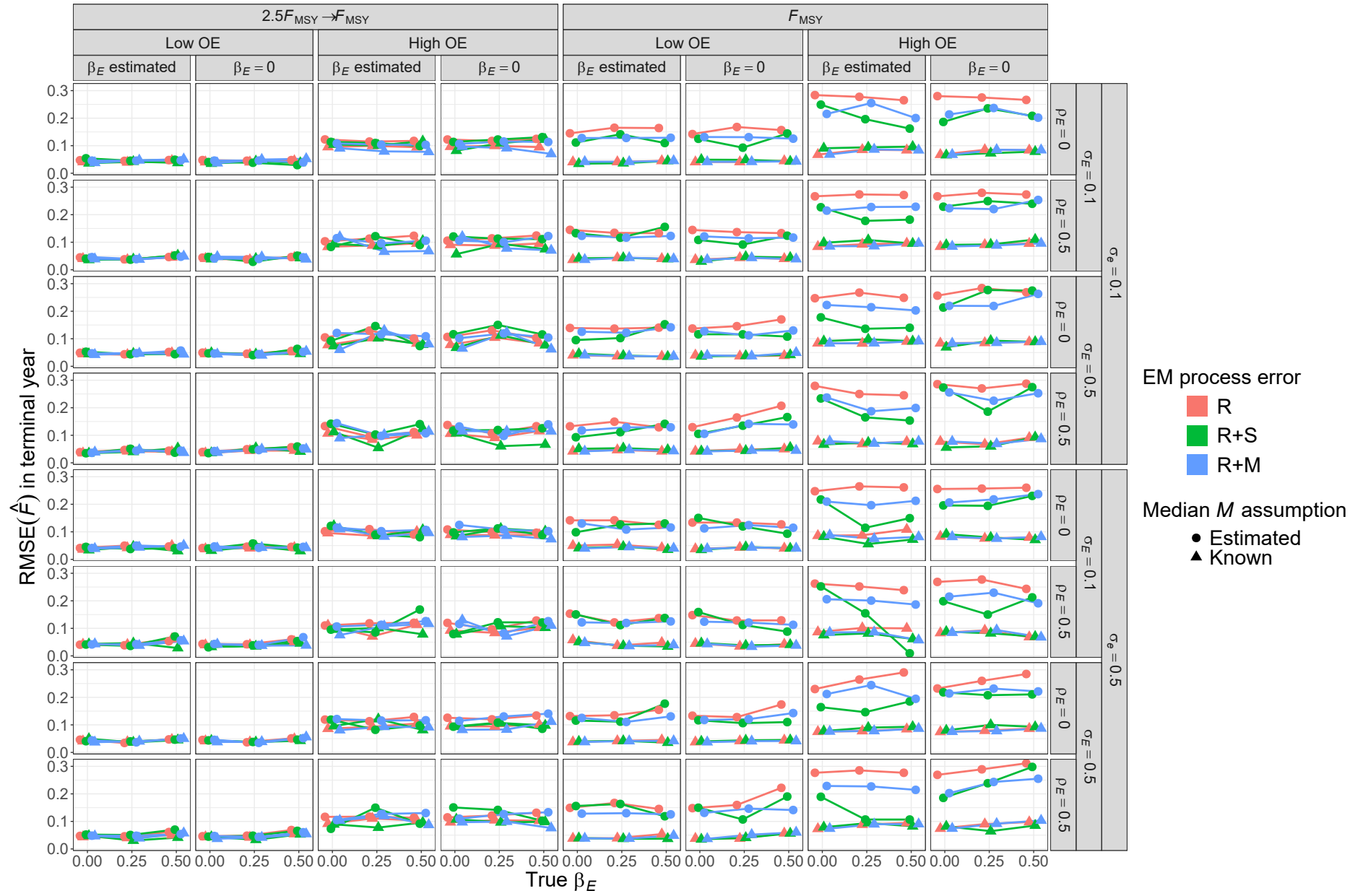


Fig. S67. For R OMs, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

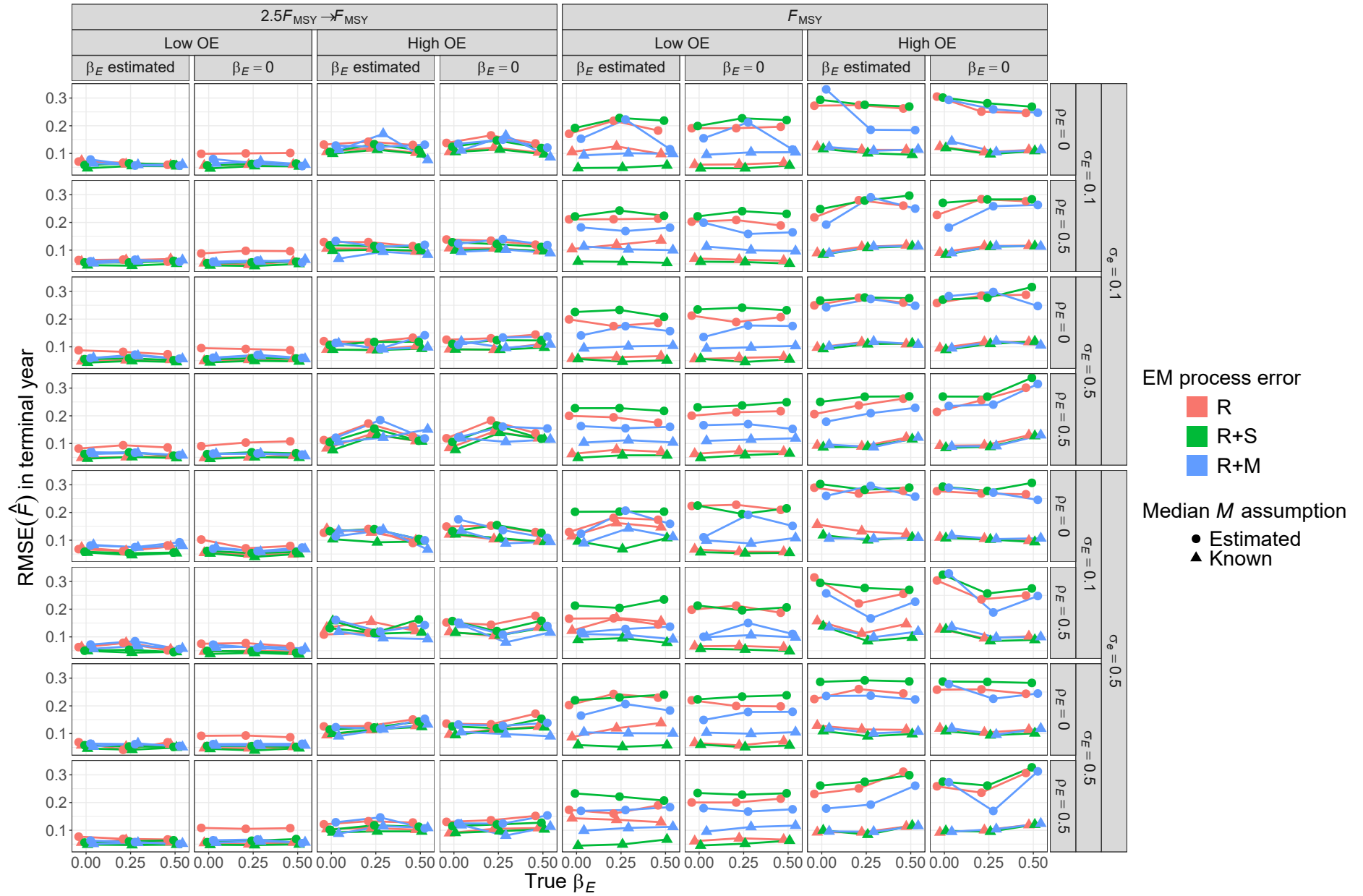


Fig. S68. For R+S OMs, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

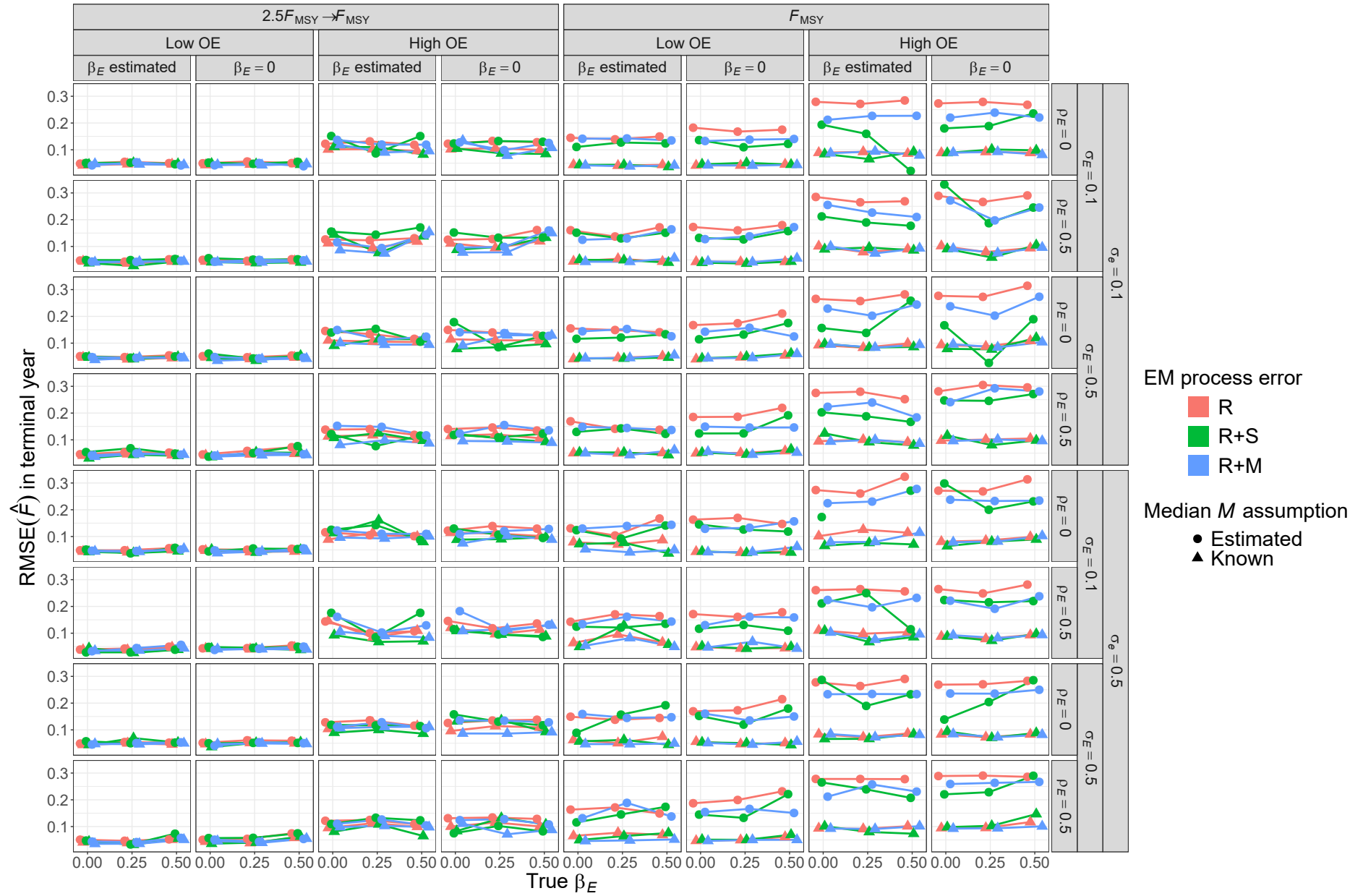


Fig. S69. For R+M OMs, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for EMs with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).